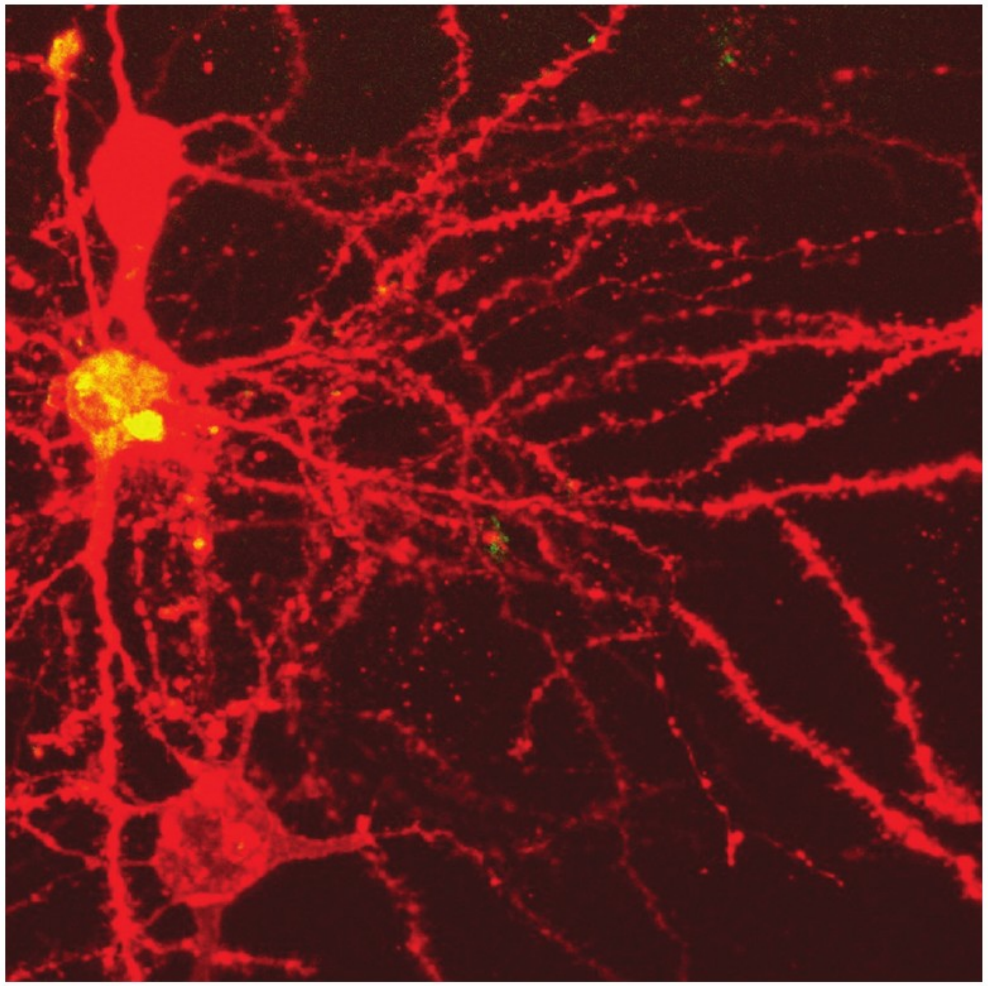


Mark N. Miller/University of California, Sf/Getty Images

Carlson and Birkett (2022), chapter opener, p. 119. Photo: Mark N. Miller / University of California, SF / Getty Images.

BEHAVIORAL NEUROSCIENCE 392

Discovery, Measurement, and Neuroengineering

# Reading and Rewriting the Brain: Tools of Discovery and Neuroengineering

Fall 2026

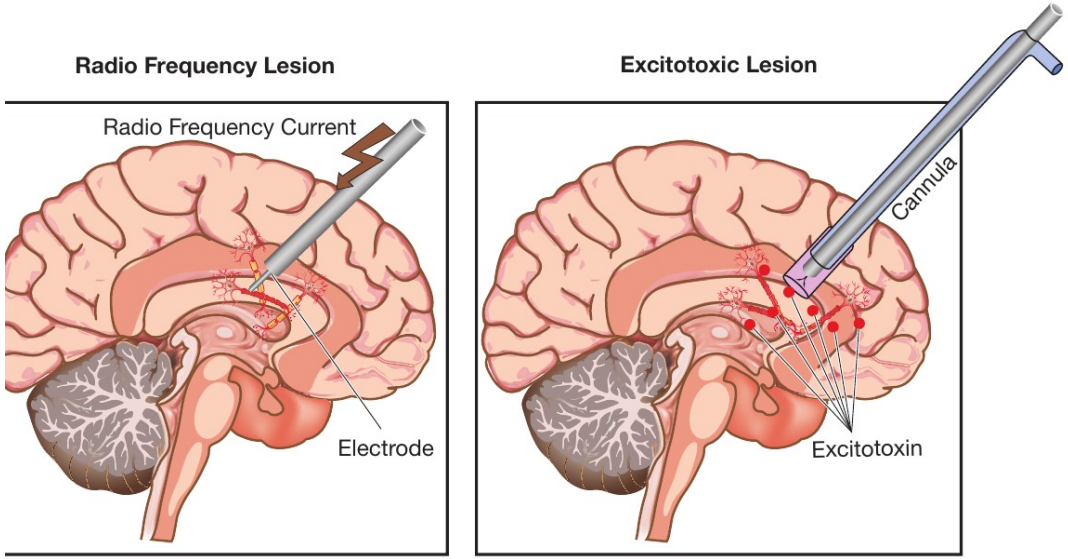
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

# Changing a circuit tests its contribution to behavior

A **lesion** is damage that interrupts the activity of nervous tissue. Researchers compare behavior before and after a lesion to identify the functions that depend on the affected cells. Difficulty reaching for food could reflect impaired movement, vision, or motivation, so the task must separate these different contributions to the same behavior.

**Recording** measures a signal produced by the nervous system, whereas **stimulation** changes the activity of the tissue. A neuron may fire during reaching because it helps plan movement, receives touch from the paw, or responds to food. Briefly interrupting its activity during a particular part of the reach tests a more specific explanation.

An experiment also needs a measurable behavior. The distance a paw travels, the time until a movement begins, and the direction of an error describe different parts of performance. Measuring these components allows researchers to distinguish a failure to initiate movement from a failure to guide an action that has already started.



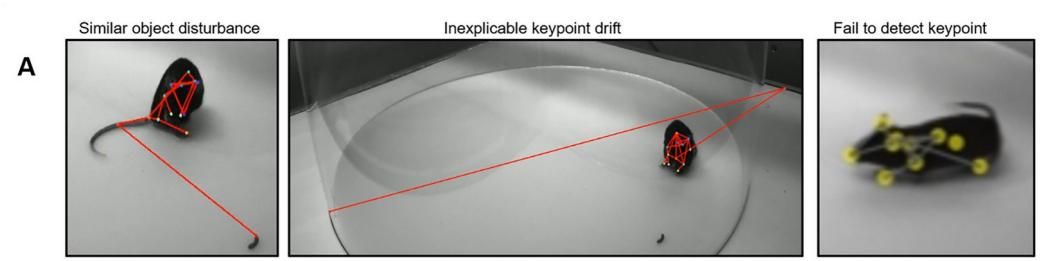

Carlson and Birkett (2022), Fig. 5.1, p. 122.

# Computer vision measures animal movement

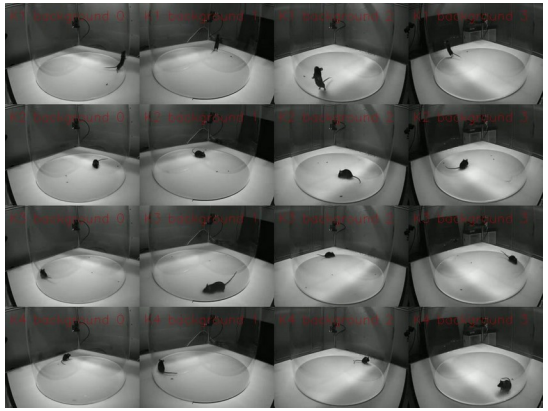
**Pose estimation** uses images to locate recognizable body parts, such as the nose, ears, and paws. Tang and colleagues developed the anti-drift pose tracker, or **ADPT**, to follow these locations during animal behavior. Their mouse recordings include changes in body position and background that can make an image-tracking system lose the intended feature.

A tracked point can suddenly jump from the animal's paw to an unrelated object. This error is called **tracking drift**. ADPT uses information about the arrangement of body parts to reduce these mistakes. A predicted paw position must fit the rest of the animal's pose, rather than simply resemble a paw in one small image patch.

Repeated position estimates form a **trajectory**, the path of a body part through time. Researchers can calculate movement speed from the distance between successive estimates and the camera's frame interval. Following individual paws separately also reveals when a mouse turns, rises onto its hindlimbs, or changes the order of its steps.




Tang et al. (2025), Fig. 1, selected panels.



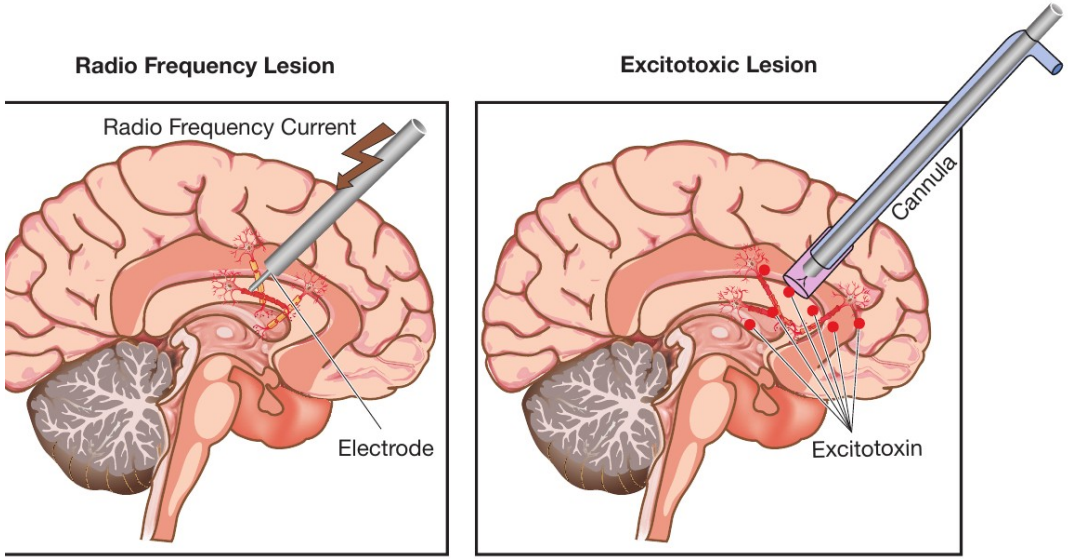
Tang et al. (2025), Fig. 3, video 1.

# A deficit can reflect several interrupted functions

**Experimental ablation** removes or destroys nervous tissue to test its contribution to a function. A damaged region may contain neurons that select an action, interpret a sensory cue, or coordinate the muscles that perform it. The resulting deficit therefore depends on which operations the task requires and which parts of the circuit remain available.

Consider an animal that stops approaching a food cue after a lesion. It might no longer recognize the cue, remember its meaning, or move toward the food. Testing whether it can detect the cue, approach another target, and eat food placed nearby separates these possible explanations through observable differences in its behavior.

A **double dissociation** provides a stronger separation between functions. Damage to one region might impair recognition while sparing movement, whereas damage elsewhere impairs movement while sparing recognition. The opposing patterns indicate that the two tasks depend differently on the affected tissue, even though both normally contribute to a successful approach toward food.



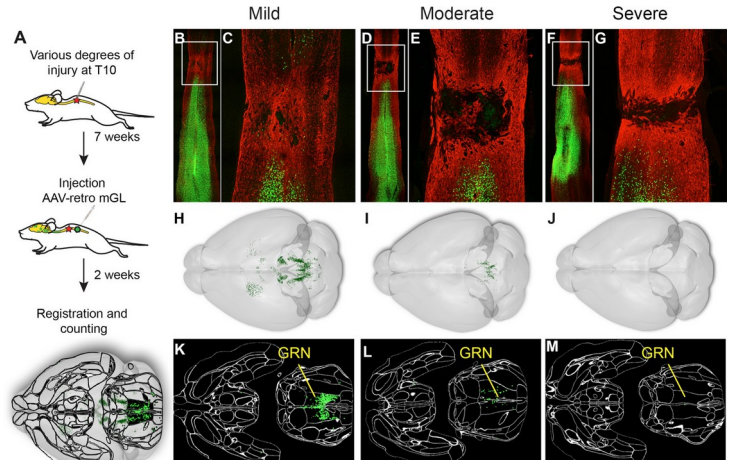

Carlson and Birkett (2022), Fig. 5.1, p. 122.

# Spinal injuries interrupt different descending populations

**Descending neurons** send axons from the brain toward the spinal cord, where local circuits help control movement and bodily functions. Wang and colleagues produced spinal injuries of different severities in mice. They then injected a tracing virus below the injury to label brain neurons whose axons still reached that lower spinal location.

The tracer entered axon endings and traveled back toward the cell bodies. Its fluorescent label therefore identified neurons with surviving access to the injection site. More severe injuries produced fewer labeled neurons across several brain regions. Clearing and imaging the whole brain allowed the investigators to locate these surviving populations in three dimensions.

The labeled populations included more than the familiar pathway from motor cortex. Brainstem and other descending systems also send signals to spinal circuits. Comparing their survival revealed that a spinal injury changes the composition of the remaining brain-to-spinal input, rather than simply weakening every descending pathway by the same proportion throughout the nervous system.




Wang et al. (2022), Fig. 5, selected panels.



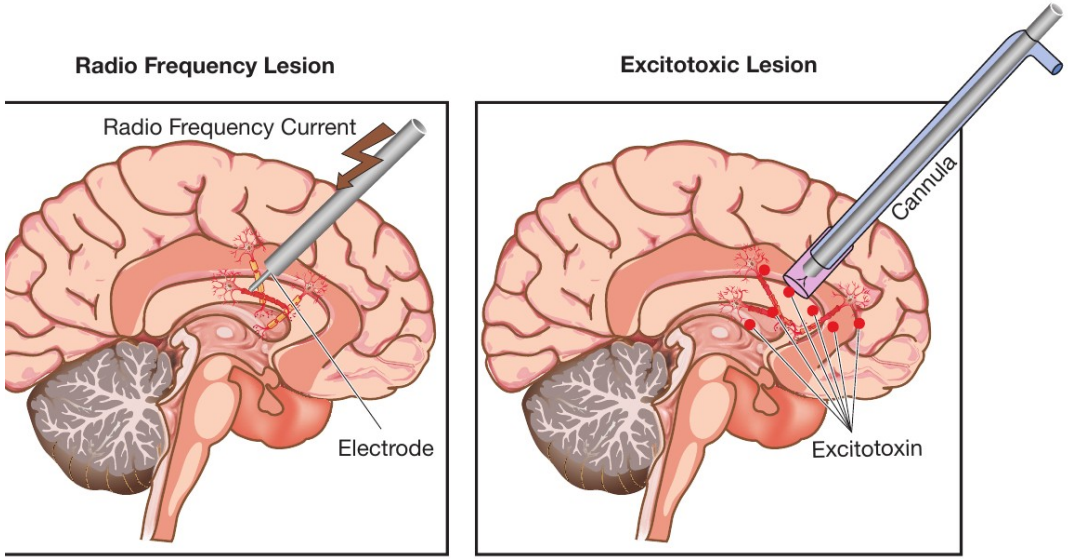

Wang et al. (2022), Fig. 3, video 2.

# Lesion methods differ in the tissue they damage

An **electrolytic lesion** uses electrical current to damage tissue near an electrode tip. The affected area can include neuronal cell bodies and axons passing through the region. If a behavioral deficit follows, the interrupted signals may have originated locally or traveled through the site from neurons located somewhere else in the brain.

An **excitotoxic lesion** uses a chemical that overstimulates receptors on neurons until the cells are damaged. This method can destroy local cell bodies while sparing many passing axons. Comparing its effects with those of an electrical lesion helps distinguish the contribution of neurons inside a region from signals traveling through that region.

**Reversible inactivation** temporarily suppresses activity using a drug or another intervention. Because the tissue can resume functioning, investigators can compare active and inactive periods within the same animal. Restricting the interruption to cue presentation, a delay, or movement execution separates contributions that would be combined in a permanent lesion extending across the entire task.



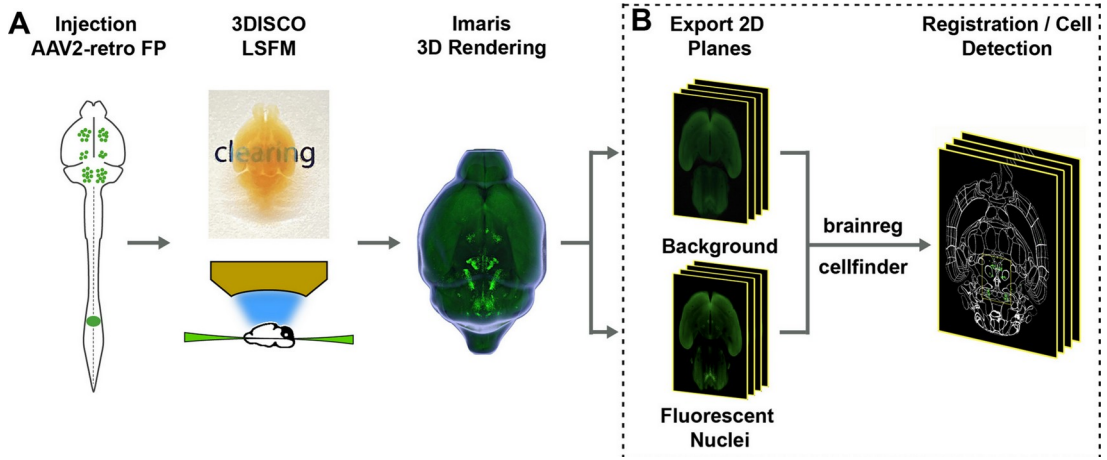

Carlson and Birkett (2022), Fig. 5.1, p. 122.

# Nuclear fluorescence makes labeled cells easier to count

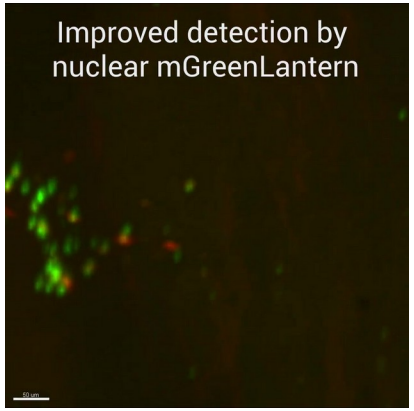
A fluorescent protein spread throughout a neuron fills its cell body and processes. In a densely labeled brain, these overlapping structures can obscure individual cells. Wang and colleagues attached the fluorescent label to **histone H2B**, a protein associated with DNA, so the signal accumulated inside each labeled neuron's nucleus rather than throughout its branches.

Each bright nucleus then provided a compact object for cell-detection software. The researchers combined this nuclear labeling with a retrograde virus injected into the spinal cord. Brain neurons became labeled because their axons reached that spinal location, and their nuclei provided separate points that could be counted in the cleared brain volume.

The workflow assigned detected nuclei to a digital anatomical atlas. This **registration** process aligns landmarks in an individual brain with corresponding landmarks in a reference brain. After alignment, the software could count labeled cells inside each named region, allowing the distribution of descending neurons to be compared across animals using the same anatomical boundaries.




Wang et al. (2022), Fig. 1, selected panels.



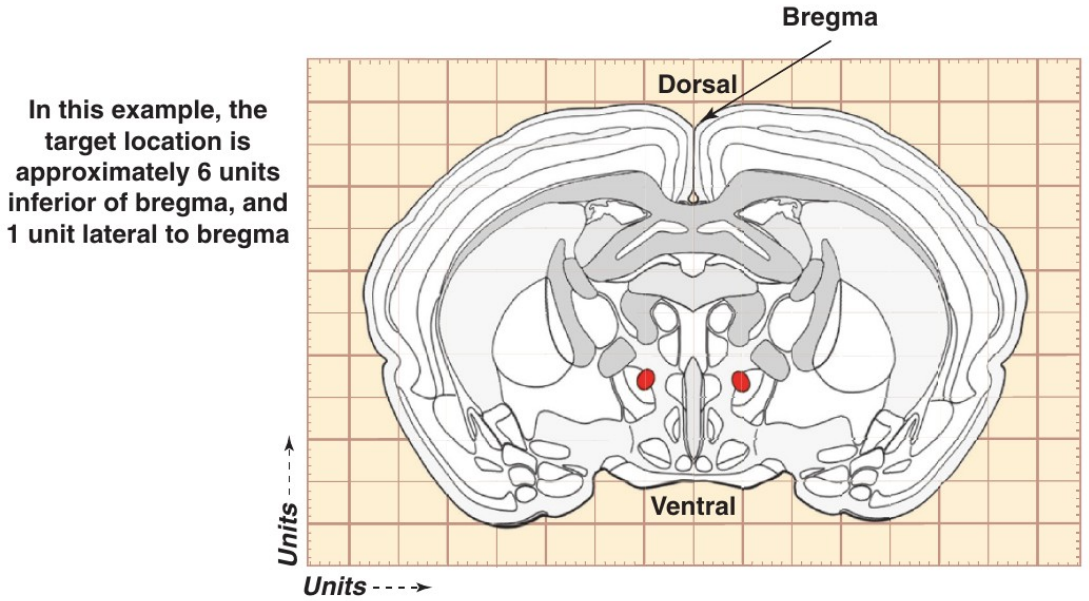

Wang et al. (2022), Fig. 1, video 1.

# Stereotaxic coordinates guide internal targeting

A **stereotaxic apparatus** holds the head in a reproducible position and moves an instrument along three axes. These describe front-to-back, side-to-side, and depth. A brain atlas relates those coordinates to internal anatomy, allowing an electrode, injection needle, or optical fiber to approach a target that cannot be seen through the intact brain surface.

A skull landmark called **bregma** marks the meeting of specific cranial sutures in rodents. Researchers measure the target's position relative to this landmark and lower the instrument to the planned depth. The atlas also shows neighboring structures, making the trajectory as important as the final location when selecting a path into deeper tissue.

After the experiment, tissue sections reveal where the instrument actually ended. A fluorescent injection or a visible probe track can be compared with anatomical landmarks in the section. This verification distinguishes a correctly targeted manipulation from one delivered to an adjacent structure, especially when neighboring cell populations produce different physiological or behavioral responses.



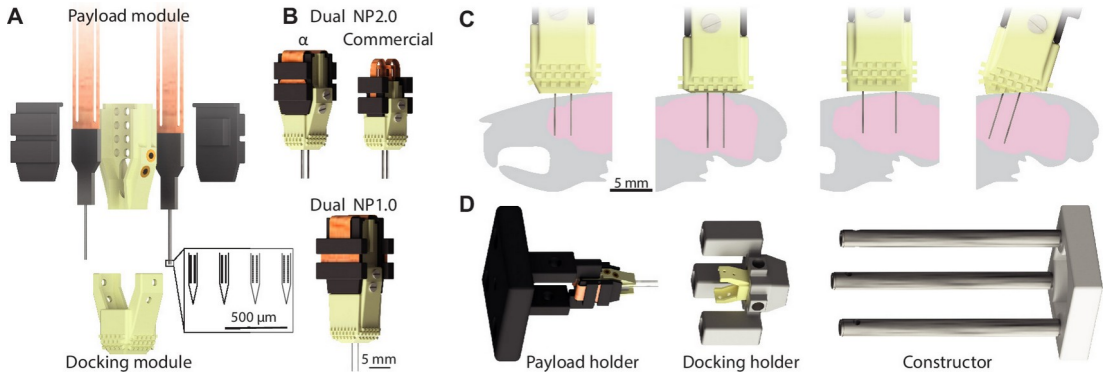

Carlson and Birkett (2022), Fig. 5.2, p. 124.

# A modular implant protects probes during chronic recording

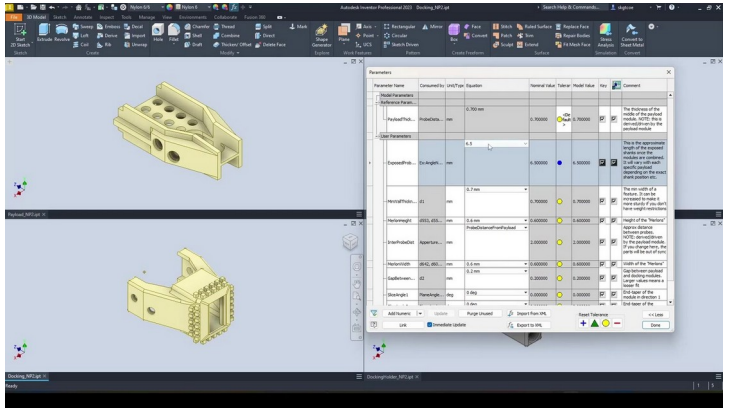
**Neuropixels probes** are narrow silicon devices containing many closely spaced electrical recording sites. Bimbard and colleagues developed the **Apollo implant** to hold these probes during repeated recordings. Its recoverable payload carries the probes, while a separate docking module attaches to the skull and stabilizes the assembly during movement and everyday housing.

The design can accommodate two probes with different insertion depths and angles. Changing these dimensions allows investigators to reach selected combinations of brain regions without redesigning the entire housing. The source demonstration shows how the implant's parameters are adjusted before fabrication, including the distance between probes and their intended paths through the brain.

Separating the payload from the skull-mounted dock also allows the probes to be recovered after an experiment. The investigators tested repeated implantation across laboratories and recorded activity over extended periods. Reusing the electronic probe requires protecting its thin shanks and flexible cables while releasing the surrounding implant, rather than pulling directly on the fragile recording surface.




Bimbard et al. (2025), Fig. 1, selected panels.



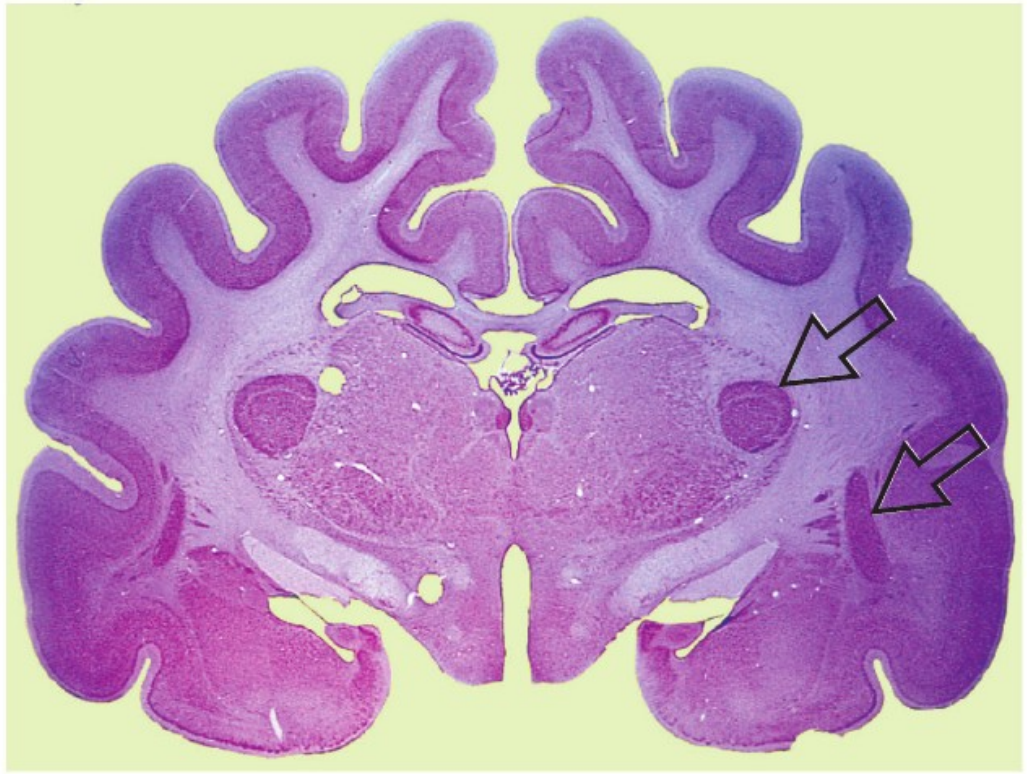
Bimbard et al. (2025), Video 1.

# Histology reveals the cells surrounding an experimental site

**Histology** examines thin sections of tissue under a microscope. After fixation preserves the tissue, a microtome or cryostat cuts sections thin enough for light to pass through. The section's orientation determines which structures appear together, so researchers match the cutting plane to the anatomical atlas used to plan the experiment.

A **Nissl stain**, such as cresyl violet, binds material concentrated in neuronal cell bodies. The resulting pattern reveals layers, nuclei, and boundaries between brain regions. A nucleus in this anatomical sense is a cluster of neurons; it differs from the cell nucleus that contains the DNA inside an individual neuron or other cell.

A lesion or instrument track can then be located relative to these landmarks. Missing cell bodies, disrupted layers, or a channel through the tissue identifies the affected area. Examining several consecutive sections reveals the site's extent in depth, which a single attractive image cannot fully describe when the manipulation spreads through a three-dimensional brain region.



Carlson and Birkett (2022), Fig. 5.6, p. 126.

# Expansion microscopy enlarges neural tissue

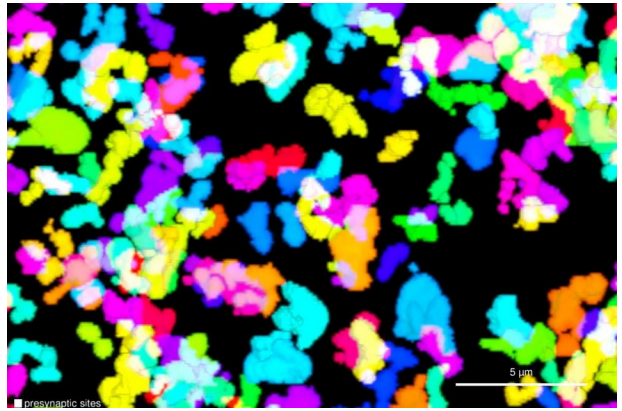
Nearby synapses can lie closer together than a light microscope can resolve. **Expansion microscopy** makes the specimen physically larger by anchoring its labeled molecules to a swellable polymer. After the tissue is processed and the polymer absorbs water, neighboring labels move farther apart, allowing a conventional optical system to distinguish smaller original separations.

Lillvis and colleagues combined tissue expansion with **lattice light-sheet microscopy** in fruit fly brains. A thin sheet of light illuminates one plane while a camera records its fluorescence. Moving through successive planes produces a three-dimensional image volume containing labeled neuronal membranes and proteins concentrated at potential sites of communication between neurons.

Their workflow used roughly eightfold linear expansion to examine presynaptic structures. A **presynaptic site** contains the machinery that releases transmitter from an axon ending. Labeling the protein Bruchpilot identified these sites, while a separate membrane label outlined the neuron carrying them. The enlarged images made it possible to assign release-site candidates to particular neuronal branches.

| | cell type specific | protein specific | synaptic resolution connectivity | fast |
|---|---|---|---|---|
| traditional light microscopy | + | + | - | + |
| electron microscopy | - | - | + | - |
| ExLLSM | + | + | + | + |

Lillvis et al. (2022), Fig. 1, selected panels.



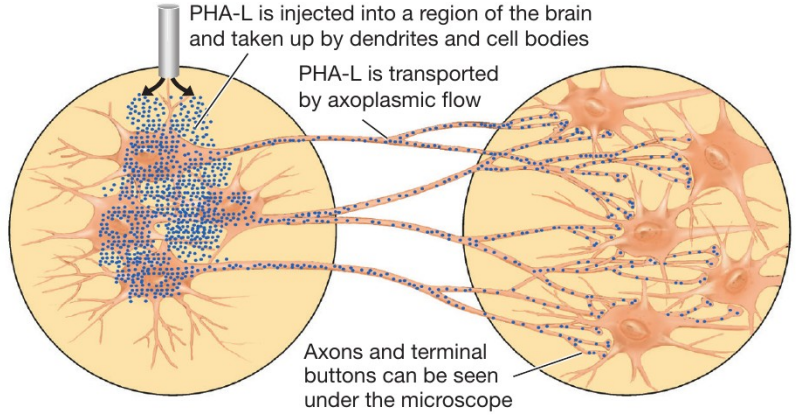

Lillvis et al. (2022), Video 1.

# Anterograde tracing follows signals away from cell bodies

An **anterograde tracer** moves from a neuron's cell body toward its axon endings. Injecting the tracer into a brain region labels some neurons near the injection and reveals where their axons terminate. Those labeled destinations identify potential output targets of the injected population, including branches that reach several different regions from the same source.

Neurons transport materials along their axons using molecular motors that move on internal protein tracks. Tracers exploit this transport instead of requiring an investigator to follow every axon visually from start to finish. After transport, histological processing or fluorescence reveals the accumulated label in distant axons and their terminal branches within the surrounding tissue.

The injection site defines which population begins labeled. A large injection may include several neighboring nuclei or cell classes, producing a mixture of output pathways. Restricting the injection and examining its borders makes the traced projection easier to interpret because the distant axon label can be assigned to a more precisely defined source population.



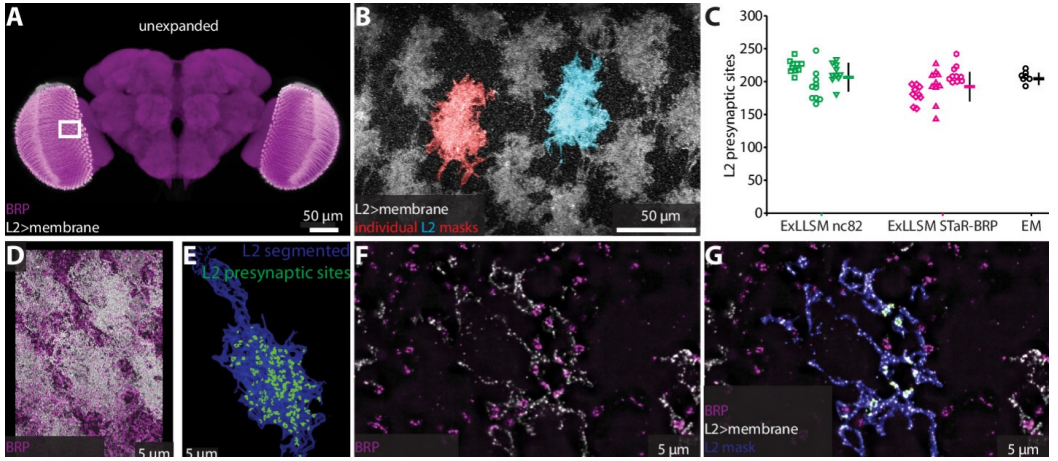

Carlson and Birkett (2022), Fig. 5.11, p. 129.

# Separate labels identify branches and release sites

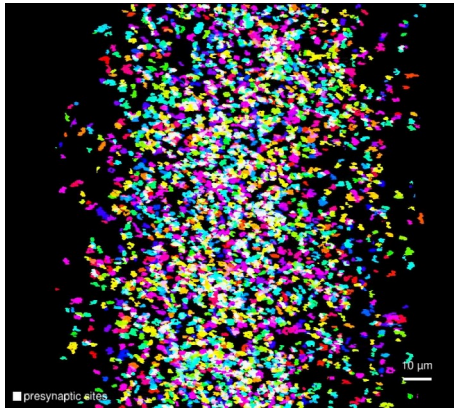
In the expanded fly brain, Lillvis and colleagues labeled the membrane of **L2 neurons**, cells in the visual system, separately from their presynaptic proteins. The membrane channel outlined each neuron's branches. The Bruchpilot channel marked small release-site structures, allowing the investigators to examine how those structures were distributed along a recognizable individual neuron.

**Segmentation** assigns image pixels or three-dimensional voxels to an object. The investigators segmented an L2 neuron to create a mask describing its volume. Presynaptic objects lying inside that mask could then be assigned to the neuron, separating its release-site candidates from nearby structures belonging to other cells in the same densely packed visual tissue.

The source rotation displays the spatial arrangement of these labeled structures along the reconstructed cell. Each object comes from fluorescent imaging of the expanded specimen rather than an artist's estimate of synapse placement. The analysis could therefore compare the number and distribution of presynaptic sites across neurons and animals while retaining each cell's anatomical identity.




Lillvis et al. (2022), Fig. 3, selected panels.




Lillvis et al. (2022), Video 2.

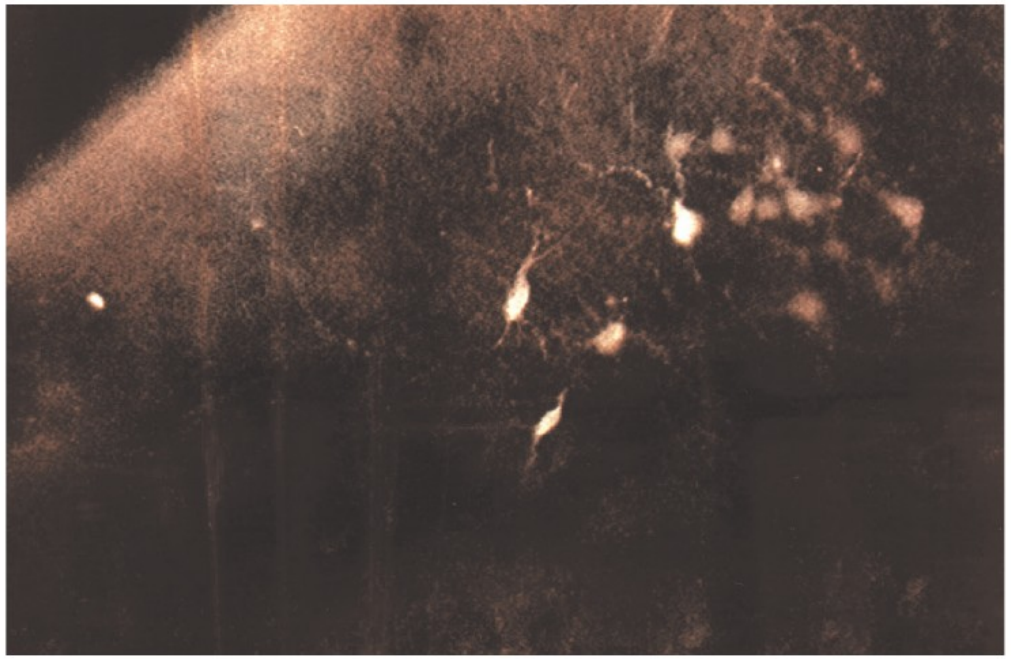
# Retrograde tracing identifies neurons that send an input

A **retrograde tracer** enters an axon and moves back toward the cell body. Injecting it into a target region reveals neurons that project to that location. This reverses the direction of anterograde tracing: the injection marks a destination, and the labeled cell bodies reveal possible sources of input to that destination elsewhere in the nervous system.

Some viruses can deliver a gene for a fluorescent protein to neurons reached through their axon endings. The infected cells then manufacture the fluorescent protein themselves. Selecting a reporter that accumulates in nuclei makes individual projecting cells easier to locate when many labeled axons and dendrites would otherwise overlap in the microscope image.

Two distinguishable tracers can test whether neurons reach two targets. A cell containing both labels has axonal access to both injection sites under the tracing conditions. A cell containing one label was detected from only one target. Comparing the labels within individual cell bodies distinguishes branching projections from neighboring neurons that project to different destinations.

Source: Courtesy of Yvon Delville, University of Massachusetts Medical School



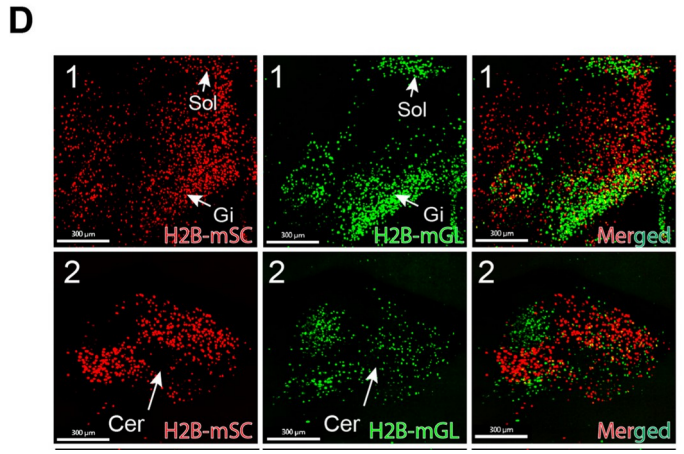
Carlson and Birkett (2022), Fig. 5.14, p. 131.

# Two spinal tracers distinguish descending populations

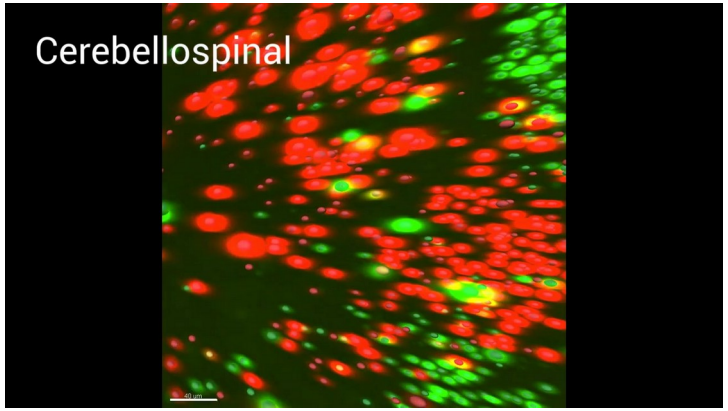
Wang and colleagues injected different fluorescent retrograde viruses into cervical and lumbar spinal cord. The **cervical** region supplies upper-body circuits, whereas the **lumbar** region contains circuits involved in hindlimb control. Brain neurons with axons reaching either injection site acquired the corresponding label, allowing the two descending populations to be compared within the same preparation.

Some nuclei contained both colors because their neurons were reached from both spinal levels. Other labeled neurons contained only one color. Three-dimensional confocal imaging allowed the researchers to inspect the nuclei across several optical planes, helping distinguish two signals inside the same nucleus from signals belonging to separate cells lying above one another.

The proportions of double-labeled neurons differed across brain regions. Descending populations therefore varied in how broadly they reached the spinal cord. This anatomical difference can influence whether a brain region coordinates several body segments or directs a more restricted spinal territory, although identifying an axon's destinations does not by itself establish its activity during a movement.




Wang et al. (2022), Fig. 3, selected panels.



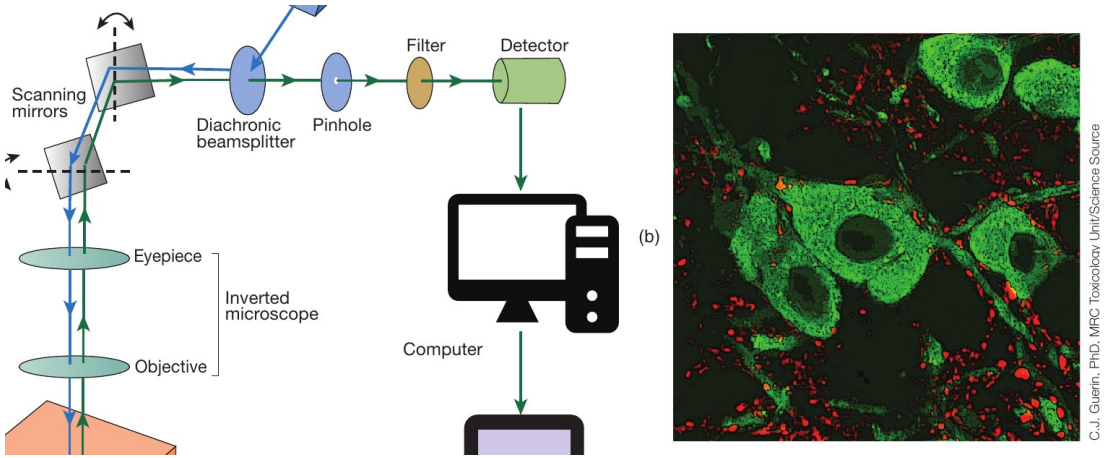

Wang et al. (2022), Fig. 3, video 1.

# Confocal microscopy rejects light from neighboring planes

A fluorescent specimen emits light from structures above and below the plane currently in focus. This out-of-focus light blurs thick tissue images. A **confocal microscope** places a small aperture, or pinhole, in the detection path so that much of this unwanted light is rejected before it reaches the detector and contributes to the final image.

The microscope scans the focused light across the specimen to build an image of one plane. Changing the focal depth produces another optical section without physically cutting the tissue again. A series of these sections forms a **z-stack**, which records how labeled structures change across depth and allows their three-dimensional arrangement to be examined.

Separate fluorescence channels can show different molecules in the same specimen. Filters direct each range of emitted wavelengths to the appropriate detector. Comparing the channels across a z-stack helps identify whether two labels occupy the same structure, such as one cell nucleus, rather than merely appearing at the same position in a flattened two-dimensional projection.



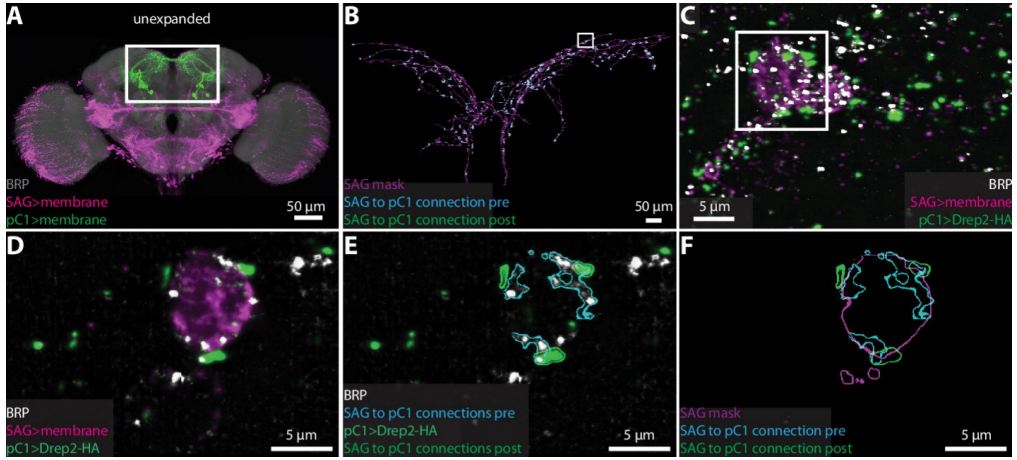

Carlson and Birkett (2022), Fig. 5.9, p. 128.

# Synaptic labels can be assigned to identified partners

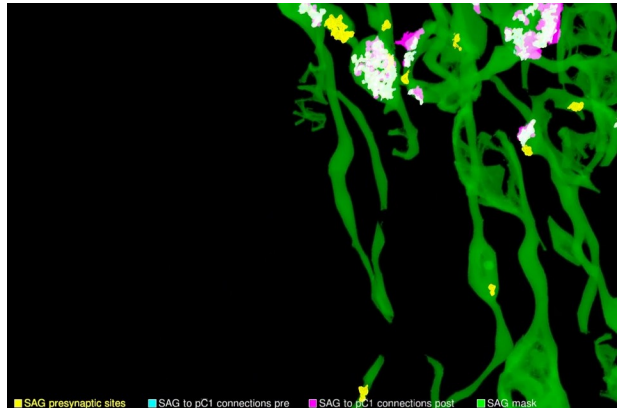
Lillvis and colleagues examined communication from **SAG neurons** to **pC1 neurons**, named cell populations in the fly nervous system. They labeled SAG membranes and presynaptic sites while marking postsynaptic structures in pC1 cells. A **postsynaptic structure** is the receiving specialization where a neuron detects chemical signals released by another cell's terminal.

The analysis first assigned presynaptic objects to the segmented SAG neuron. It then searched for nearby pC1 postsynaptic labels, rather than treating every close pair of neuronal branches as a connection. The expanded tissue separated these small structures sufficiently for their positions to be compared in the same three-dimensional image volume after segmentation.

Using partner-specific labels allowed the investigators to count candidate contacts between these populations. They compared those counts with electron-microscopy reconstructions and obtained similar estimates for the tested connections. Keeping the molecular labels visible also allowed the optical workflow to compare identified circuit elements across multiple individual animals rather than reconstructing an entire brain each time.




Lillvis et al. (2022), Fig. 4, selected panels.



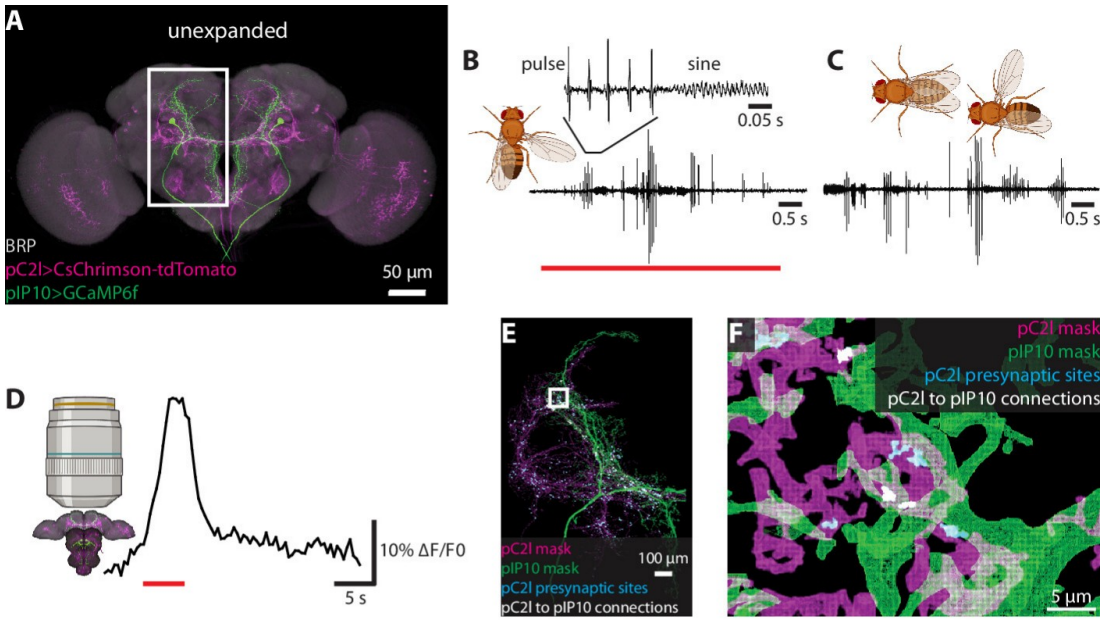

Lillvis et al. (2022), Video 3.

# A circuit can be measured structurally and functionally

A circuit's **structural connectivity** describes its anatomical contacts, whereas **functional connectivity** describes how one population's activity changes when another is active or stimulated. Lillvis and colleagues examined both properties in identified fly neurons. They measured a calcium response during stimulation and subsequently used expanded tissue to quantify labeled synaptic structures in the same circuit.

Calcium fluorescence increased in the receiving population when the upstream neurons were stimulated under the tested conditions. The optical response provided a measure of the circuit's functional effect. Anatomical labeling then identified the relevant neurons and their presynaptic structures, allowing the physiological measurement to be compared with a structural measurement from the corresponding preparation.

The study examined individual variation in these circuit properties and related them to behavior. The measurements remained distinct: a synaptic-object count described anatomy, a calcium change described cellular activity, and the behavioral assay described the animal's actions. Pairing these measurements in identified neurons reduced the uncertainty about whether they came from the same circuit components.



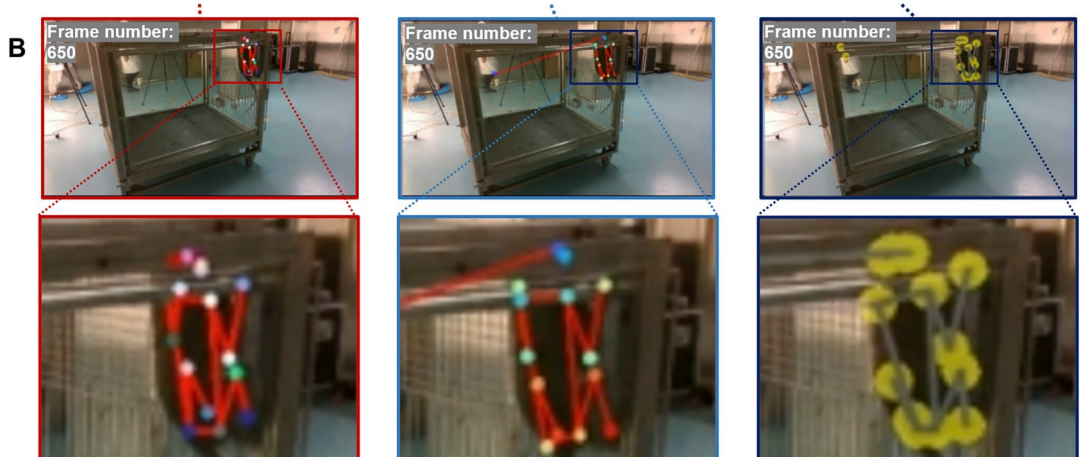

Lillvis et al. (2022), Fig. 6, selected panels.

# Cross-species tracking uses body geometry

Tang and colleagues evaluated ADPT on monkey movements as well as mouse recordings. The system estimated positions of body landmarks across video frames. A monkey's longer limbs and different posture change the arrangement of these landmarks, so tracking across species requires a model to recognize the relevant anatomical configuration rather than memorize one animal's silhouette.

The investigators compared predicted landmark positions with manually annotated positions. A manual annotation identifies where a person judged the body part to be in a recorded frame. Measuring the distance between a predicted and annotated point reveals whether an apparently smooth trajectory follows the correct feature or consistently follows the wrong part of the animal.

Changes in camera view and background can also shift the image features used by a tracker. ADPT's tests included movement sequences in which other systems drifted away from the intended landmarks. The resulting comparison examined whether body-part estimates stayed on the animal as its pose changed, which is necessary before using those coordinates to describe movement timing or coordination.




Tang et al. (2025), Fig. 4, selected panels.



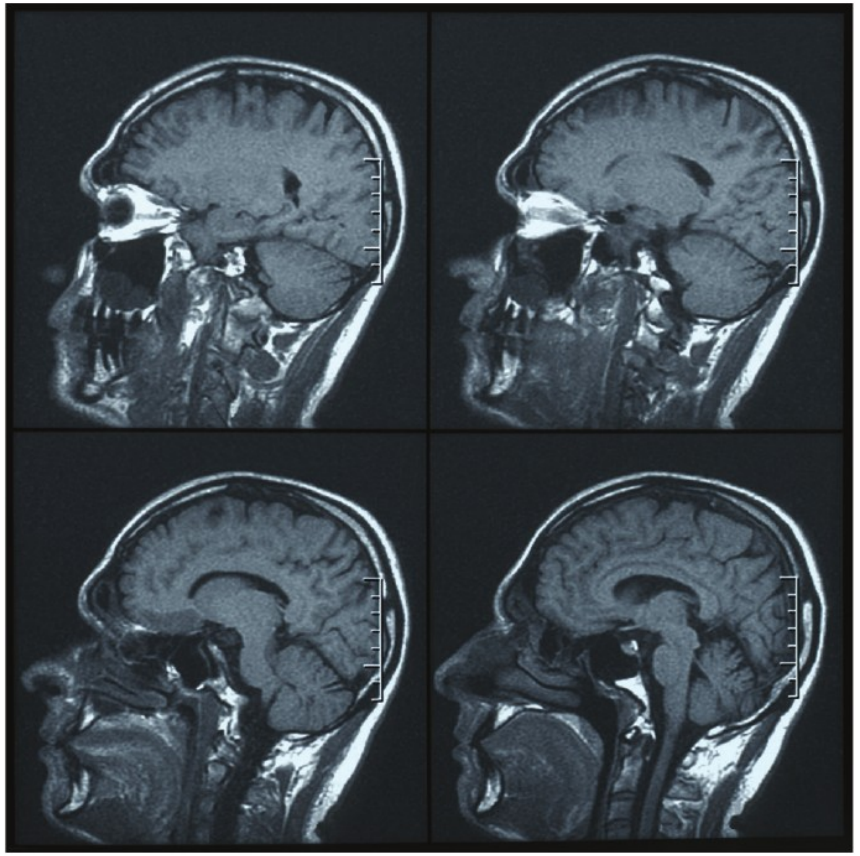
Tang et al. (2025), Fig. 4, video 1.

# MRI maps tissue using signals from hydrogen nuclei

**Magnetic resonance imaging**, or **MRI**, uses a strong magnetic field and radio-frequency pulses to measure signals from atomic nuclei, especially hydrogen in water. After a pulse, the nuclei return toward their previous state and emit a detectable signal. Differences in this relaxation process help distinguish tissues that contain different amounts and environments of water.

Magnetic-field gradients make the signal depend on location. The scanner changes these gradients during acquisition, and a computer reconstructs the spatial pattern into slices or a volume. Unlike an optical microscope, MRI can collect information from deep inside an intact head without requiring light to pass through the skull and intervening brain tissue.

A **structural MRI** emphasizes anatomical differences, including the boundaries between gray matter, white matter, and fluid-filled spaces. The acquisition sequence determines which tissue properties dominate the image contrast. Researchers can use the resulting anatomy to plan an intervention, locate a lesion, or align functional measurements with the participant's own brain structures rather than a generic outline.



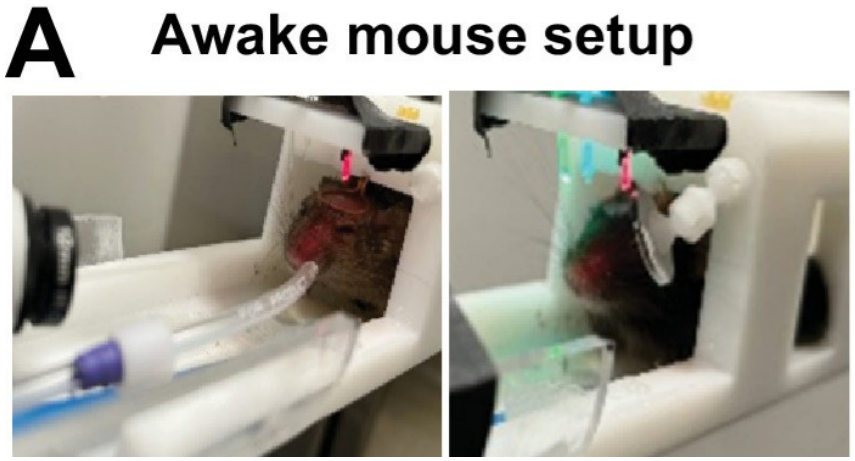

Carlson and Birkett (2022), Fig. 5.18, p. 133.

# An implanted coil improves awake mouse brain imaging

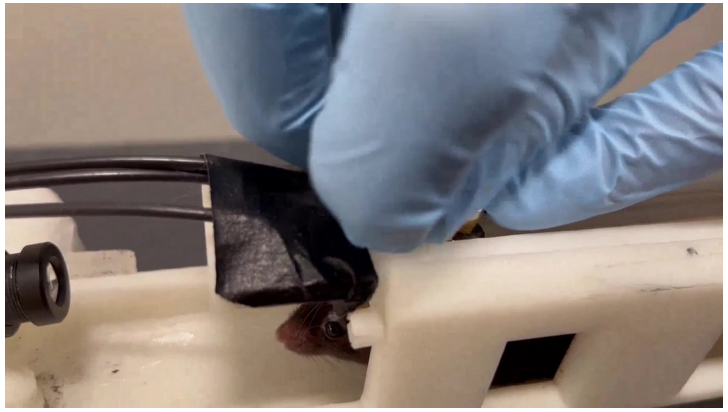
Hike and colleagues developed an implanted **radio-frequency coil** for awake mouse MRI. This coil receives the magnetic-resonance signal close to the brain and also serves as part of the head-fixation assembly. Its shape reduces image distortion associated with the boundary between air and tissue while positioning the head consistently inside the scanner during acquisition.

The mice gradually became accustomed to the scanner environment and restraint. This acclimation matters because head movement changes which tissue contributes to an image location. A movement during acquisition can distort the reconstructed brain or produce abrupt signal changes that resemble a response even though they result from the animal shifting its head within the scanner.

The system used a 14-tesla scanner to acquire small brain volumes repeatedly during sensory stimulation. The investigators measured responses to visual signals and air puffs directed at the whiskers. Their acquisition combined fine spatial sampling with repeated measurements over time, allowing activity-related blood-oxygen changes to be mapped across both sensory pathways and additional association regions.




Hike et al. (2025), Fig. 2, selected panels.



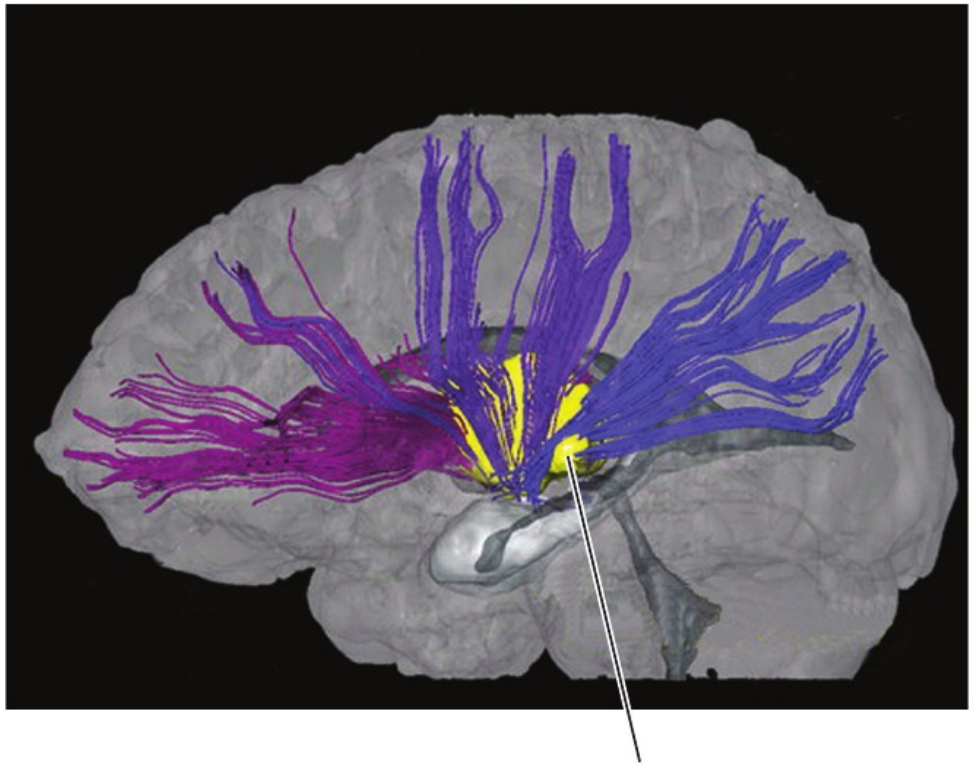
Hike et al. (2025), Video 1.

# Diffusion imaging estimates white-matter orientation

Water molecules move through tissue by **diffusion**, their random thermal motion. In a large axon bundle, membranes and surrounding structures restrict movement differently along and across the fibers. Diffusion-sensitive MRI measures this directional difference, providing information about the orientation of tissue organization within each small three-dimensional image element, called a **voxel**.

**Tractography** follows estimated orientations from voxel to voxel to reconstruct possible white-matter pathways. The resulting streamlines summarize the directions favored by the diffusion measurements. They help describe how distant brain regions may be joined by bundles of axons, especially when those bundles are too deep to examine directly in a living human with light microscopy.

A streamline represents a computational path through the measured orientation field. It does not identify an individual axon, its synapses, or the direction in which it carries signals. Retrograde tracing supplies a different kind of information because a label is transported inside projecting neurons; diffusion imaging infers tissue organization from the motion of water throughout each voxel.



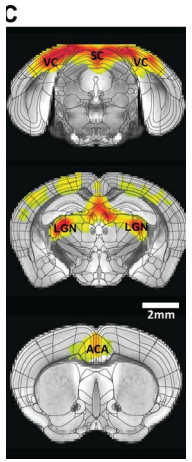
Carlson and Birkett (2022), Fig. 5.19, p. 134.

# Awake fMRI follows sensory responses across the brain

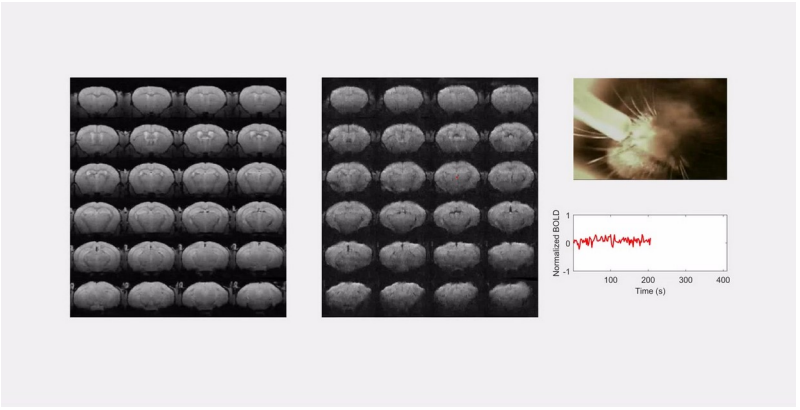
During visual stimulation, Hike and colleagues detected blood-oxygen changes in several parts of the mouse visual pathway. These included the **lateral geniculate nucleus**, a thalamic relay, the **superior colliculus**, a midbrain structure involved in orienting, and visual cortex. Repeated images allowed the response within each anatomical region to be followed over time during stimulation.

The researchers also stimulated the whiskers with air puffs. Responses appeared in the thalamus and **barrel cortex**, a cortical area organized around whisker inputs, as well as additional regions. Presenting the stimulus at known times allowed changes during stimulation to be compared with the preceding baseline and the period after the stimulus had ended.

The accompanying raw acquisition illustrates why motion must be examined alongside these responses. Head movement changes the location and shape of tissue within the image, creating abrupt fluctuations. Aligning successive volumes and identifying motion-contaminated periods reduces these effects before a response time course is assigned to a specific anatomical structure in the functional map.




Hike et al. (2025), Fig. 3, selected panels.



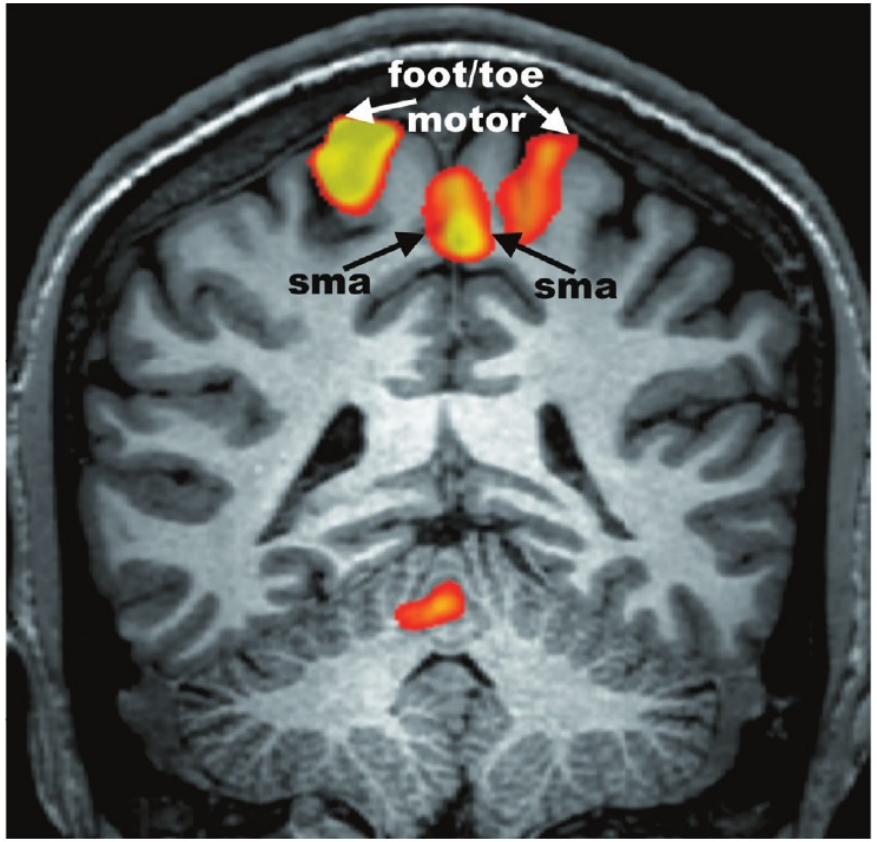

Hike et al. (2025), Video 2.

# The BOLD signal reflects changes in blood oxygenation

**Functional MRI**, or **fMRI**, often measures the **blood-oxygen-level-dependent signal**, abbreviated **BOLD**. Oxygenated and deoxygenated hemoglobin affect the local magnetic environment differently. When neural activity changes blood flow and oxygen use, the resulting shift in the balance between these forms of hemoglobin changes the magnetic-resonance signal measured from nearby tissue within the scanner.

Active neurons and surrounding cells consume energy, and local blood vessels supply additional oxygenated blood. The blood-flow increase often exceeds the immediate rise in oxygen consumption, reducing the local concentration of deoxygenated hemoglobin. Under common acquisition conditions, this produces a positive BOLD response that develops more slowly than the electrical events initiating the tissue's metabolic demand.

A BOLD response therefore describes a **hemodynamic** change, meaning a change involving blood and its circulation. It is not a direct recording of action potentials. The response integrates activity and vascular processes across a tissue volume, so its timing and spatial extent differ from an electrode measurement made near an individual neuron during the same event.




Living Art Enterprises/Science Source

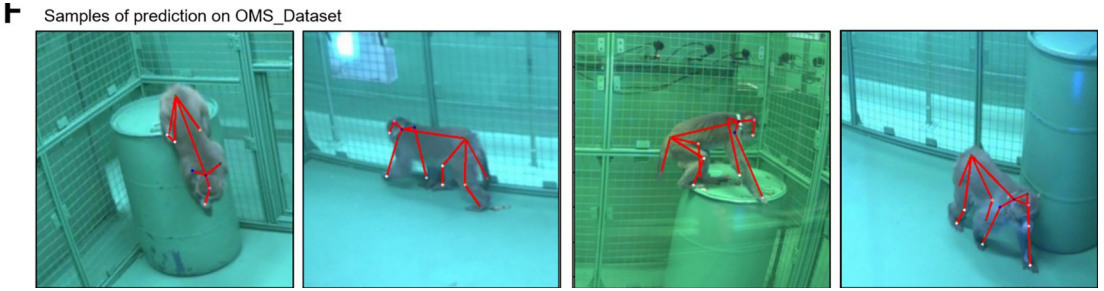
Carlson and Birkett (2022), Fig. 5.26, p. 140.

# Accurate landmarks are needed to measure movement

Tang and colleagues also evaluated pose estimation on dog recordings and other public animal datasets. Different coats, body proportions, and backgrounds change the visual features available to the model. Tracking the same anatomical landmark across these conditions requires identifying its position relative to the animal's body rather than following one familiar color or patch of texture.

The investigators compared ADPT with other tracking approaches across these datasets. Their analysis measured how well the predicted positions matched annotated body parts, including situations in which image noise caused drift. Such comparisons test whether a method continues locating the intended anatomical feature when the appearance of the animal or the surrounding scene changes during a recording.

A small location error can become a large speed error when positions are differentiated across closely spaced frames. A predicted point that jumps from the paw to the floor appears to make an extremely fast movement. Inspecting landmark trajectories and the corresponding source frames helps distinguish actual rapid movements from discontinuities introduced by the tracking algorithm itself.




Tang et al. (2025), Fig. 5, selected panels.



Tang et al. (2025), Fig. 5, video 1.

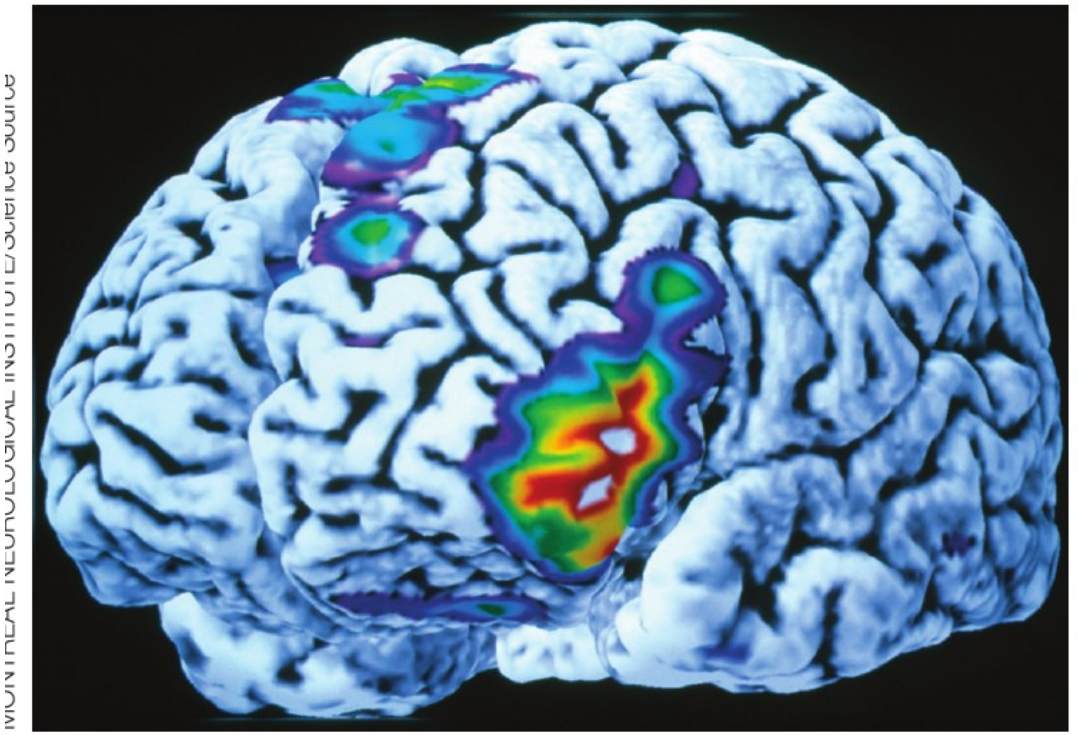
# PET locates radioactive tracers within living tissue

**Positron emission tomography**, or **PET**, detects radiation from an injected tracer. A tracer contains a radioactive isotope attached to a molecule with a particular biological fate. Depending on the molecule, its distribution can reveal aspects of glucose use, receptor availability, or chemical processing rather than simply showing the tissue boundaries visible in structural imaging.

When the isotope emits a positron, that particle encounters an electron and produces two photons traveling in nearly opposite directions. Detectors surrounding the head register paired events. A computer uses many such detections to reconstruct where the tracer accumulated, producing a spatial map of the labeled molecule's distribution within the tissue during the measurement period.

A receptor-binding tracer occupies a different experimental role from a glucose-related metabolic tracer. The former can estimate the availability of its molecular binding sites, whereas the latter reflects uptake and processing related to tissue metabolism. Interpreting a PET image therefore requires knowing what molecule was administered and what physiological process determines its concentration at the scanned location.

ndicating increased activity.



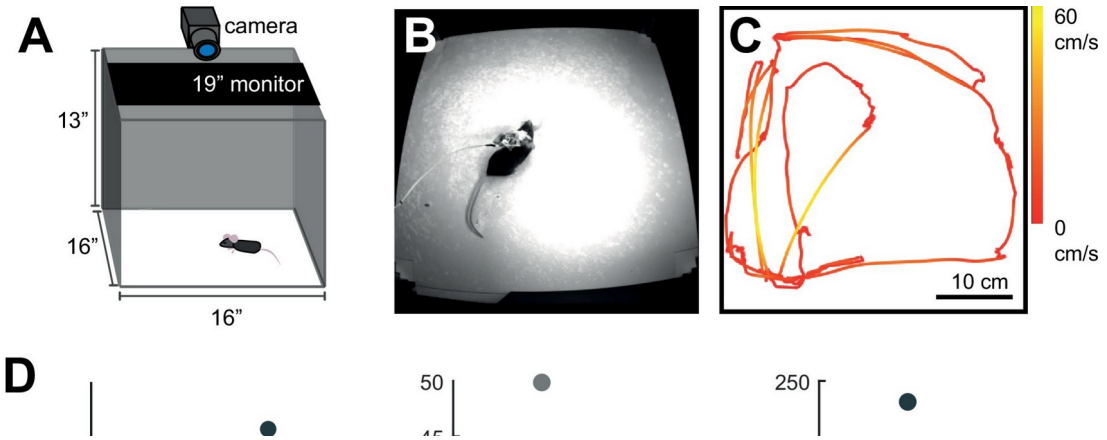
Carlson and Birkett (2022), Fig. 5.25, p. 140.

# Chronic probes record activity during free movement

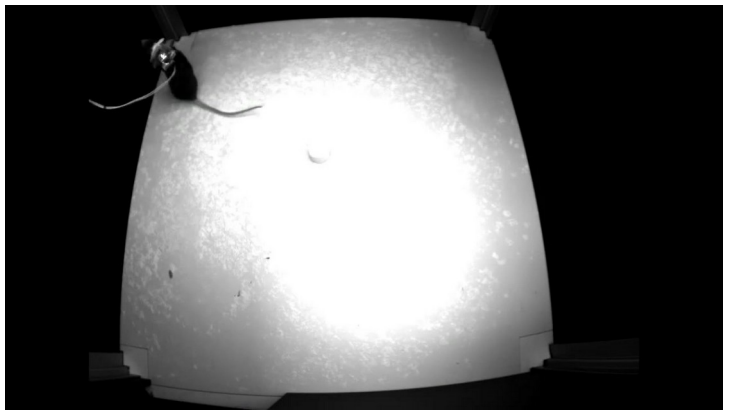
Juavinett and colleagues developed an implant that allowed Neuropixels probes to remain in freely moving mice. A protective housing stabilized the probe while a tether carried its electrical signals to the recording system. The mice could explore an arena, allowing activity to be collected during movements that would be prevented by rigid head fixation during recording.

Their behavioral comparison examined locomotion, exploration, and responses to looming visual stimuli. A **looming stimulus** expands in the visual field and can evoke defensive movement. Measuring these behaviors in implanted and unimplanted animals assessed whether the recording assembly substantially altered the actions that investigators wanted to study in the freely moving preparation under the tested conditions.

The probe sampled cells along a narrow path through several structures. Closely spaced recording sites captured different mixtures of extracellular electrical signals depending on their depth and proximity to neurons. The resulting recordings could therefore include activity from cortex and deeper regions while the same animal moved through the arena and encountered changes in its visual environment.




Juavinett et al. (2019), Fig. 3, selected panels.



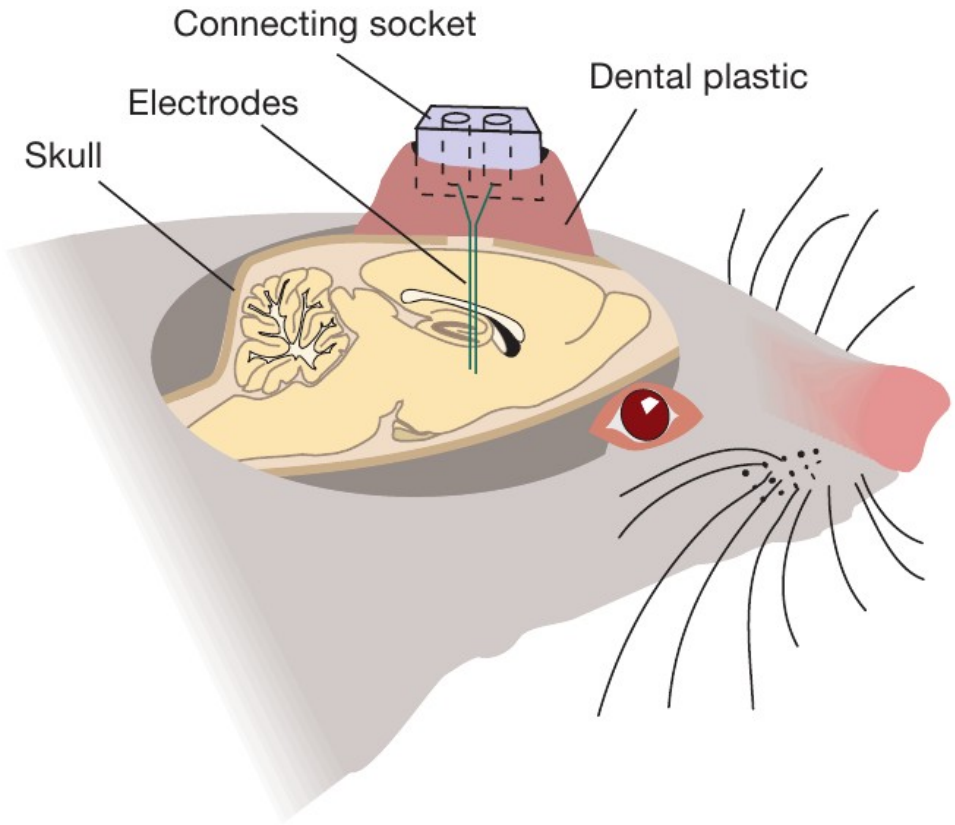
Juavinett et al. (2019), Video 2.

# Extracellular electrodes detect nearby voltage changes

An **extracellular electrode** measures voltage in the fluid outside cells relative to a reference electrode. When a nearby neuron fires, ions cross its membrane and produce small changes in the surrounding electrical field. The electrode detects a brief waveform associated with that action potential without entering the cell or measuring its full membrane voltage directly.

A nearby electrode may detect spikes from several neurons. **Spike sorting** groups recorded waveforms according to their shape and their distribution across recording sites. Repeated events with a similar spatial pattern are assigned to a candidate unit, allowing the activity of several nearby neurons to be separated from a mixed extracellular recording over time.

The signal's size depends on the distance and orientation between the neuron and the recording sites. A small tissue movement can therefore change a waveform even when the same neuron continues firing. High-density probes record each event at several neighboring locations, providing a spatial pattern that can help follow the signal as its position shifts along the probe.



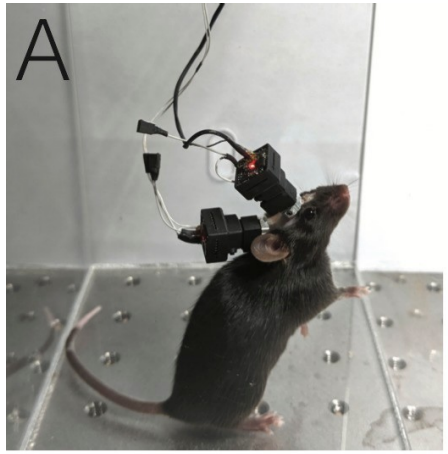

Carlson and Birkett (2022), Fig. 5.20, p. 136.

# Miniature microscopes preserve access to active behavior

De Groot and colleagues built **NINscope**, a small fluorescence microscope that can be carried on a mouse's head. Its light source excites a fluorescent indicator inside neurons, and its image sensor records the emitted light. The microscope moves with the head, allowing neural fluorescence to be collected while the animal walks, rears, and changes direction.

The researchers mounted two microscopes to record from separate brain regions in the same animal. The housing, lens arrangement, and mounting angles determined whether the two devices could fit without colliding. Their behavioral measurements compared rearing under single- and dual-microscope conditions, examining whether carrying the additional assembly disrupted this recognizable action in the tested preparation.

A source recording shows a mouse jumping while wearing both devices. The optical equipment remains attached as the head accelerates, but movement can still shift tissue within each image. Simultaneous video and an accelerometer, a sensor that measures changes in motion, provide a record of the animal's actions that can be aligned with the fluorescence data.




de Groot et al. (2020), Fig. 3, selected panels.



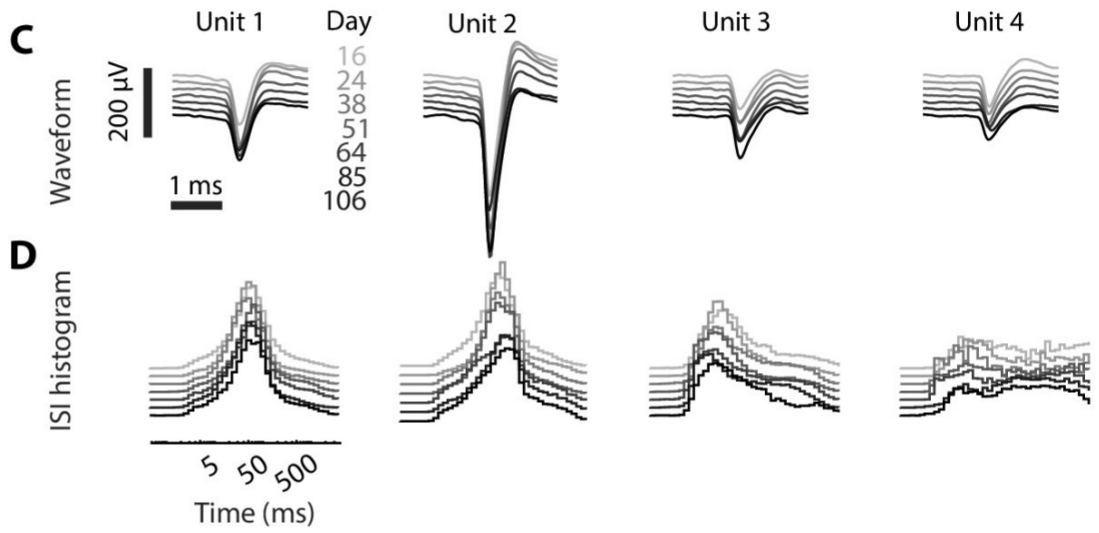
de Groot et al. (2020), Fig. 3, video 1.

# Long recordings require matching neurons across days

Bimbard and colleagues used the Apollo implant to collect Neuropixels recordings over weeks and months. They compared the spatial arrangement and shapes of recorded spikes across sessions. A unit that reappears with a similar waveform at corresponding recording locations is a candidate for the same neuron, rather than simply another neuron firing at a similar average rate.

They also compared **interspike intervals**, the times separating successive spikes. Different neurons can have distinctive patterns of short and long intervals because their intrinsic properties and inputs differ. Agreement in both waveform and spike timing provides more information for matching units across days than either a shared anatomical depth or an average firing rate considered alone.

The source examples include neurons tracked across a thirteen-week period. Maintaining a recognizable electrical signature across this interval allows investigators to ask how individual cells change during learning or repeated experience. Counting the total units in each session answers a separate question: an unchanged population size does not guarantee that every recorded neuron retained its identity across sessions.



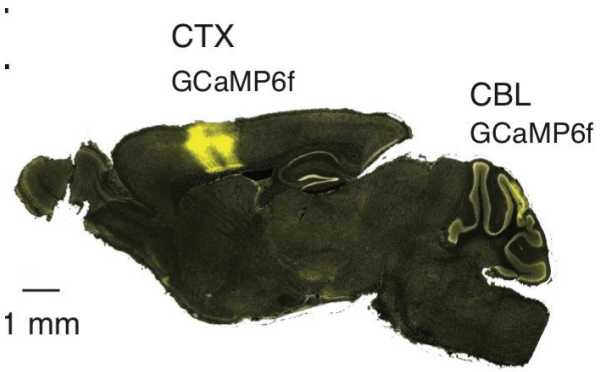

Bimbard et al. (2025), Fig. 5, selected panels.

# Image stabilization separates motion from fluorescence

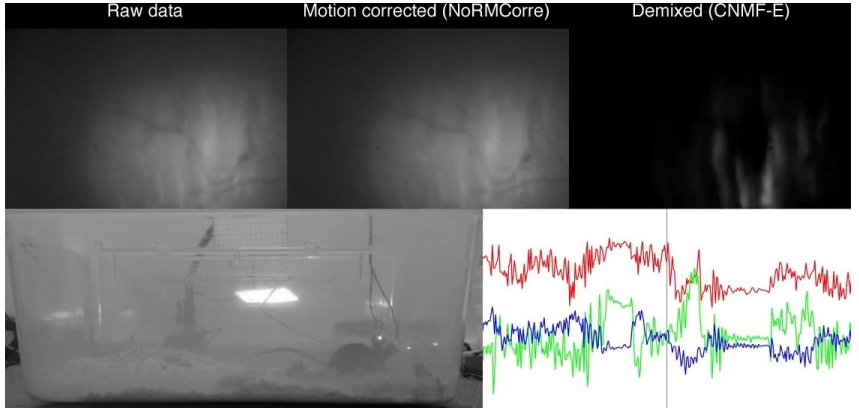
When an animal moves, the microscope image may shift even though the fluorescent cells retain their positions relative to one another. **Motion correction** estimates this displacement and aligns each frame with a reference. De Groot and colleagues used this processing before extracting activity from the small fluorescent structures recorded through the head-mounted microscopes during behavior.

After alignment, **source separation** identifies spatial patterns that belong to individual cells or dendritic arbors. The analysis then estimates how each pattern's fluorescence changes over time. This separates overlapping cellular signals from diffuse background fluorescence, which can arise from surrounding tissue and other labeled structures outside the focal plane during the same imaging session.

The dual-microscope example displays raw images, corrected images, extracted cellular activity, and the animal's movement measurements together. A shift affecting many bright structures at once suggests image displacement, whereas a localized fluorescence change can follow one cellular pattern. Examining these channels together helps establish which fluctuations remain after the recorded field has been aligned across successive frames.




de Groot et al. (2020), Fig. 3, selected panels.



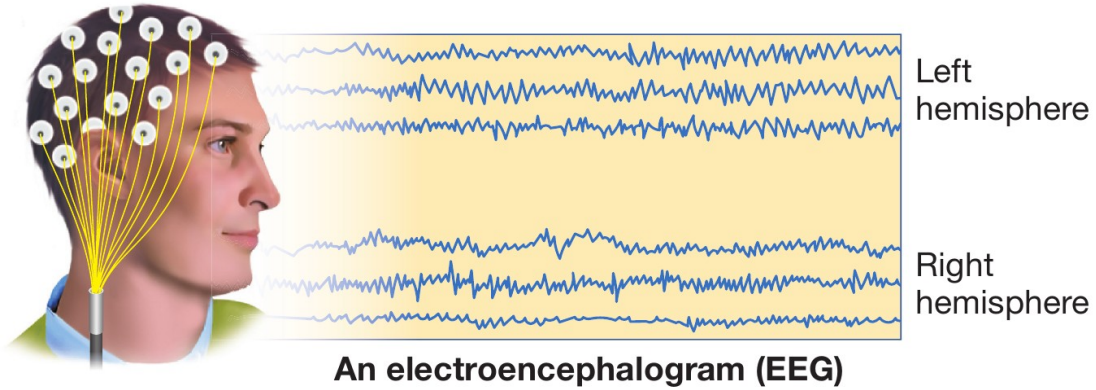

de Groot et al. (2020), Fig. 3, video 2.

# EEG measures coordinated population currents

**Electroencephalography**, or **EEG**, records voltage differences between electrodes placed on the scalp. Much of this signal reflects summed currents associated with synaptic activity in aligned cortical neurons. Many cells must contribute in a coordinated way for their electrical fields to remain detectable after spreading through brain tissue, skull, and scalp before reaching the electrodes.

The changing voltage can be examined as a waveform or separated into frequency components. Slow and fast rhythms differ in how often the voltage pattern repeats each second. Sleep stages, waking behavior, and some seizure states produce recognizable changes in these population patterns, making EEG useful for following rapid changes in the brain's overall electrical organization.

Scalp electrodes detect mixtures of signals from multiple sources. The same underlying brain source can contribute to several electrodes because electrical fields spread through conductive tissue. Locating the source therefore requires a model of the head and the likely generators, whereas identifying the time of a voltage change can be much more precise in the recorded waveform.



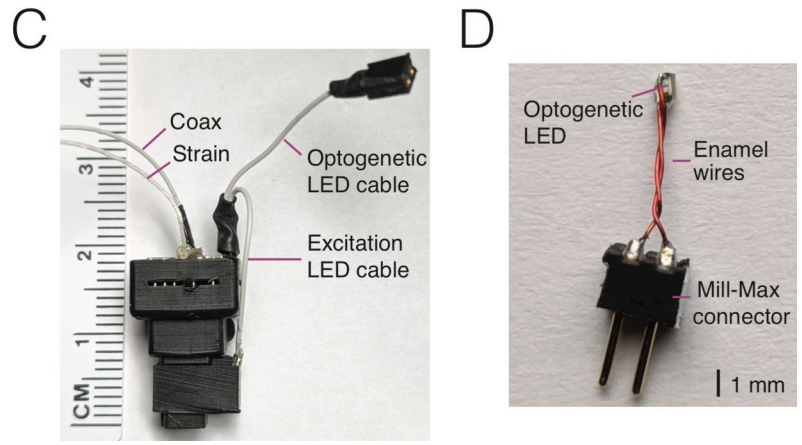

Carlson and Birkett (2022), Fig. 5.21, p. 137.

# Fluorescent beads expose movement artifacts

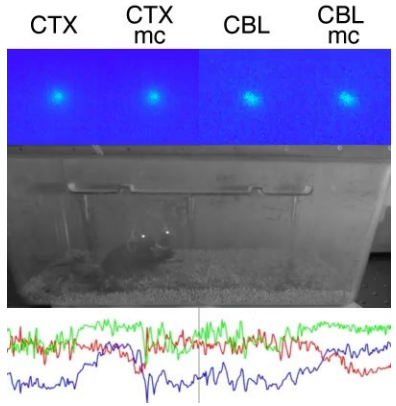
De Groot and colleagues imaged fluorescent beads placed in tissue to examine motion artifacts in the dual-microscope preparation. A bead fluoresces but does not generate a neuronal calcium response. Changes in its apparent position or brightness during movement therefore reveal contributions from the imaging geometry and processing rather than changes in a neuron's physiological activity.

The bead recordings were displayed before and after motion correction, alongside the animal's behavior and accelerometer measurements. Comparing these channels showed how movements translated into shifts in the optical image. Correcting the displacement reduced image movement while retaining the underlying fluorescent objects, providing a direct check on the stabilization step applied to the neuronal recordings.

This approach examines one specific source of fluorescence variability using a physically stable label. A calcium indicator changes brightness when calcium binds, whereas a bead's fluorescence is not controlled by neuronal calcium. The comparison therefore separates movement-related imaging effects from the biological process that a calcium-sensitive protein is designed to report inside the recorded cells.




de Groot et al. (2020), Fig. 1, selected panels.




de Groot et al. (2020), Fig. 3, video 3.

# MEG detects magnetic fields generated by neural currents

**Magnetoencephalography**, or **MEG**, measures the small magnetic fields produced by electrical currents in the brain. Synaptic currents flowing through populations of similarly oriented neurons generate fields that sensitive detectors can record outside the head. As with EEG, coordinated population activity is especially important because fields from differently oriented or asynchronously active cells can partly cancel.

MEG and EEG respond differently to source orientation and the tissues between the source and detector. Electrical potentials are altered by the conductivity of the skull and scalp, whereas magnetic fields pass through these tissues with different distortions. Combining the measurements with structural MRI can help estimate where the generators lie within a participant's own cortical anatomy.

Both methods provide a rapidly changing signal that can be aligned with a stimulus or movement. Repeated presentations allow responses occurring at a consistent delay to be examined across trials. The resulting time course can distinguish early sensory responses from later processing stages, even though assigning each component to an exact small cell population requires additional anatomical and modeling information.



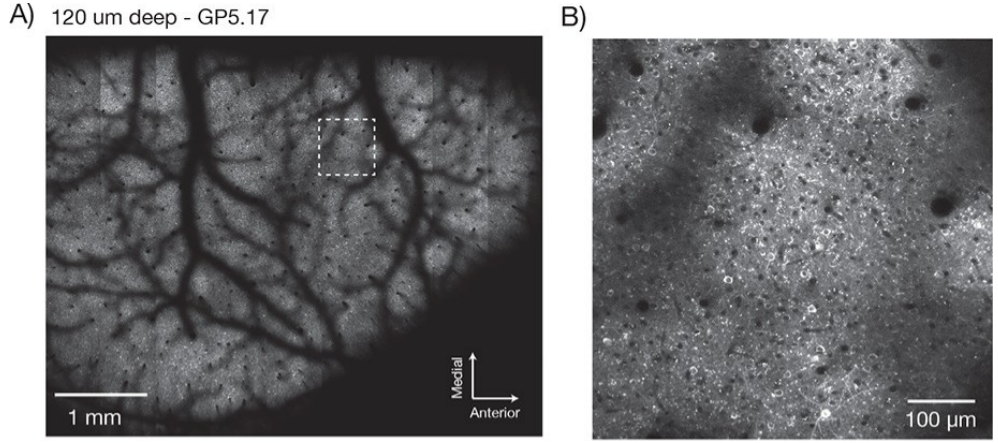
Carlson and Birkett (2022), Fig. 5.22, p. 138.

# A mesoscope spans separated cortical regions

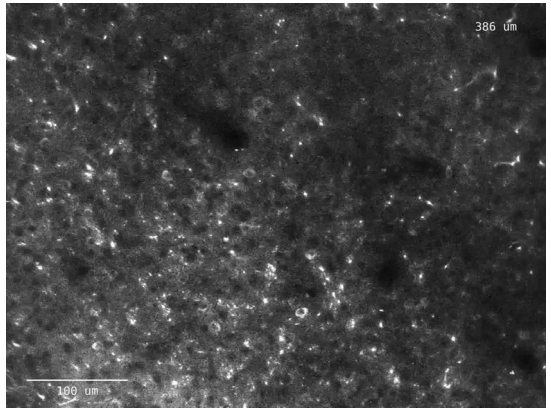
Sofroniew and colleagues built a **two-photon mesoscope** that combines a large field of view with the ability to resolve small cellular structures. A **mesoscope** images an intermediate spatial scale, larger than the field of a typical cellular microscope. Their system allowed distant parts of mouse cortex to be examined within the same optical preparation.

The source image stack spans several levels of detail, from broad cortical anatomy to recognizable neurons and smaller processes. Acquiring a stack means changing the focal depth through the tissue and collecting successive optical sections. Researchers can then choose recording sites using the visible anatomy rather than treating the entire field as one undifferentiated fluorescence signal during imaging.

Covering a wider field also changes how measurements are sampled. A scanner spends time collecting light from each selected location, so scanning more tissue can reduce how often any one location is revisited. The instrument allowed recording patterns to be selected according to whether an experiment required broad anatomical coverage or faster sampling from a smaller set of regions.




Sofroniew et al. (2016), Fig. 7, selected panels.



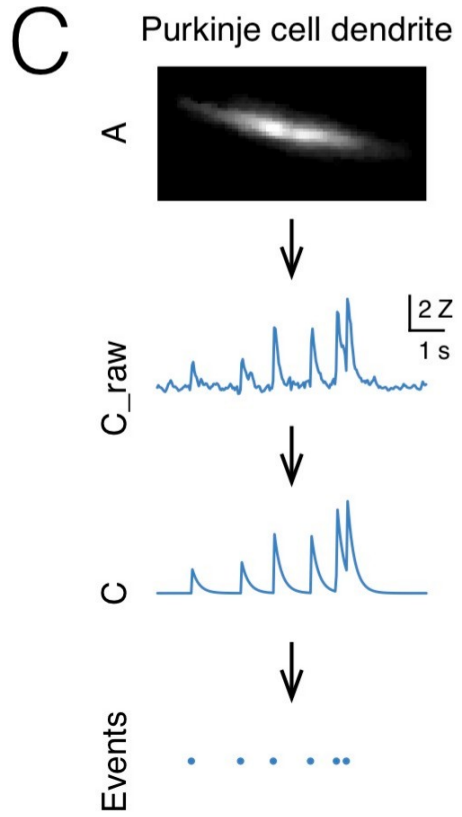

Sofroniew et al. (2016), Video 1.

# Calcium indicators turn ion changes into fluorescence

**Calcium indicators** are molecules whose fluorescence changes when they bind calcium ions. Neuronal activity can open calcium-permeable channels, increasing the calcium concentration inside a cell or its processes. A genetically encoded indicator such as **GCaMP** combines a fluorescent protein with calcium-binding components, allowing this intracellular change to be observed through an optical recording system.

A neuron produces GCaMP after receiving DNA that encodes the indicator. The protein is distributed within the targeted cells, where its brightness changes as calcium binds and is subsequently cleared. An optical trace therefore rises and falls according to both the cell's calcium dynamics and the indicator's own binding and fluorescence properties period.

An **action potential** is a rapid voltage event, while a calcium transient is a chemical concentration change that often outlasts the initiating electrical activity. Several closely spaced spikes may contribute to one prolonged fluorescent response. Interpreting a GCaMP trace requires keeping these two processes distinct, especially when comparing the timing of a movement with the rise of fluorescence.



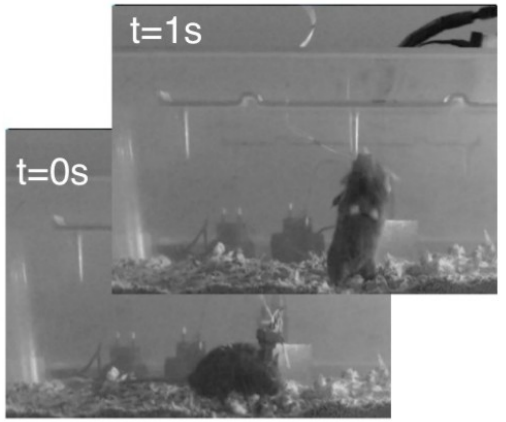

de Groot et al. (2020), Fig. 2, selected panels.

# Purkinje-cell calcium changes accompany movement

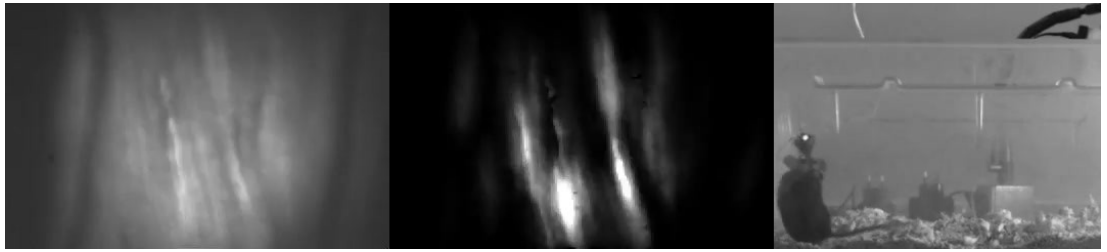
De Groot and colleagues expressed GCaMP6f in **Purkinje cells**, large inhibitory neurons in the cerebellar cortex. Their branching dendrites produced recognizable fluorescent patterns under NINscope. The investigators separated these spatial patterns and extracted their calcium traces, allowing activity in several dendritic arbors to be followed while the mouse explored its environment and performed spontaneous movements.

The microscope included an accelerometer that measured movement along three axes. The researchers aligned calcium changes with acceleration and with behaviors visible in the accompanying camera recording. Some Purkinje-cell dendritic signals increased around rearing, when the mouse lifted its body, allowing the optical activity to be examined relative to the distinct phases of the animal's movement.

They also recorded a rapid movement evoked by a loud clap and aligned the fluorescence to acceleration onset. The resulting traces described how dendritic calcium changed around that event. Keeping the camera, acceleration, and calcium measurements synchronized made it possible to compare the animal's visible action with cellular signals recorded from a particular region of the cerebellar cortex.




de Groot et al. (2020), Fig. 2, selected panels.



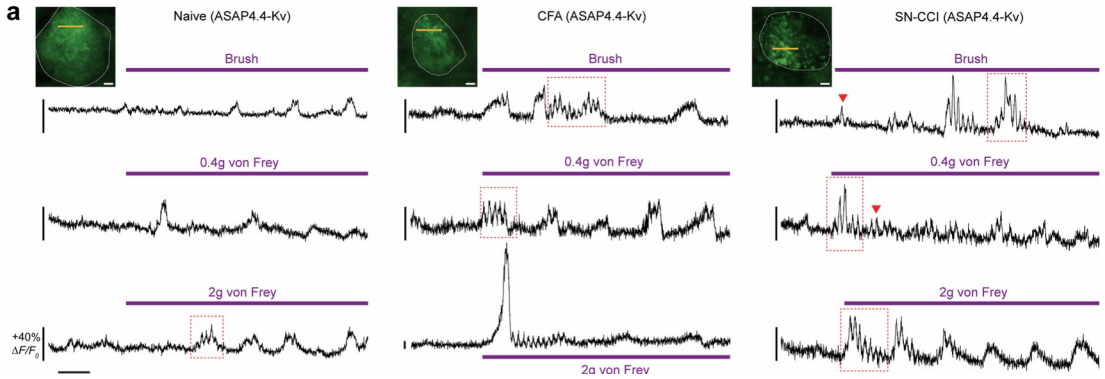
de Groot et al. (2020), Fig. 2, video 1.

# Voltage imaging follows fast sensory-neuron responses

Zhang and colleagues used a fluorescent **voltage indicator** called ASAP4.4-Kv to record sensory neurons in living mice. These neurons have cell bodies in the **dorsal root ganglia**, clusters beside the spinal cord. The indicator changes brightness with membrane voltage, allowing rapid electrical events to be detected optically without inserting a recording electrode into each cell.

The investigators compared responses to mechanical stimulation using voltage imaging and calcium imaging. A voltage trace can resolve brief events and faster changes that become blended in a slower calcium signal. The two indicators therefore report different parts of the response: the electrical change at the membrane and the resulting changes in intracellular calcium concentration after stimulation.

The study examined untreated animals and animals with inflammation or nerve injury. The recorded sensory responses differed across these conditions, including changes in spontaneous electrical activity and responses to touch. A fast optical indicator allowed these changes to be examined at the level of individual sensory-neuron cell bodies while retaining the anatomical position of the recorded cells.



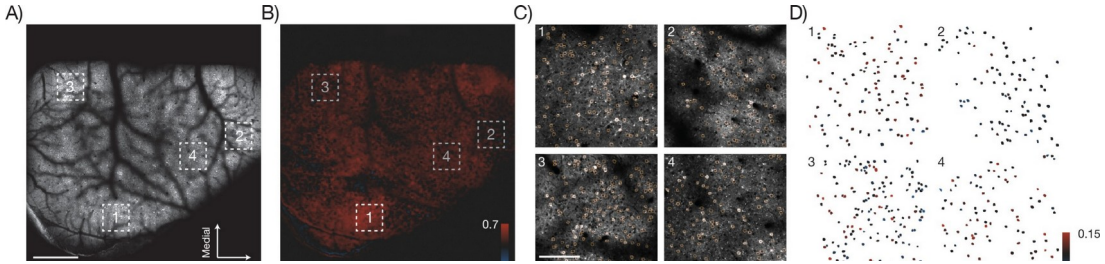

Zhang et al. (2025), Fig. 2, selected panels.

# A large optical field samples distant cortical populations

Sofroniew and colleagues used the mesoscope to record calcium fluorescence across a large expanse of mouse cortex. Individual cells remained visible within this broad field. The acquisition therefore sampled spatially separated neuronal populations within one preparation instead of requiring every measurement to come from a small patch under a conventional high-magnification microscope objective.

The instrument could also revisit selected regions at higher speed. Restricting the scan to chosen locations reduced the time spent traversing tissue that was not needed for the current measurement. This allowed the investigators to collect faster signals from multiple cortical sites while retaining the ability to place those sites within the larger anatomical field observed through the same optics.

A fluorescence map preserves the relative positions of the sampled neurons, but the measured time course still depends on the indicator. The population traces reflect calcium dynamics rather than direct membrane-voltage recordings. Comparing responses across distant cortical sites can reveal coordinated activity while keeping the slower optical measurement separate from the millisecond events recorded by extracellular electrodes.



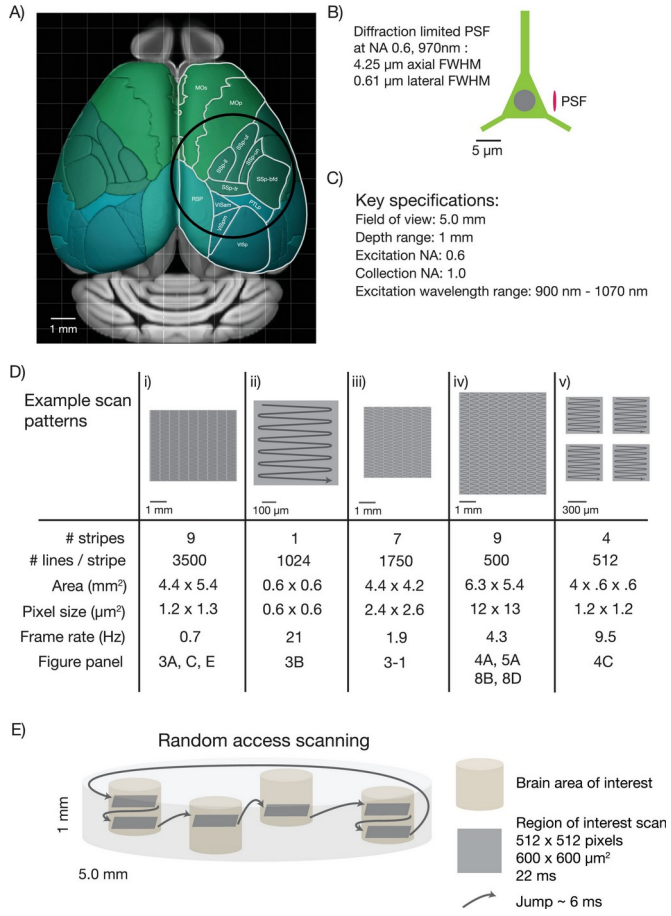

Sofroniew et al. (2016), Fig. 8, selected panels.

# Two-photon excitation confines the fluorescent signal

In **two-photon microscopy**, a fluorescent molecule absorbs two lower-energy photons almost simultaneously. This event is most likely where the laser is tightly focused and photon density is highest. Fluorescence is therefore generated mainly within a small focal volume, reducing excitation of tissue above and below the plane currently being recorded inside the specimen.

Longer-wavelength excitation light can penetrate scattering brain tissue more effectively than shorter-wavelength visible light under typical imaging conditions. Scanning the focus across a plane builds a cellular image, and moving the focus in depth produces an optical stack. The detected fluorescence still depends on how much emitted light escapes the tissue and reaches the collection optics.

Collecting more photons improves the ability to distinguish a small fluorescence change from random fluctuations in detected light. Increasing illumination also deposits more energy in the tissue. Microscope design therefore balances collection efficiency, field size, scan timing, and illumination so that recognizable cellular signals can be measured repeatedly.



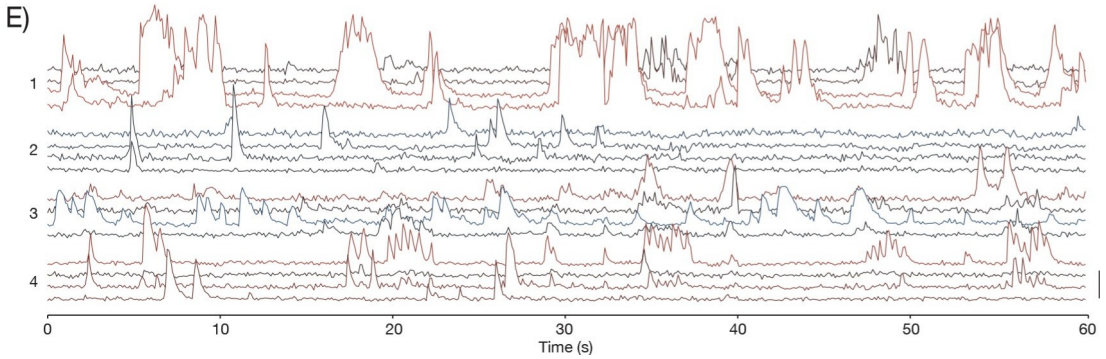

Sofroniew et al. (2016), Fig. 1, selected panels.

# Targeted scanning revisits chosen sites more often

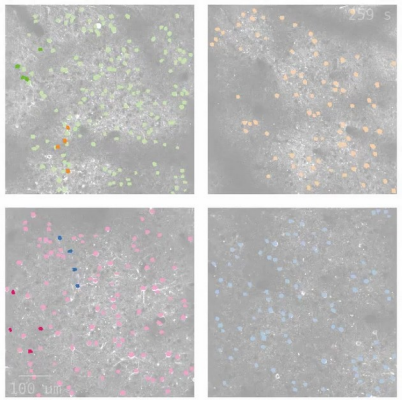
The mesoscope demonstration begins with a broad cortical view and then switches to selected recording sites. This changes the sampling pattern while keeping the instrument over the same preparation. Visiting fewer chosen regions allows each site to be sampled more frequently than it would be during a continuous scan covering every position across the entire field.

The source recording displays calcium activity from the selected regions using the authors' original processing and playback timing. Differences in fluorescence reveal changing intracellular calcium in labeled neurons. Comparing the sites helps identify whether population events occur together or in different parts of the recorded field, provided the scan timing for each region is included in the analysis.

A scanner does not necessarily measure all locations at precisely the same instant. The acquisition assigns a time to each visit, and successive regions can be sampled at slightly different moments. Using those timestamps preserves the order of the measurements when comparing activity across sites, rather than treating an assembled image as a perfectly simultaneous electrical snapshot of cortex.




Sofroniew et al. (2016), Fig. 8, selected panels.



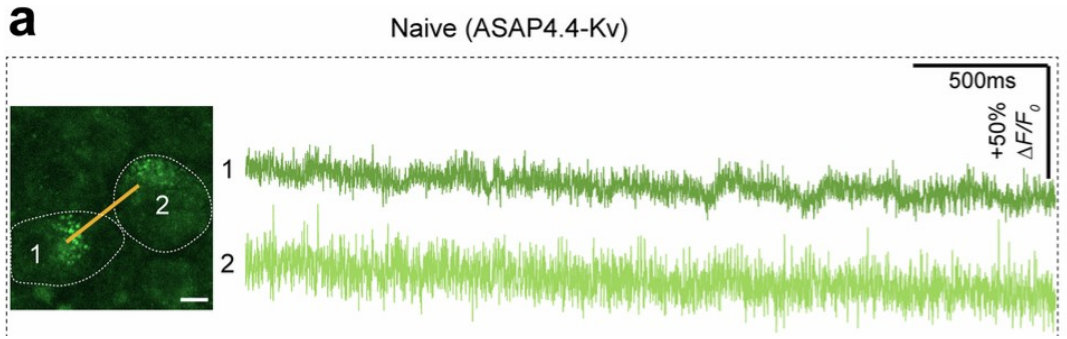
Sofroniew et al. (2016), Video 3.

# Voltage and calcium sensors measure different variables

A **genetically encoded voltage indicator** contains a component that responds to the electrical field across the cell membrane. Changes in that component alter the attached fluorescent signal. Because the sensor is associated with the membrane, its fluorescence can follow depolarization and repolarization more directly than a reporter that waits for intracellular calcium to accumulate after an electrical event.

A calcium indicator instead binds ions inside the cell. Its response depends on calcium entry, release from internal stores, buffering, and removal, as well as the reporter's binding kinetics. A large calcium transient may therefore reflect several electrical events or a prolonged chemical response, whereas a voltage trace can reveal separate rapid fluctuations during the same interval in the cell.

The detector must sample quickly enough to capture the variable of interest. Measuring a brief spike requires frequent observations and enough photons during each short exposure. Faster sampling gives less time to collect light per measurement, making sensor brightness and light collection especially important when resolving individual voltage events in living tissue with an optical instrument.



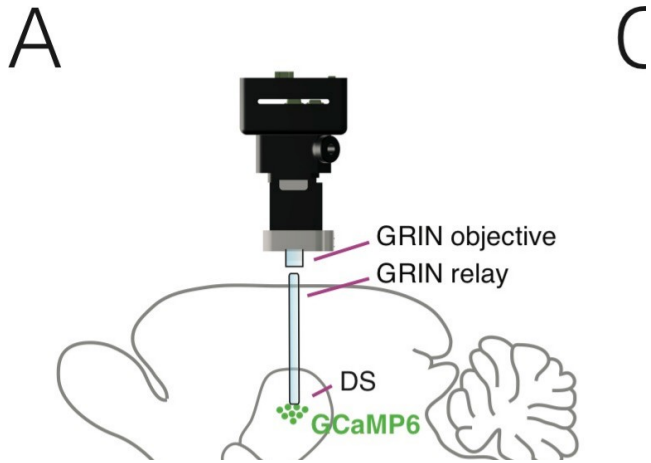

Zhang et al. (2025), Fig. 1, selected panels.

# A relay lens provides optical access to deep brain cells

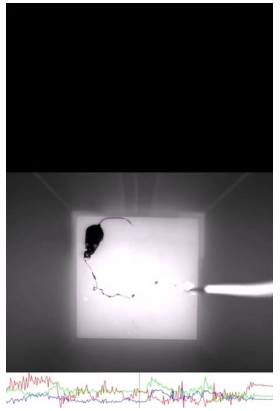
The **dorsal striatum** lies beneath cortex, beyond the direct optical access available at the brain surface. De Groot and colleagues used a **GRIN relay lens**, which guides light through a cylindrical element with a changing refractive index. The implanted lens brought an image of deep GCaMP-labeled neurons to the head-mounted microscope while the mouse moved freely.

The investigators aligned striatal fluorescence with the mouse's turning movements. In recordings from right dorsal striatum, the population response increased around turns to the left, the opposite side. Camera tracking and accelerometer signals provided independent descriptions of the movement, allowing calcium activity to be compared with the direction and onset of an identifiable action during exploration.

The optical image also preserved differences among individual cells. Extracting separate spatial footprints and traces allowed researchers to examine which neurons changed during a turn rather than averaging all fluorescence from the field. Tissue sections documented the relay-lens track and the surrounding indicator-expressing cells, tying the deep optical recording to a specific anatomical location in the striatum.




de Groot et al. (2020), Fig. 5, selected panels.



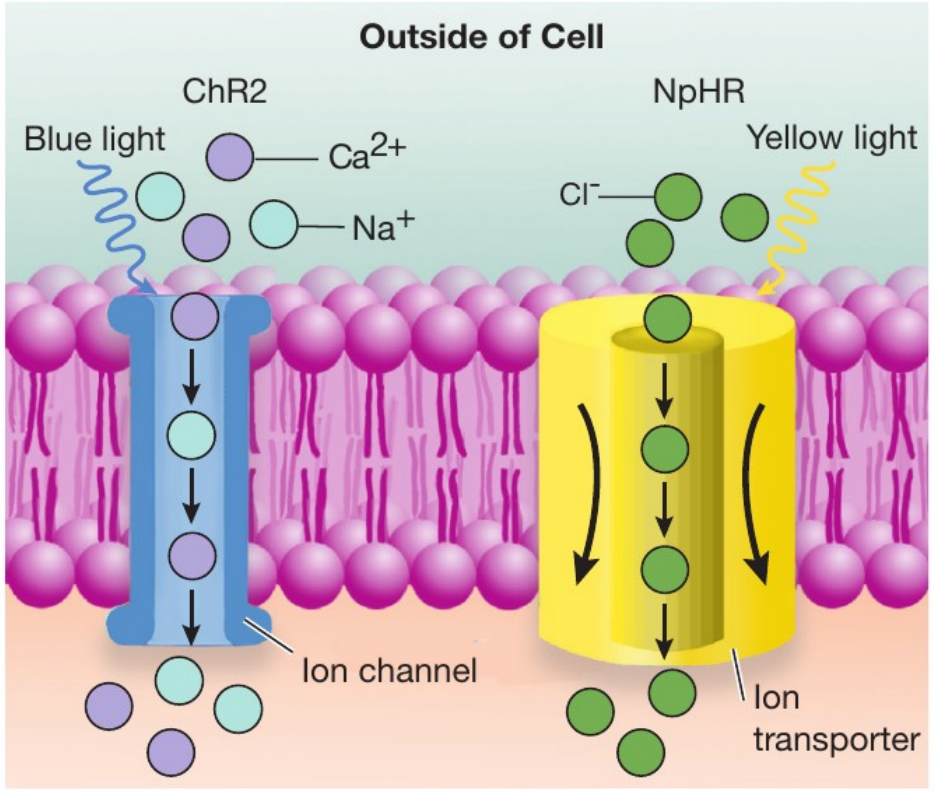
de Groot et al. (2020), Fig. 5, video 1.

# Optogenetics changes membrane currents with light

**Optogenetics** uses genetically introduced light-sensitive proteins to change the activity of selected cells. **Channelrhodopsin** forms a membrane channel that opens in response to an appropriate wavelength. Positive ions then move through the open channel according to their electrochemical gradients, often depolarizing the neuron enough to increase firing while the light is delivered to the tissue.

Other light-sensitive proteins suppress activity by moving ions in a direction that opposes excitation. **Halorhodopsin** pumps chloride into the cell, while other tools can increase potassium conductance or inhibit transmitter release. The biological response depends on which protein is expressed, where it is located, and how its light-driven ion movement changes the membrane's electrical state.

An optical fiber or implanted light source delivers illumination to the target. The light reaches a volume containing both labeled and unlabeled cells, but only cells carrying the relevant light-sensitive protein acquire its intended direct response. Combining the optical location with genetic targeting therefore restricts the manipulation more precisely than light delivery or gene expression considered alone.



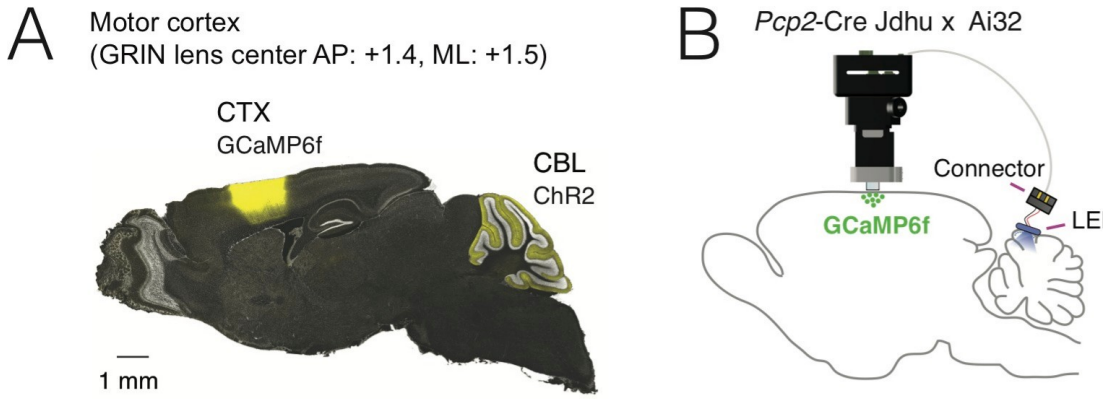

Carlson and Birkett (2022), Fig. 5.29, p. 142.

# Stimulating cerebellar cells changes activity in motor cortex

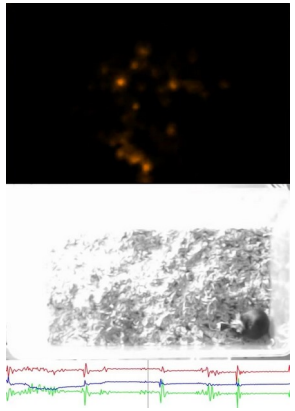
De Groot and colleagues expressed channelrhodopsin in cerebellar Purkinje cells while recording GCaMP fluorescence from motor cortex. Brief blue-light pulses activated the targeted Purkinje population. The microscope then measured changes in a separate cortical population, allowing the investigators to follow an effect of cerebellar stimulation beyond the illuminated tissue in the same behaving animal.

Purkinje cells release the inhibitory transmitter **GABA**, but their effect on distant cortex depends on intervening circuits. Altering their firing changes the output of cerebellar targets, which influence additional relays on the way to cortex. The cortical response therefore reflects activity transmitted through a network rather than blue light directly exciting the cortical neurons being imaged.

The investigators observed increased cortical calcium responses and movement responses following stimulation at selected cerebellar sites. Stimulation on opposite sides could evoke head movements in different directions. Comparing the light timing with the accelerometer and cortical fluorescence separated the rapid visible movement from the slower calcium signal recorded in the receiving cortical population.




de Groot et al. (2020), Fig. 4, selected panels.



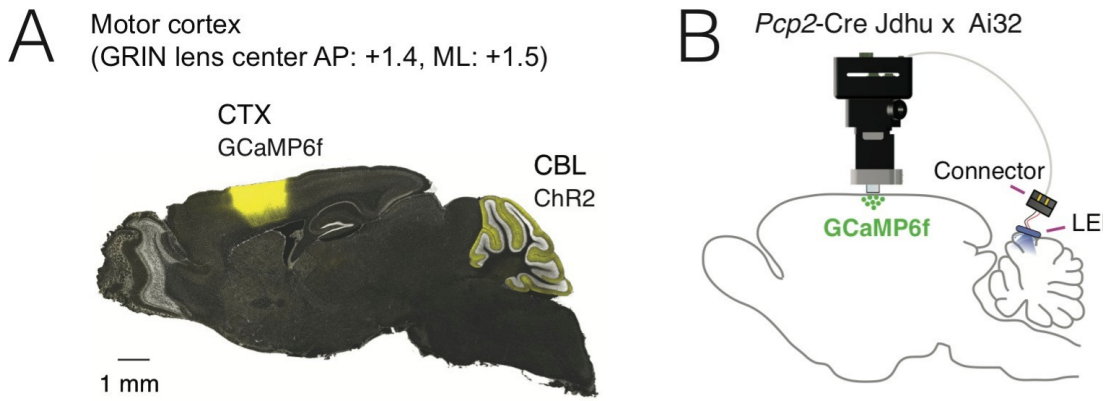
de Groot et al. (2020), Fig. 4, video 1.

# Genetic targeting selects the cells that respond to light

A **promoter** is a DNA sequence that helps control where a gene is expressed. Researchers can use promoters associated with particular cell populations to restrict a light-sensitive protein or fluorescent reporter. The result depends on the expression pattern: a broadly expressed construct reaches a different set of cells from one restricted to a defined neuronal population.

The **Cre–lox system** provides another targeting step. Cre is an enzyme that rearranges DNA at short lox recognition sequences. A reporter construct can be arranged so that its gene becomes active only after this rearrangement. Crossing a mouse expressing Cre in a selected population with a compatible reporter line restricts expression to cells where Cre acts.

In the NINscope study, a Purkinje-cell Cre line was combined with an optogenetic reporter line to express channelrhodopsin in those cells. A separate viral delivery introduced GCaMP into motor cortex. The two populations therefore had different experimental roles: cerebellar cells received the direct light-driven manipulation, while cortical cells produced the fluorescence used to measure a downstream response.



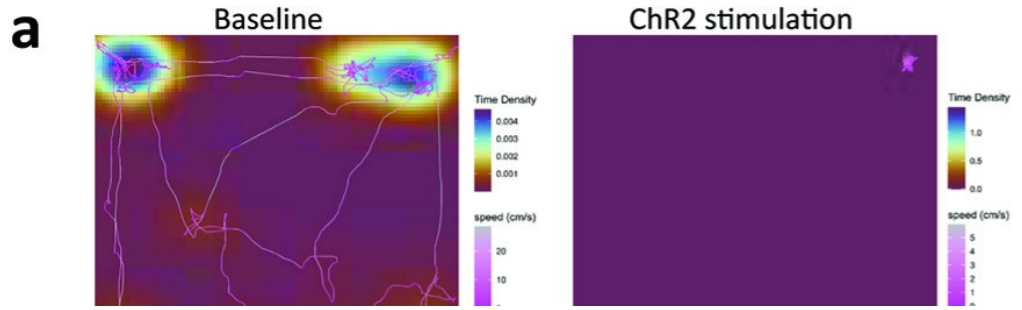

de Groot et al. (2020), Fig. 4, selected panels.

# Hypothalamic terminal activation interrupts movement

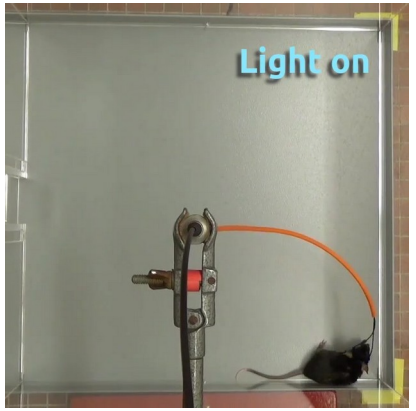
Cola and colleagues targeted neurons expressing **Foxb1**, a gene used to identify populations in the hypothalamus. Their axons reach the **periaqueductal gray**, or **PAG**, a midbrain region involved in defensive and autonomic responses. Expressing channelrhodopsin in these neurons allowed the investigators to activate their axon terminals by illuminating a selected part of the PAG.

Illuminating terminals in the rostral dorsolateral PAG produced an abrupt reduction in movement. The mice often remained largely still during stimulation and moved again after it ended. In one source example, the limbs remained relatively immobile while the mouse continued moving its head, showing that different components of the animal's visible behavior could change by different amounts.

The investigators also verified where the optical fibers ended in the brain. Terminal stimulation produced the characteristic response at particular PAG locations rather than uniformly throughout the midbrain. Relating the behavioral change to the fiber location and labeled projection narrowed the manipulation to a defined pathway instead of treating the entire hypothalamus or PAG as one functional unit.




Cola et al. (2024), Fig. 3, selected panels.



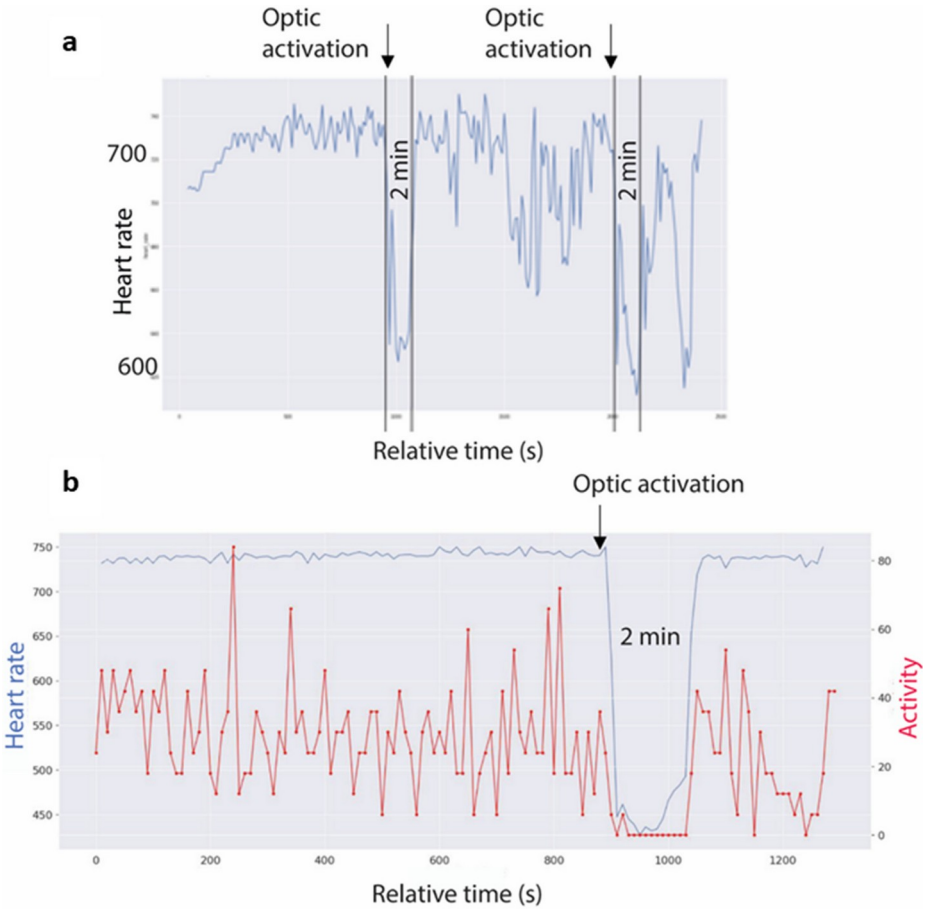

Cola et al. (2024), Video 1.

# Pathway stimulation changes heart rate and breathing

Cola and colleagues measured heart rate during optical activation of the hypothalamic projection to the PAG. **Bradycardia** means a slowing of the heart. Their recordings showed heart-rate reductions during stimulation at effective sites, adding an autonomic measurement to the visible reduction in locomotion produced by activation of the targeted axonal endings in the midbrain.

The heartbeat was measured using an implanted telemetric system that transmitted physiological signals while the animal moved in an arena. Aligning this signal with light onset identified when the cardiac response developed relative to the manipulation. The method followed an internal physiological variable that would not be apparent from an ordinary video of the animal's posture or location.

Chemogenetic activation of the Foxb1-expressing population also changed breathing. **Tachypnea** means an increase in breathing rate, which the researchers measured using pressure changes in a respiratory chamber. Movement, heart rate, and respiration therefore supplied distinct outputs from the manipulated system, allowing the experiment to describe changes in both overt action and internal bodily regulation in the same general pathway.



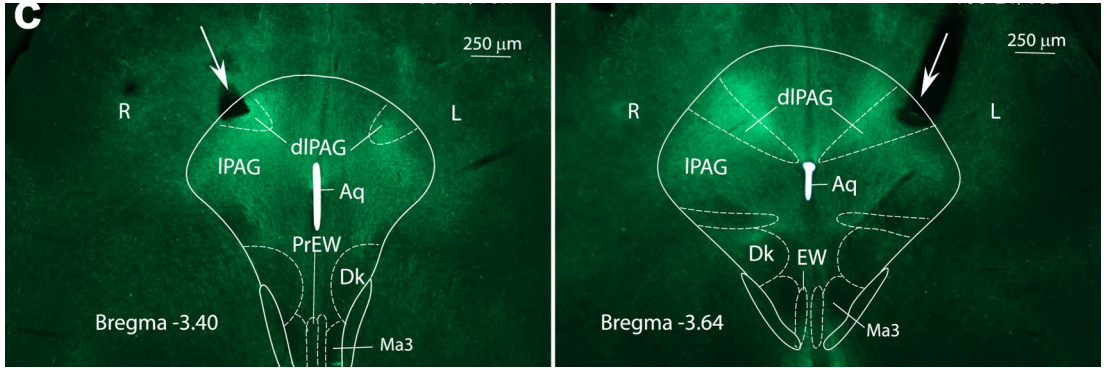

Cola et al. (2024), Fig. 6, selected panels.

# Fiber position determines which terminals receive light

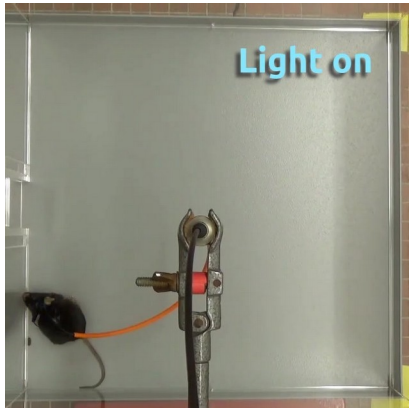
A light-sensitive axon responds only where illumination reaches its membrane with sufficient intensity. Cola and colleagues placed fibers above terminals of Foxb1-expressing neurons in the PAG. The amount and location of illuminated tissue depended on the fiber tip, the pattern of axonal labeling, and the scattering of light through the tissue surrounding the optical target during stimulation.

Their anatomical reconstructions placed the effective sites in the rostral dorsolateral PAG. These locations produced pronounced immobility during the stimulation interval. The source recordings show the animal before, during, and after illumination, preserving the transition into and out of the altered behavior rather than presenting a still animal without the timing of the manipulation that preceded it.

Stimulating axon terminals and stimulating all cell bodies need not produce identical responses. A cell body can send branches to multiple targets, whereas light delivered near one terminal field initially acts on a more spatially restricted axonal population. Describing the manipulated compartment specifies which portion of the neuron's anatomy received the light-driven change in membrane current during that experiment.




Cola et al. (2024), Fig. 7, selected panels.



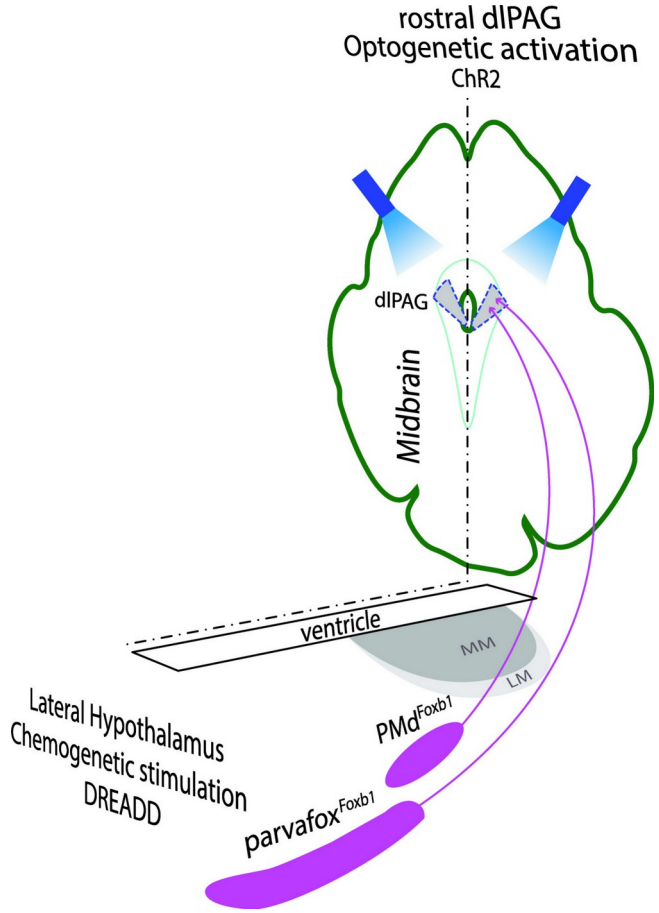

Cola et al. (2024), Video 2.

# Chemogenetic receptors convert drugs into cellular signals

**Chemogenetics** introduces an engineered receptor whose activation changes cellular signaling. A commonly used activating receptor, **hM3Dq**, couples to a signaling protein that stimulates enzymes inside the cell. These reactions can increase neuronal excitability and alter firing after an appropriate activating compound reaches the receptor, without requiring light to be delivered continuously to the target tissue during the manipulation.

An inhibitory receptor such as **hM4Di** engages a different signaling pathway that can reduce excitability or transmitter release. These engineered receptors are often called **DREADDs**. The acronym identifies the tool family, but the relevant biological distinction is the intracellular response produced by each receptor and how that response changes the activity or output of the cells carrying it.

The administered compound must be absorbed, distributed, and cleared, so chemogenetic effects usually develop and persist on a slower timescale than brief optogenetic pulses. In the Foxb1 study, the researchers compared physiological responses after drug administration in animals expressing activating or inhibitory receptors and in comparison animals, measuring how receptor-specific signaling altered breathing over the subsequent recording period.



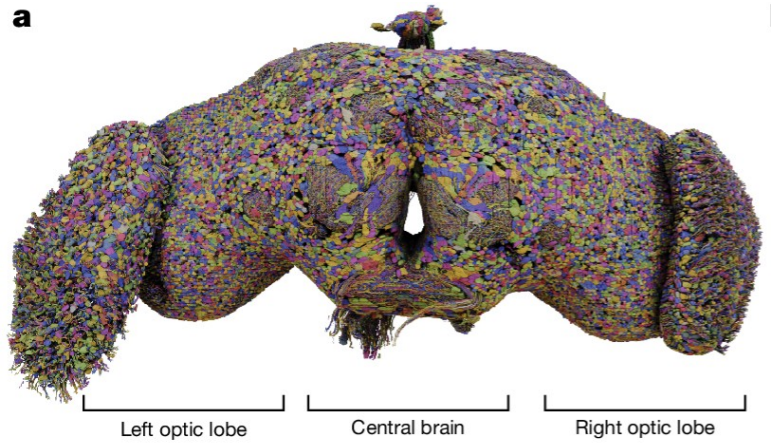

Cola et al. (2024), Fig. 1.

# Electron microscopy reconstructs a whole adult fly brain

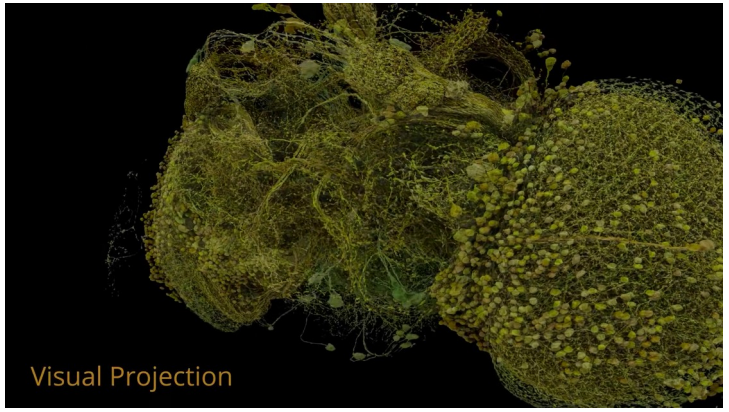
Dorkenwald and colleagues assembled a **connectome**, a map of neuronal connections, from electron-microscopy images of an adult female fruit fly brain. Their reconstruction contained 139,255 neurons and approximately fifty million chemical synapses. Electron microscopy resolves the small membrane structures needed to follow branches through densely packed tissue and identify where presynaptic and postsynaptic elements meet within the specimen.

Computational segmentation grouped image regions into candidate neuronal branches. Human proofreading then corrected mistaken joins and breaks so that a branch would be assigned to the appropriate neuron. A wrong join can falsely merge two cells, while a missed continuation can split one cell into several fragments, changing the apparent pathways through the reconstructed network of neurons and synapses.

The source rendering displays neurons reconstructed from this imaged brain. The geometry preserves the arrangement of cell bodies and branches identified in the specimen. Each reconstructed connection refers to an anatomical observation; predicting what signal travels through it still requires information about transmitter identity, receptor effects, and the activity of the participating cells during a defined sensory event or behavior.




Dorkenwald et al. (2024), Fig. 1, selected panels.



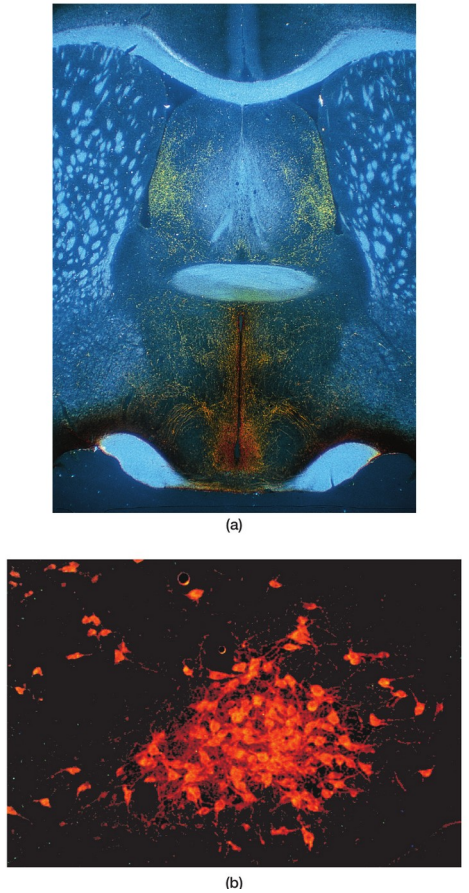

Dorkenwald et al. (2024), Supplementary Video 1.

# Antibodies reveal proteins inside identified cells

**Immunohistochemistry** uses antibodies to locate particular proteins in tissue. An antibody binds a molecular feature on its target, and a fluorescent or chemically visible label reveals where binding occurred. Researchers can label an enzyme used to manufacture a transmitter, a receptor that detects it, or another protein associated with a particular cell population in a brain section.

**Choline acetyltransferase**, or **ChAT**, is the enzyme that synthesizes acetylcholine from choline and an acetyl donor. Labeling this enzyme identifies cells with part of the machinery needed to produce acetylcholine. The resulting image marks a chemical feature of the cells rather than directly measuring the amount of acetylcholine released during a particular behavior or sensory response in the animal.

Multiple antibodies can be combined when their labels can be distinguished. One channel might identify a neuronal population while another marks a receptor or activity-associated protein. Comparing their distributions within the same tissue reveals which labeled cells contain both molecular features, adding chemical identity to the anatomical location and shape observed in the section under the microscope during examination.



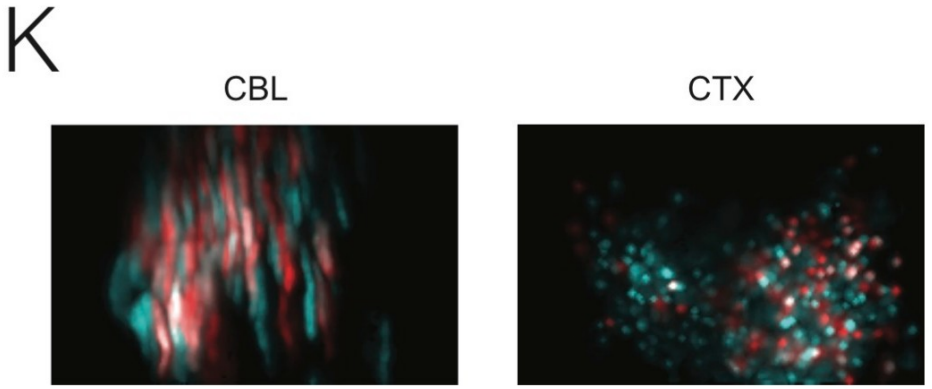

Carlson and Birkett (2022), Fig. 5.30, p. 145.

# Two microscopes measure activity across separate regions

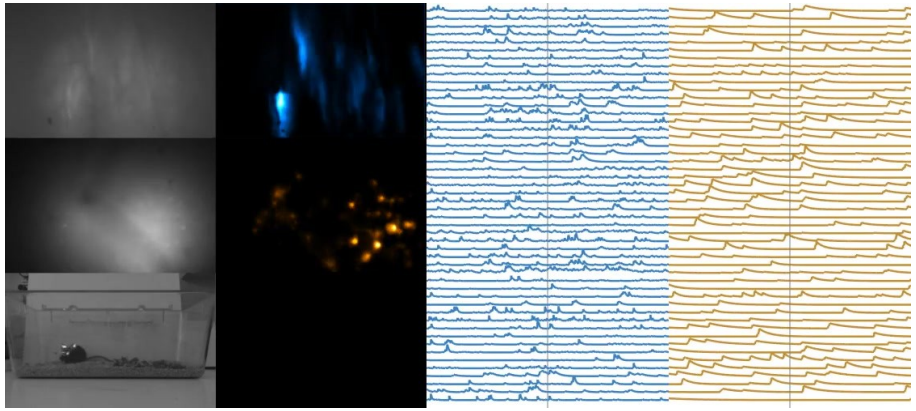
De Groot and colleagues recorded calcium activity from cerebellar Purkinje cells and motor-cortex neurons simultaneously using two NINscopes. Each microscope sampled a different region while the animal moved freely. The resulting datasets preserved separate cellular signals in both locations, allowing changes within one population to be compared with changes in another during the same period of behavior and recording.

The investigators identified episodes when groups of cells became active together, both within a region and across the two regions. They aligned these population events with acceleration measured by the animal's motion sensor. Some coordinated events preceded acceleration, whereas others occurred around movement or without a pronounced accompanying acceleration in the measured channel during that specific episode in the recording.

A shared event can arise because regions interact or because both receive a common input. The two recordings describe coordination in time; the optogenetic experiments in the same study separately tested an imposed change in one circuit. Combining these approaches distinguishes observing a spontaneous population pattern from measuring a response after investigators deliberately change the activity of a targeted neuronal population.




de Groot et al. (2020), Fig. 3, selected panels.



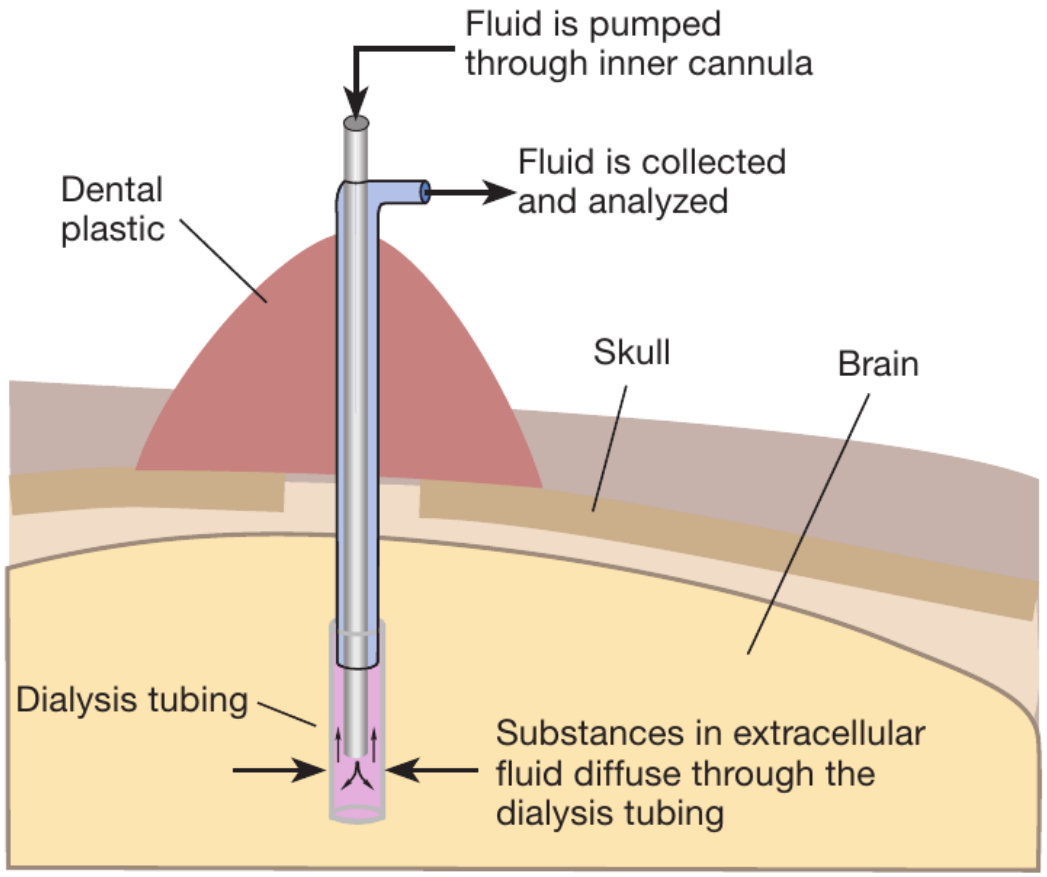
de Groot et al. (2020), Fig. 3, video 4.

# Microdialysis samples chemicals from extracellular fluid

**Microdialysis** collects molecules from the fluid surrounding cells. A small probe contains a membrane that allows selected small molecules to diffuse across it. Artificial fluid flows through the probe, picks up substances from the surrounding extracellular space, and leaves through an outlet, where the collected sample can be analyzed to determine which chemicals it contains and at what concentration.

Molecules move across the membrane according to concentration differences and the membrane's permeability. The collected concentration also depends on the flow rate and the recovery efficiency of the probe. A slow chemical measurement can therefore integrate release, diffusion, uptake, and metabolism over a larger interval than the brief events detected by an electrode near a firing neuron in the same tissue.

The collected samples can be separated and measured using chemical analysis. Researchers might compare transmitter levels before and after a drug or behavioral event. Because the probe samples extracellular fluid, this measurement differs from staining a transmitter-producing enzyme inside fixed cells: one follows molecules outside cells over time, while the other reveals where a molecular component is located within preserved tissue.



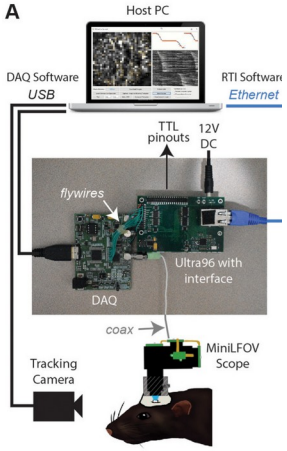

Carlson and Birkett (2022), Fig. 5.31, p. 146.

# Real-time image processing prepares neural feedback

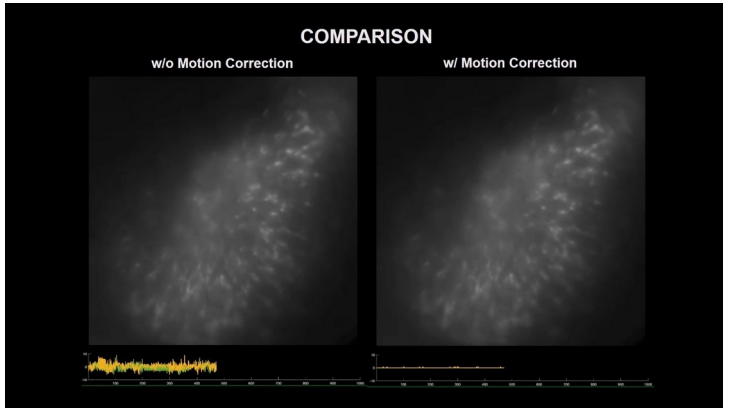
Chen and colleagues developed **DeCalciOn**, hardware that processes miniscope images as they arrive. The device receives video data, corrects image movement, enhances cellular fluorescence, and extracts activity measurements. Processing these steps during acquisition makes the resulting signals available for an experiment that must respond to the animal's current neural activity instead of waiting for analysis after the session has ended.

The system uses a **field-programmable gate array**, or **FPGA**, an electronic device whose circuits can be configured to perform operations in parallel. Different stages of image processing can therefore proceed as the data stream moves through the hardware. This design reduces the delay between receiving the optical image and producing an output that an external experimental device can use.

The source motion-correction demonstration compares raw and stabilized images with estimates of their displacement. Aligning the frames helps prevent the same neuron from moving into and out of an extraction region during behavior. The subsequent calcium measurement can then follow changing fluorescence within a consistent part of the imaged field rather than repeatedly sampling different tissue as the animal moves during recording.




Chen et al. (2023), Fig. 1, selected panels.



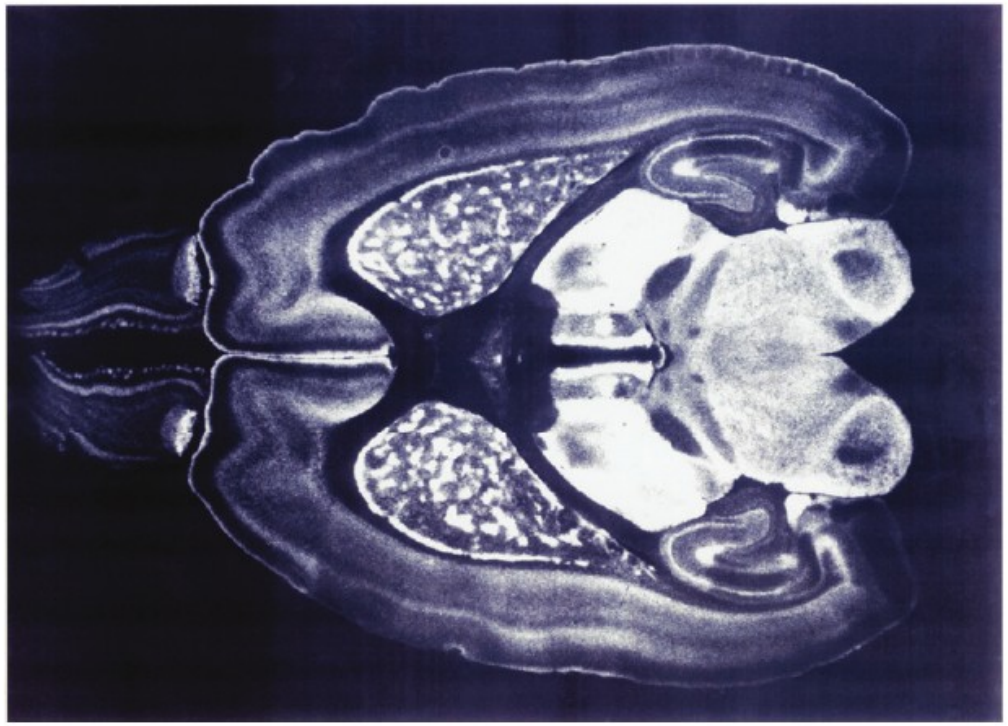

Chen et al. (2023), Video 1.

# Receptor labeling maps molecular binding sites

A **ligand** is a molecule that binds a receptor or another molecular target. A labeled ligand can reveal where its binding sites are concentrated in tissue. Researchers select a ligand with an appropriate affinity for the receptor of interest, apply it under controlled conditions, and detect the label remaining after unbound material has been removed from the specimen during processing.

In **autoradiography**, the label is radioactive. Emissions from the labeled tissue expose a detector or photographic material, producing an image of the distribution of bound tracer. Regions with stronger labeling contain more detected tracer under the binding conditions, allowing the anatomical distribution of a receptor-associated signal to be compared across different parts of the nervous tissue in the section.

Binding-site density and chemical signaling describe different variables. A receptor map shows where a tracer can bind, whereas microdialysis samples extracellular molecules and a calcium indicator reports an intracellular ion change. Describing the measured variable prevents a bright receptor image from being interpreted as evidence that the transmitter was being released or that those cells were firing rapidly.



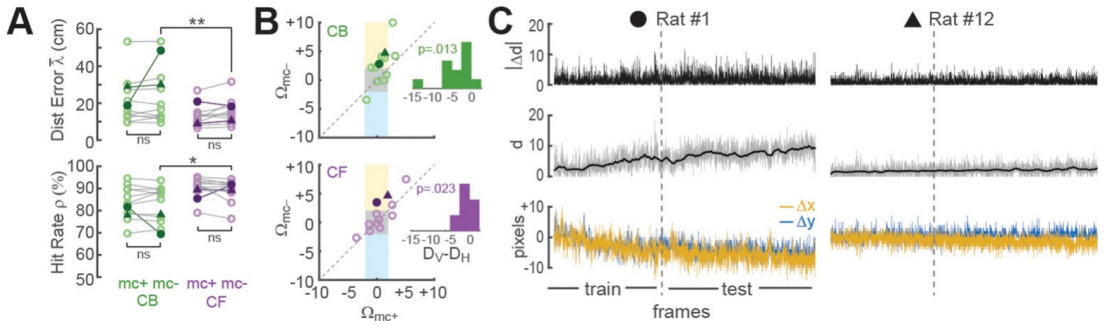
Carlson and Birkett (2022), Fig. 5.23, p. 139.

# Image regions yield separate fluorescence traces

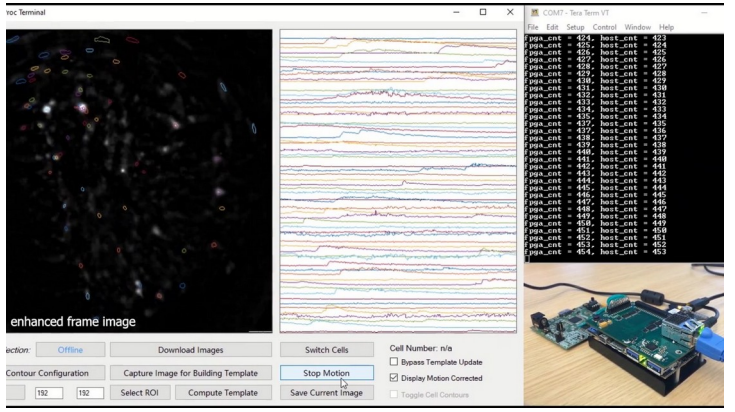
DeCalciOn can extract calcium traces from **contours**, boundaries placed around fluorescent cellular patterns. For each frame, the system sums or combines the pixel values within a selected region. Repeating this calculation produces a time series describing how that region's fluorescence changes as the sensor records successive images of the labeled neural tissue.

Chen and colleagues also implemented a **contour-free** approach that samples the image through a set of pixel masks. These masks need not correspond exactly to manually identified cells. A decoder can use the resulting population of fluorescence measurements to estimate a behavioral variable, provided the measurements contain a reproducible pattern associated with the variable being predicted.

The system decoded a rat's position on a linear track from hippocampal calcium measurements. During training, recorded positions were paired with fluorescence patterns so the decoder could learn their association. During online use, the incoming pattern generated an estimated position, which could be compared with the independently tracked position to measure how well the neural data predicted the animal's location during movement.




Chen et al. (2023), Fig. 4, selected panels.



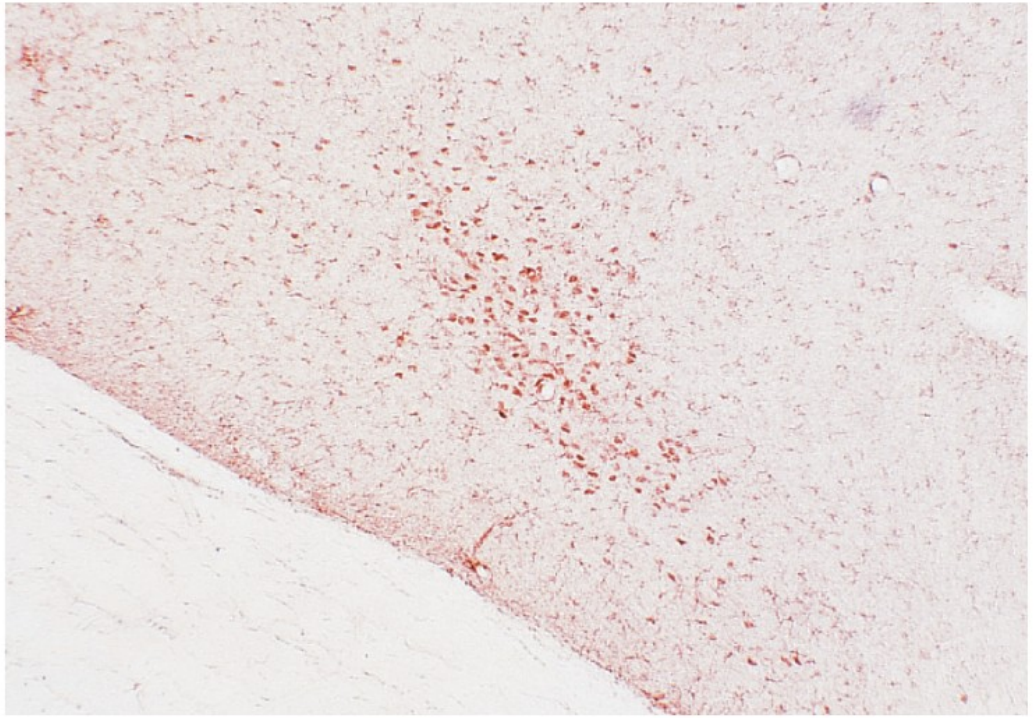

Chen et al. (2023), Video 2.

# Activity-associated genes leave a slower molecular record

Some patterns of neuronal activity activate genes whose protein products can be detected later. **Fos** is a protein made after activation of the immediate-early gene **c-fos** under many experimental conditions. Staining for Fos can reveal populations that underwent this gene-expression response during a preceding period, even when an investigator did not record those neurons electrically while the event was taking place.

The process includes transcription of the gene into RNA and translation of that RNA into protein. These steps take time, so the detectable protein distribution develops more slowly than the action potentials that may have initiated the cellular response. The tissue therefore contains an accumulated molecular record associated with prior activation rather than a moment-by-moment replay of the neuron's electrical firing pattern during the experience.

Researchers can combine Fos labeling with a marker of cell identity to ask which populations underwent the response. The overlap might identify transmitter-producing cells or neurons projecting to a particular target. Because different cell types and signaling conditions regulate gene expression differently, interpreting the overlap requires specifying the marker, the experience, and the interval between that experience and tissue collection for histological analysis.



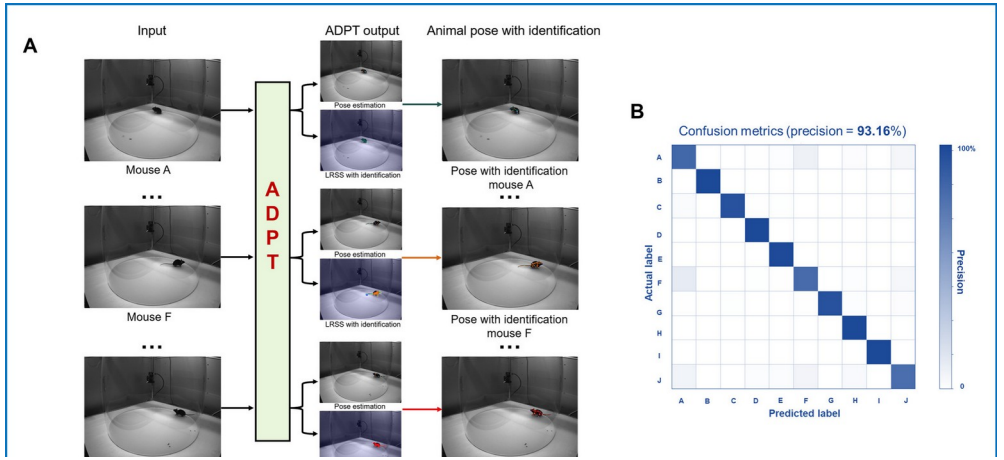
Carlson and Birkett (2022), Fig. 5.24, p. 139.

# Tracking separates body position from animal identity

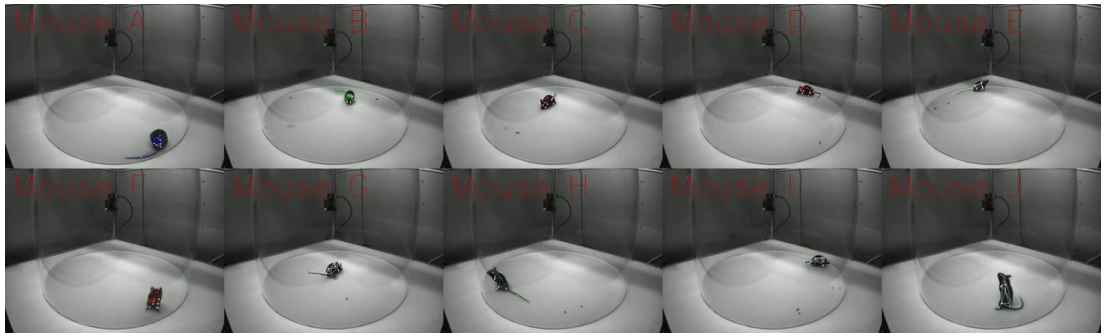
Following a body landmark and identifying the animal carrying it are different computational problems. Tang and colleagues developed ADPT to combine pose tracking with individual identification. In recordings containing several unmarked mice, the system estimated body-part positions while attempting to preserve which positions belonged to which individual animal as they moved across the camera's field.

An identity error can occur when animals cross or overlap. A trajectory may remain on a real mouse yet switch from one individual to another. The investigators evaluated identification alongside landmark accuracy, allowing these errors to be examined separately instead of assuming that accurate body-part coordinates automatically preserve the identity of the animal carrying the tracked feature.

Preserving identity makes it possible to assign successive actions to the same individual. Researchers can examine whether one mouse approaches another, how long their positions remain close, and which animal moves away afterward. These descriptions come from the spatial and temporal sequence of tracked bodies; a claim about motivation or social meaning requires an additional behavioral definition appropriate to the specific experimental task being analyzed.




Tang et al. (2025), Fig. 7, selected panels.



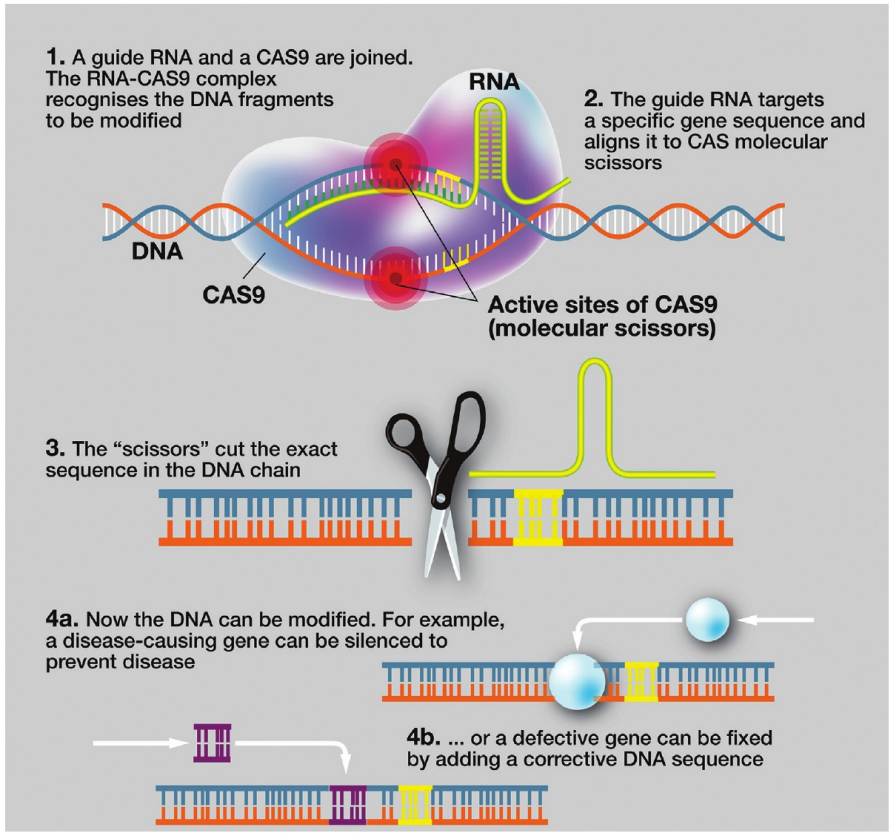

Tang et al. (2025), Fig. 7, video 1.

# Inherited variants differ from experimental mutations

**Twin studies** compare similarity between identical twins, who share nearly all inherited sequence variation, and fraternal twins, who share less on average. **Adoption studies** compare traits with biological and adoptive relatives. These designs estimate contributions from inherited and environmental differences to variation in a population; they do not identify a complete molecular pathway inside a particular neuron during the behavior being studied.

A **genome-wide association study**, or **GWAS**, tests whether common DNA variants occur at different frequencies among people with different measured traits. An associated variant may influence a gene directly or lie near a functional variant inherited with it. Following the signal into a biological mechanism requires additional evidence about the affected cells, the gene product, and its effects on physiology within those cells.

An experimental mutation changes a selected gene in a controlled model. Comparing altered and unaltered cells can test the gene's contribution to a particular cellular response. Restricting the manipulation to adulthood or a defined cell population helps separate an immediate physiological effect from changes that arose earlier because the gene also influenced brain development, cell survival, or circuit formation in the animal.



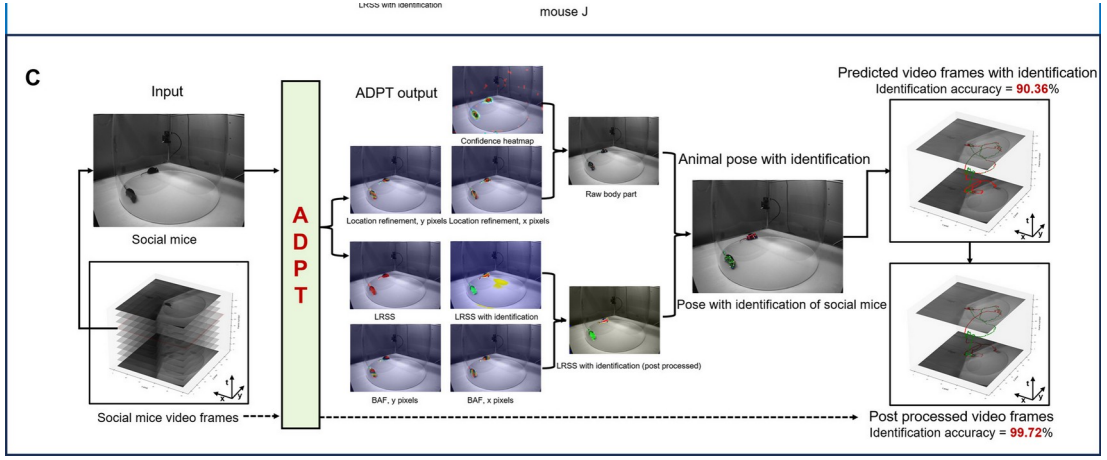

Carlson and Birkett (2022), Fig. 5.34, p. 151.

# Social recordings track several interacting bodies

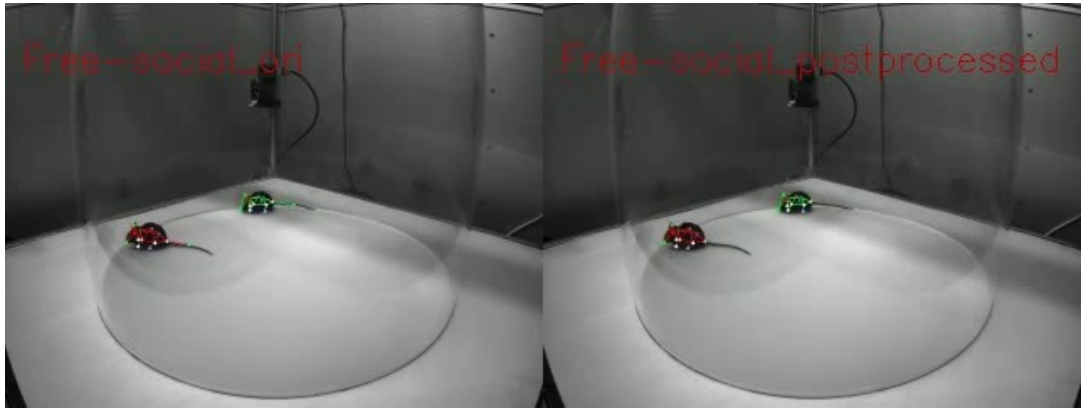
Tang and colleagues tested ADPT on freely interacting mice, where animals frequently approach, overlap, and obscure one another. The source recordings preserve the animals' movements with their estimated landmarks and identities. Maintaining these assignments allows an investigator to describe how the positions of two bodies change relative to one another rather than analyzing each visible fragment as an independent object in every frame.

The method combined image features used for identity with estimates of body pose. When one animal partly blocked another, the available landmarks and image appearance changed. Evaluating the resulting assignments against annotated examples measured whether the system recovered the appropriate individual as the animals separated, rather than silently exchanging their identities after an overlap or close encounter.

A social-distance measure depends on which landmarks define the distance. Nose-to-nose separation describes a different spatial arrangement from the distance between body centers. Choosing a measure that matches the behavior allows researchers to distinguish orientation toward another animal from simple proximity, particularly when both animals remain near each other while turning their heads or changing the alignment of their bodies during the interaction.




Tang et al. (2025), Fig. 7, selected panels.



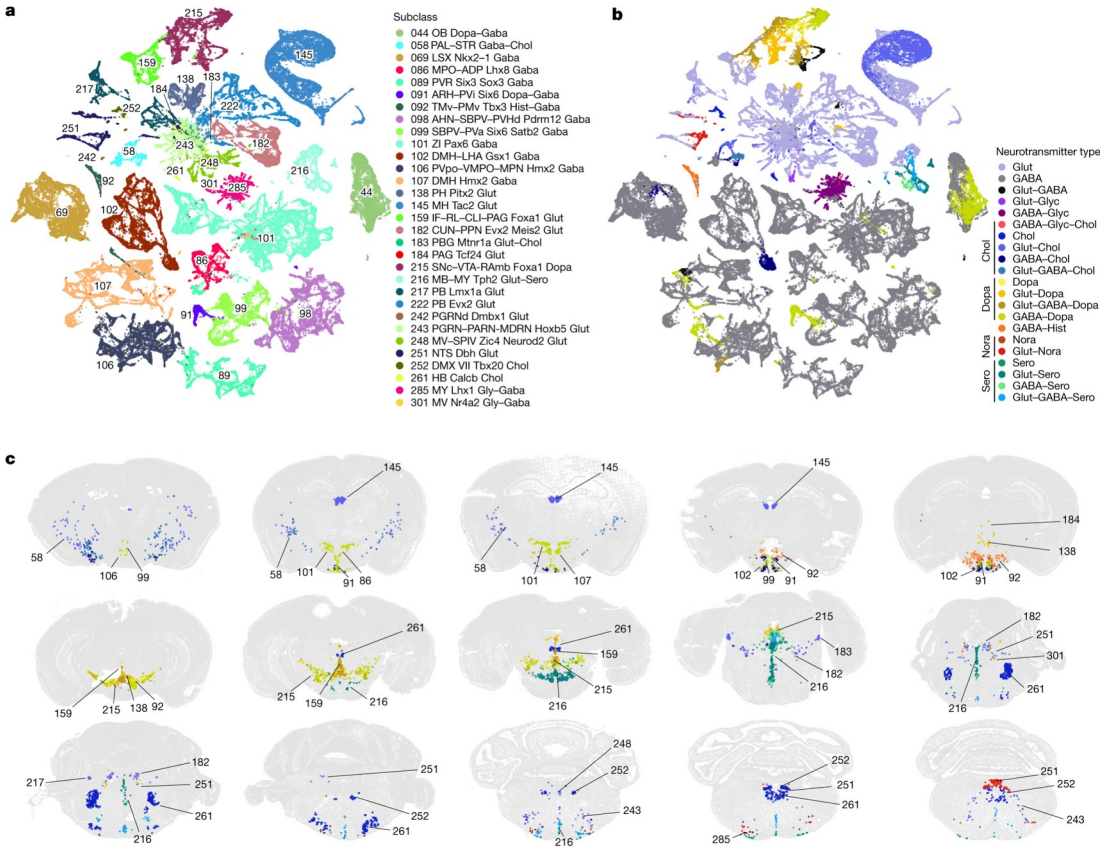

Tang et al. (2025), Fig. 7, video 2.

# RNA profiles distinguish neuronal chemical identities

Yao and colleagues built a mouse brain cell atlas using **single-cell RNA sequencing** and spatial measurements. RNA sequencing identifies transcripts, the RNA molecules produced when genes are expressed. Measuring many transcripts within each cell provides a molecular profile that separates populations with different chemical machinery, including differences that are difficult to recognize from cell shape alone.

The investigators examined genes associated with transmitter synthesis, packaging, and other cellular functions. Their atlas revealed diverse patterns of transmitter and neuropeptide gene expression. A **neuropeptide** is a short protein-like signaling molecule. Cells can express machinery for several signaling substances, producing combinations more specific than a single broad transmitter category assigned to an entire brain region.

Spatial measurements located these molecularly classified populations within the brain. This revealed whether related profiles occurred in neighboring cells or distant structures. Combining identity and location described the organization of chemically distinct populations across regions, while keeping gene expression separate from the release of a transmitter during a particular movement or behavioral state in the living animal.



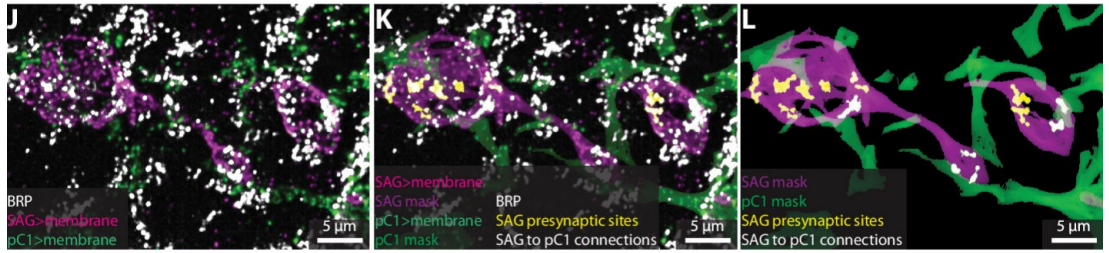

Yao et al. (2023), Fig. 3, selected panels.

# Cell-specific labels preserve identity in dense tissue

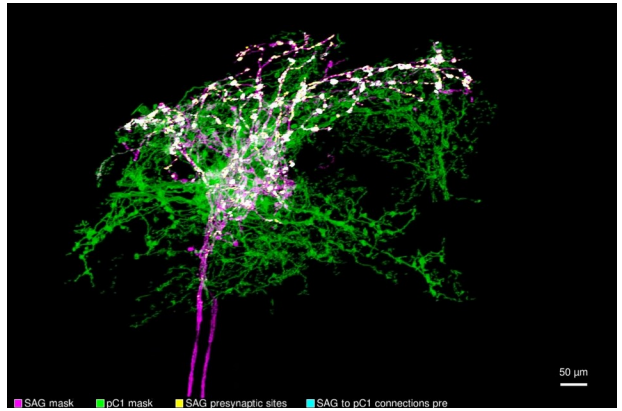
Lillvis and colleagues used genetic tools to label selected fly neurons and their synaptic components. A membrane reporter outlined the chosen cells, while additional labels identified release sites and receiving structures. Restricting expression to a recognizable population made it possible to follow that population through dense tissue containing many unlabeled or differently labeled branches belonging to neighboring cells.

Their analysis compared contacts identified from postsynaptic labels with contacts estimated from a partner neuron's membrane. A postsynaptic label marks receiving machinery, whereas a membrane label outlines the cell surface. The first count requires a detected receiving specialization; the second can include contacts with regions of the receiving cell's membrane where that postsynaptic marker was not detected.

The resulting three-dimensional volumes preserve the locations of the assigned objects along the selected cells. A rotated view makes it easier to inspect whether an apparent overlap persists across depth or separates when viewed from another angle. The underlying contact counts still come from the recorded channels and segmentation criteria, while the source visualization makes the spatial arrangement of the labeled structures easier to examine during instruction.




Lillvis et al. (2022), Fig. 4, selected panels.




Lillvis et al. (2022), Video 4.

# Single-cell sequencing organizes cells by expressed genes

Yao and colleagues profiled millions of mouse brain cells to construct a hierarchy of molecular cell types. Each cell contributed measurements of many RNA transcripts. Cells with related expression patterns were grouped into broader and narrower categories, producing an organization that distinguishes large classes while preserving finer differences among the populations within those classes.

**Clustering** groups observations according to similarities in their measured features. For RNA data, those features include the abundance of different transcripts. Genes involved in transmitter handling, receptor signaling, and regulation of other genes can help distinguish populations. The resulting cluster describes a reproducible expression pattern rather than a single cell's momentary firing rate during a sensory stimulus or behavioral action in the living animal.

The atlas combines this molecular classification with independently measured spatial organization. Similar cell types can then be compared across regions, while neighboring cells can be separated by their transcript profiles. This organization helps select genetic markers for a subsequent experiment, such as targeting a reporter to a narrower population instead of expressing it in every neuron within a broadly defined anatomical region of the brain.



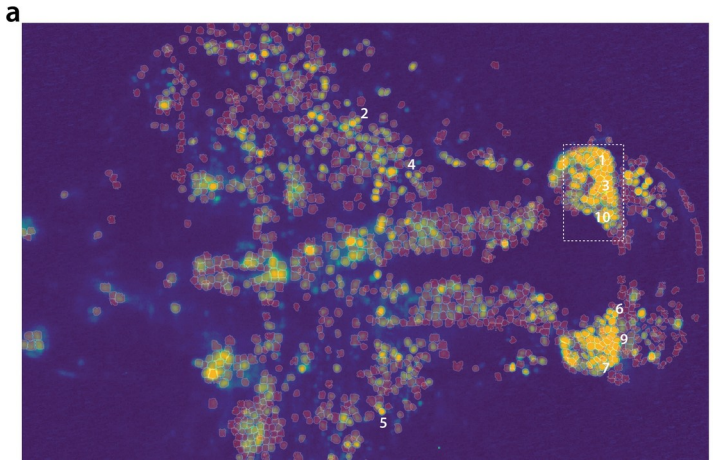

Yao et al. (2023), Fig. 1, selected panels.

# A reporter gene makes a selected cellular process visible

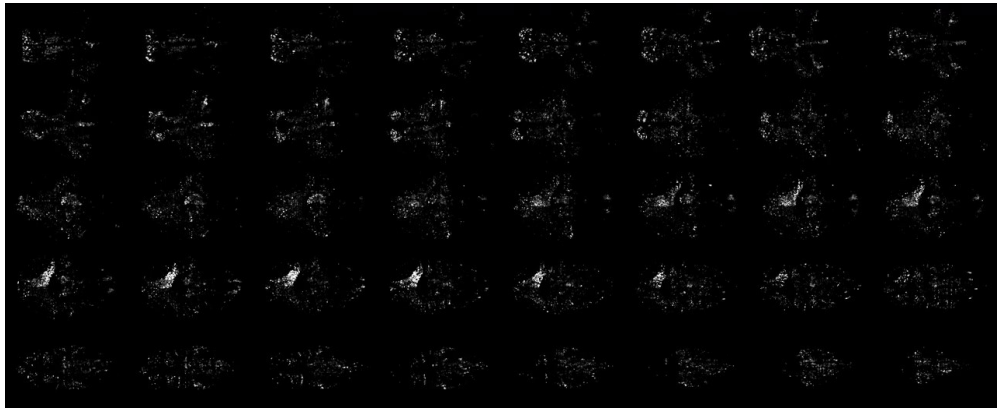
A **reporter gene** encodes a product that researchers can detect, such as a fluorescent protein. Delivering the gene allows selected cells to manufacture the reporter themselves. A stable fluorescent label reveals cell position and shape, whereas a responsive reporter changes its signal when a particular cellular variable, such as calcium concentration, changes.

Gene delivery and gene editing perform different operations. A viral vector can introduce an additional reporter without deliberately cutting a selected genomic sequence. **Gene editing** changes a specified DNA target within the cell's genome. The fluorescent cells in an imaging experiment therefore do not automatically indicate that an endogenous gene was removed or repaired in those cells.

In calcium imaging, the expressed protein combines molecular recognition with light emission. Calcium binding changes the reporter's fluorescence, and a camera converts the emitted photons into image values. The CaImAn source recording visualizes activity extracted from labeled zebrafish brain tissue, illustrating how genetically introduced reporters provide optical access to neuronal populations throughout a recorded volume of nervous tissue.




Giovannucci et al. (2019), Fig. 6, selected panels.



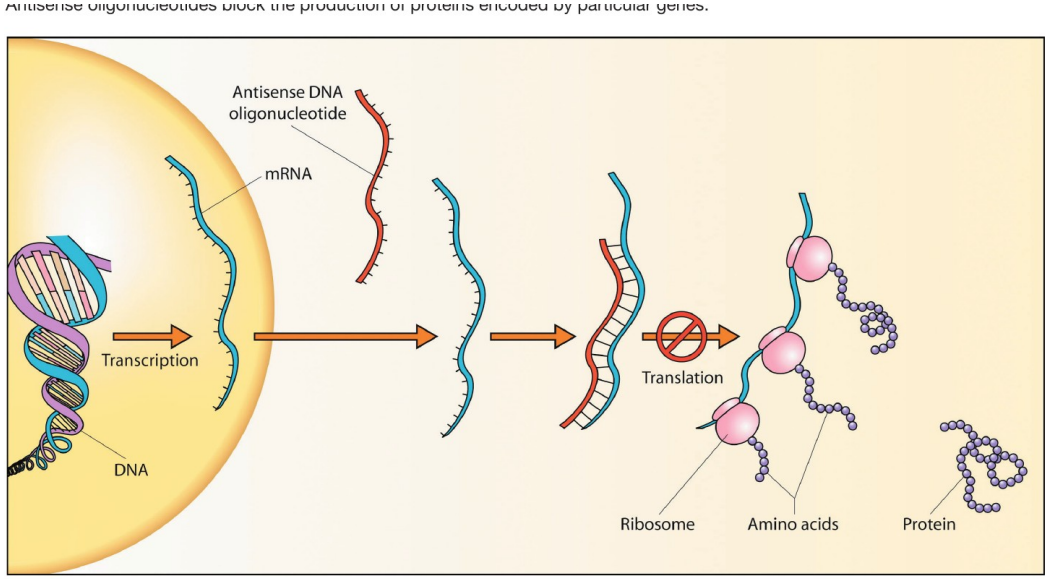
Giovannucci et al. (2019), Fig. 6, video 1.

# Antisense molecules change how a selected RNA is used

An **antisense oligonucleotide**, or **ASO**, is a short synthetic strand designed to bind a complementary RNA sequence. Complementary bases pair, allowing the ASO to recognize a selected transcript among many RNA molecules. Once bound, an ASO can alter the RNA's processing or availability, changing how much functional protein is produced from that transcript inside the targeted cells.

Some ASOs recruit enzymes that degrade the bound RNA. Others alter **splicing**, the process that removes selected RNA segments before the mature message is translated. Changing splicing can restore a useful protein sequence or reduce an unwanted product, depending on the RNA target and where the ASO binds relative to the sequences controlling RNA processing.

An ASO acts at the RNA level rather than necessarily changing the underlying DNA sequence. Its effect therefore depends on delivery to the relevant cells, binding to the transcript, and the time required for existing proteins to turn over. A rapid change in RNA abundance can precede a slower change in cellular physiology when the original protein remains functional for some time.



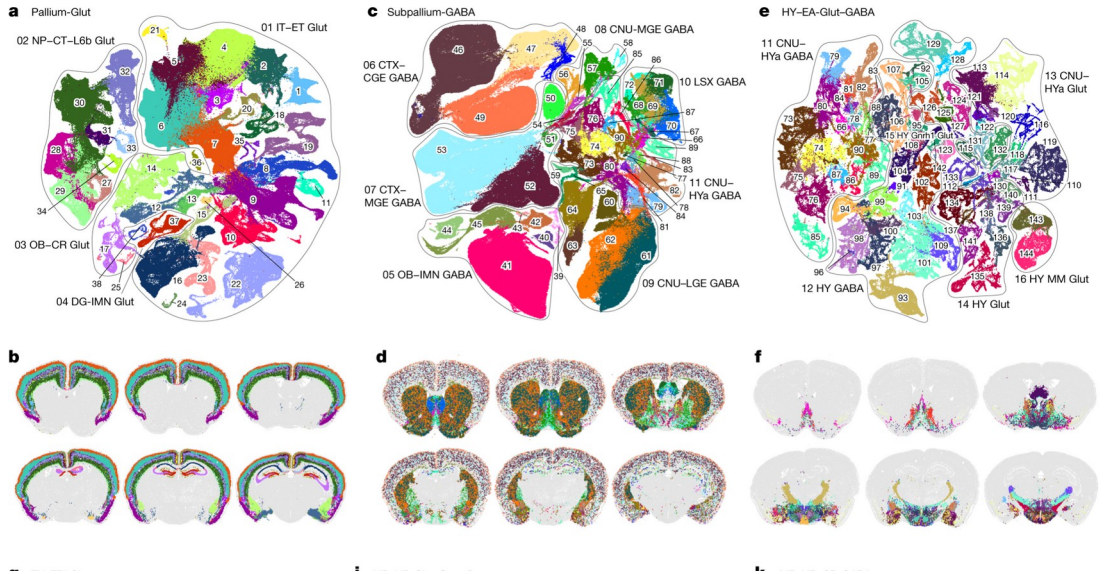

Carlson and Birkett (2022), Fig. 5.33, p. 150.

# Spatial RNA measurements locate molecular cell types

Yao and colleagues combined RNA sequencing with **MERFISH**, a method that detects selected RNA molecules at their original locations in tissue. Fluorescent probes bind complementary RNA sequences. Successive rounds of imaging assign a pattern of detected signals to each molecular target, allowing many genes to be measured while preserving the positions of the cells within the brain section.

The measured transcripts formed molecular profiles that could be compared with the sequencing-based cell groups. A cell's spatial location was therefore paired with an expression-based identity. This allowed the investigators to determine where particular populations occurred, including mixtures of molecularly distinct cells within a single anatomical region that would appear more uniform in an image showing only cell-body shape or tissue boundaries.

The resulting maps revealed region-specific organization of neuronal and non-neuronal populations. Some expression-defined populations were restricted to particular territories, while related populations occupied broader areas. Retaining the spatial coordinates made it possible to examine boundaries and neighboring cell types, rather than inferring anatomical position from a detached cell's RNA profile after it had been removed from the surrounding tissue for sequencing.



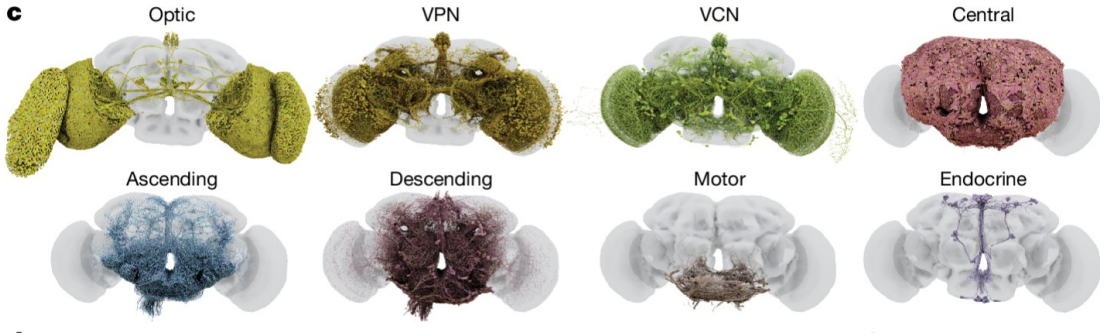

Yao et al. (2023), Fig. 2, selected panels.

# A connectome traces sensory inputs to brain outputs

Dorkenwald and colleagues grouped reconstructed fly neurons according to their position and direction of information flow. **Afferent neurons** bring signals into the brain from sensory organs or other regions, while **efferent neurons** carry signals outward. Intrinsic neurons remain within the brain, where their branches provide intermediate processing between incoming sensory pathways and outgoing commands to other parts of the animal.

The reconstruction distinguished populations reaching the ventral nerve cord, the insect structure containing many motor circuits. Descending neurons carry signals from the brain toward these circuits, while ascending neurons bring information in the opposite direction. The labeled source renderings preserve these anatomical categories, allowing students to recognize how neurons with different projection patterns occupy the same connected nervous system.

The researchers traced synaptic pathways from selected visual inputs toward descending outputs. Each step followed identified connections in the reconstructed specimen rather than proximity alone between two visible branches. This established candidate anatomical routes through which visual information could influence movement, with intermediate neurons contributing additional processing before signals reached populations projecting out of the brain toward the nerve cord.




Dorkenwald et al. (2024), Fig. 2, selected panels.



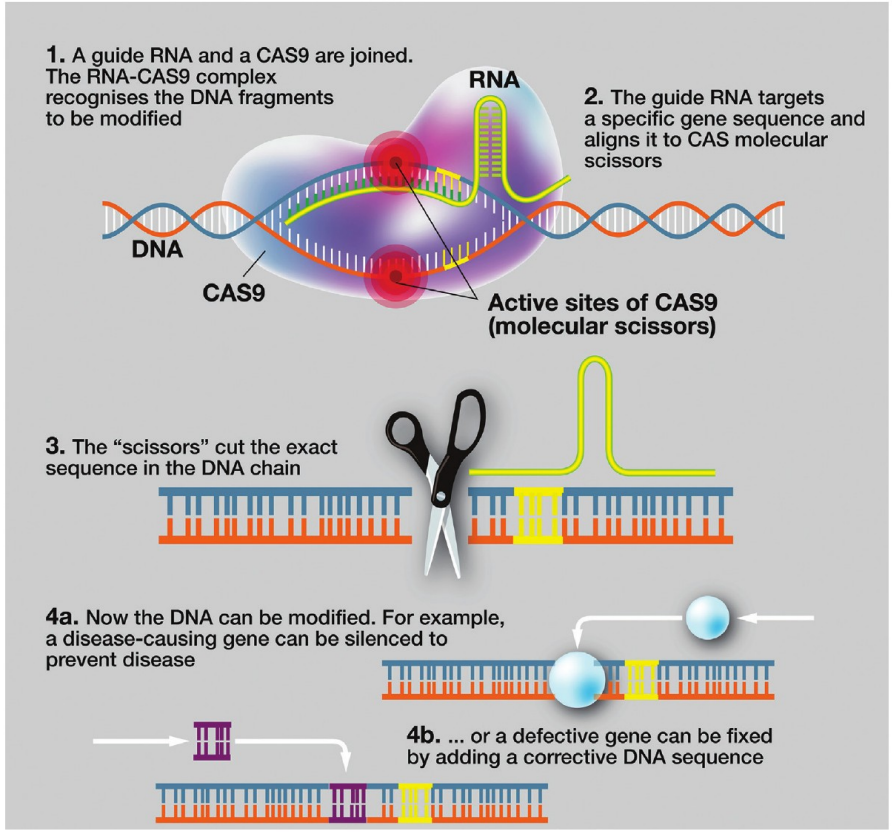
Dorkenwald et al. (2024), Supplementary Video 2.

# CRISPR directs molecular tools to a chosen DNA sequence

**CRISPR–Cas systems** use a guide RNA to direct a protein to a complementary DNA sequence near a suitable recognition motif. In common editing approaches, **Cas9** cuts the DNA at that location. The cell then repairs the break, and the repair process can change the sequence, interrupt a gene, or incorporate a supplied change under appropriate conditions in the targeted cells.

A small insertion or deletion can disrupt the instructions needed to produce a normal protein. Studying the resulting cell tests how that gene product contributes to its physiology. A **conditional manipulation** restricts the intervention to selected cells or a selected time, helping separate the gene's role in a mature neuron from its earlier contribution to the development of the nervous system.

Modified Cas proteins can also bind DNA without cutting it. When attached to components that repress or activate transcription, they can reduce or increase expression of a selected gene. **CRISPR interference**, often called **CRISPRi**, uses this targeted repression, allowing gene regulation to be changed through recruitment of transcriptional machinery rather than a permanent sequence change produced by a double-strand break.



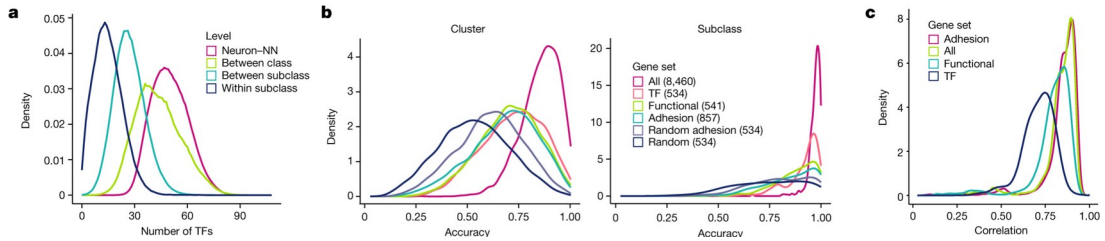

Carlson and Birkett (2022), Fig. 5.34, p. 151.

# Regulatory gene combinations distinguish cell identities

A **transcription factor** is a protein that binds DNA and influences the expression of other genes. Yao and colleagues examined these regulators across the mouse brain cell atlas. Their analysis found combinations of transcription-factor expression that distinguished molecular populations, providing a description of cellular identity based on a coordinated gene-expression pattern rather than one isolated marker in a single brain region.

Cells with related functions can share some expressed genes while differing in others. Comparing combinations makes it possible to separate populations that would be merged by a broad marker such as a transmitter-synthesis enzyme. The atlas organized these differences at several nested levels, preserving both general identities and narrower cell groups with more specific expression profiles across the brain.

These profiles can guide the design of future targeting strategies. A researcher seeking one population can examine which regulatory or marker genes distinguish it from neighboring cells. The atlas identifies candidates for this targeting; an experimental delivery system must still be tested to establish where the reporter or manipulator is actually expressed before its physiological effect is assigned to the intended population.



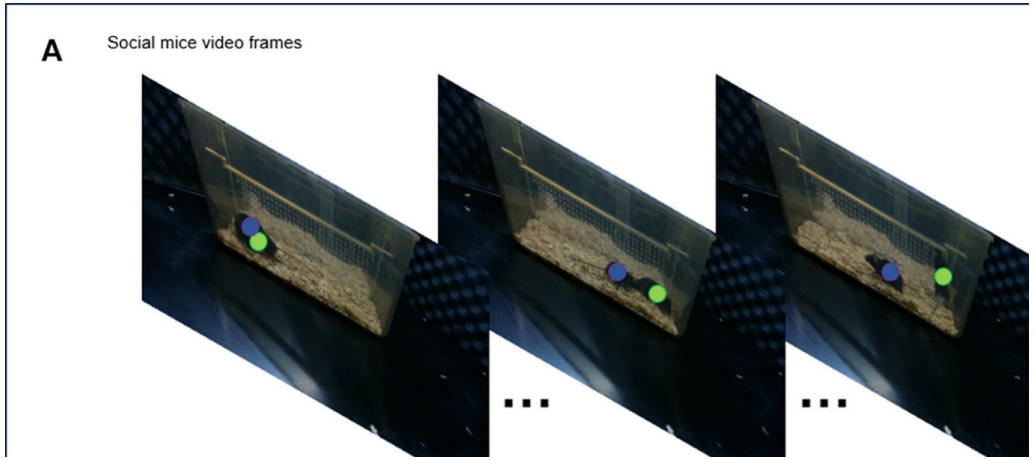

Yao et al. (2023), Fig. 5, selected panels.

# Homecage tracking follows naturally occurring behavior

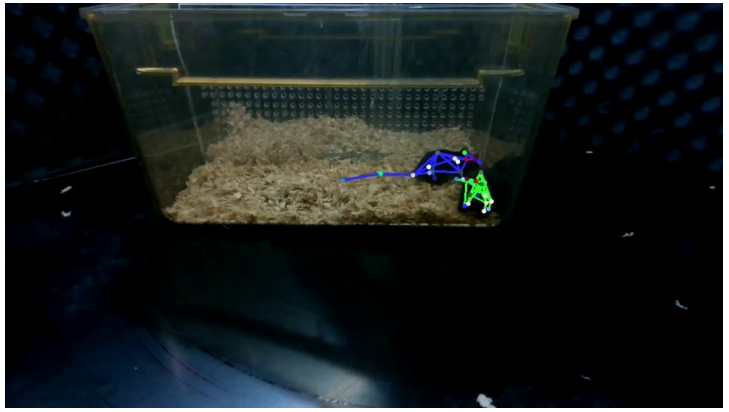
Tang and colleagues tested ADPT on mice interacting in a homecage setting. The camera observed actions occurring over time rather than only movements triggered by a brief laboratory cue. Tracking body parts and identities across these recordings allowed positions from different individuals to be distinguished even as the animals moved close together or changed their orientation relative to the camera.

The researchers compared predicted locations with annotated positions and examined performance across individuals. A method that works in a sparse arena can encounter new visual problems in a cage containing several bodies and background objects. Their homecage evaluation therefore tested whether the same tracking approach could preserve landmark placement under a different combination of scene complexity and social interaction.

Continuous positional data can describe when an animal approaches another, remains still, or moves between parts of the cage. Aligning those events with a neural recording requires shared timing between the camera and recording system. An event timestamp can then identify the corresponding electrical or optical samples, allowing naturally occurring actions to be examined without assuming that all behavior begins at an experimenter's cue.




Tang et al. (2025), Fig. 8, selected panels.



Tang et al. (2025), Fig. 8, video 1.

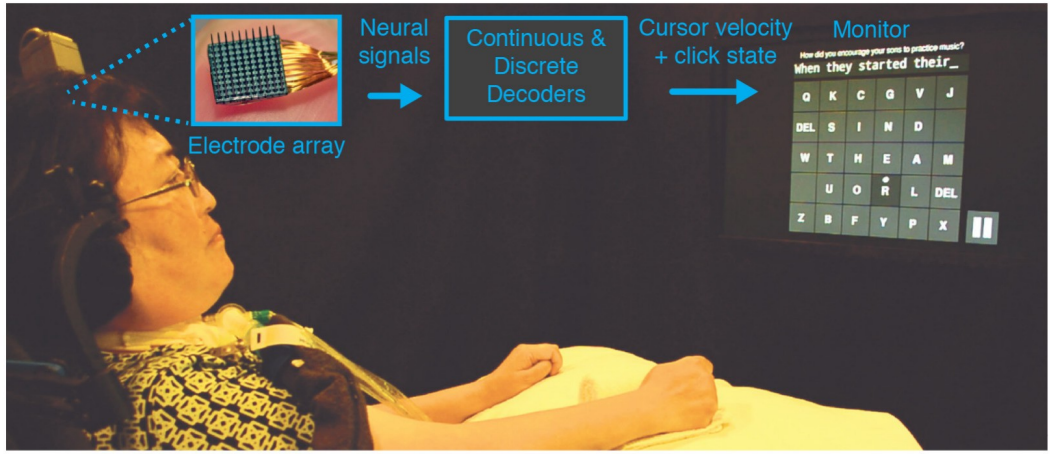
# A decoder turns neural signals into device commands

A **brain–computer interface**, or **BCI**, uses measured neural activity to operate an external device. A recording system captures electrical or optical signals, and a **decoder** converts patterns in those measurements into an output. The output might specify cursor velocity, a selected letter, or a stimulation command delivered to a second part of the nervous system.

The decoder is calibrated using examples that pair a neural pattern with a known action or intention-related task. During later use, it receives a new pattern and estimates the appropriate output. A cursor controller might map the population's current activity onto horizontal and vertical motion, allowing the user to influence the cursor by changing the recorded neural pattern over time.

The person sees the cursor move and can adjust their next attempt. This feedback completes a **closed loop** between neural activity, decoder output, visible device behavior, and subsequent neural activity. The recording electrodes do not need to activate a muscle directly because the external computer receives the command that would otherwise have contributed to an action through the body's motor pathways.

a



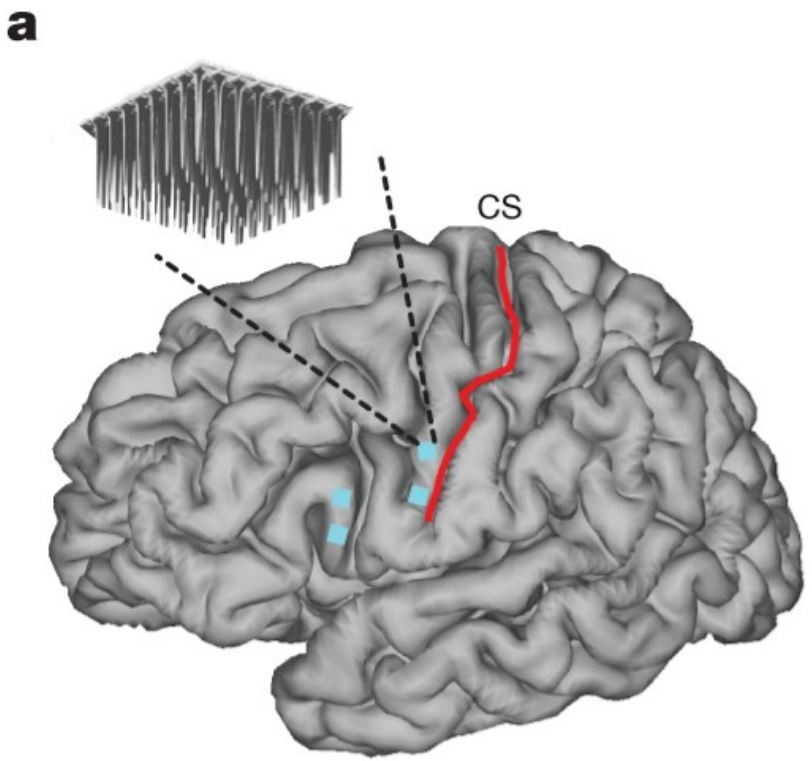

Pandarinath et al. (2017), Fig. 1, selected panels.

# Attempted speech retains distinct motor-cortex patterns

Willett and colleagues implanted microelectrode arrays in a participant with amyotrophic lateral sclerosis, or **ALS**, whose speech had become unintelligible. The arrays recorded activity in cortical regions involved in speech production. The participant attempted spoken sounds and words while the researchers measured how the recorded neuronal population changed across different movements of the mouth and different attempted speech items.

Producing speech normally requires coordinated movements of the lips, tongue, jaw, and larynx. Cortical populations contribute patterns related to these **articulators**, the structures shaping the speech sound. In this participant, patterns distinguishing attempted articulatory actions remained detectable despite the severe impairment of intelligible vocal output, allowing the experiment to access the cortical activity preceding an unsuccessful spoken response.

The recorded patterns were distributed across many electrodes, with different sounds and movements producing different population responses. A decoder could use these differences rather than requiring a separate electrode for every word. The experiment therefore supplied a neural input that retained information about attempted speech even when listening to the participant's natural vocalizations did not provide an intelligible version of the intended sentence.



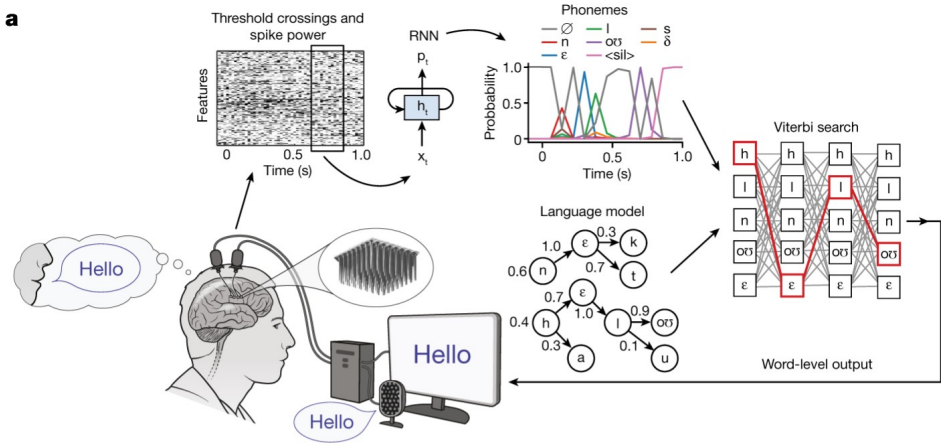

Willett et al. (2023), Fig. 1, selected panels.

# A speech decoder estimates successive sound units

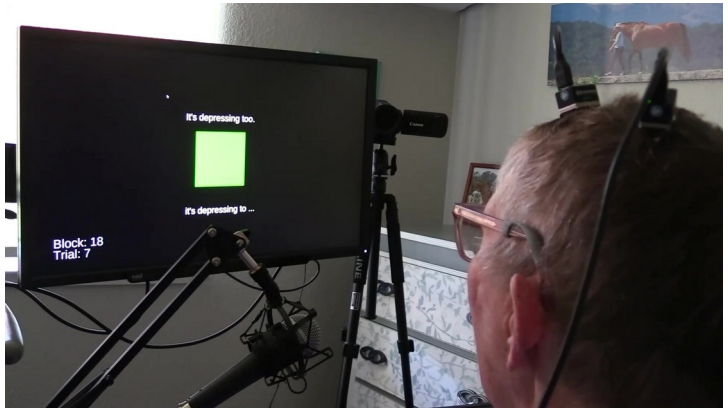
The speech decoder first converted electrode signals into measurements such as threshold-crossing counts and spike-band power. A **threshold crossing** marks an electrical event that exceeds a selected detection level. These measurements summarized the activity available from the implanted arrays during successive short intervals while the participant attempted to speak sentences displayed on the screen.

A recurrent neural network then estimated a sequence of **phonemes**, the sound units that distinguish words in a language. A language model used those estimates to identify likely word sequences. The neural model therefore supplied information about the attempted sound pattern, while the language model supplied constraints about plausible combinations of sounds and words in the final decoded sentence.

In the real-time demonstration, the participant prepared a sentence, attempted to speak when cued, and viewed the decoder's output. The system then read the resulting text aloud using synthetic speech. The study reported attempted-speech decoding at 62 words per minute, a measurement of the interface's output during this task rather than a restoration of normal vocal movement through the participant's speech muscles.




Willett et al. (2023), Fig. 2, selected panels.



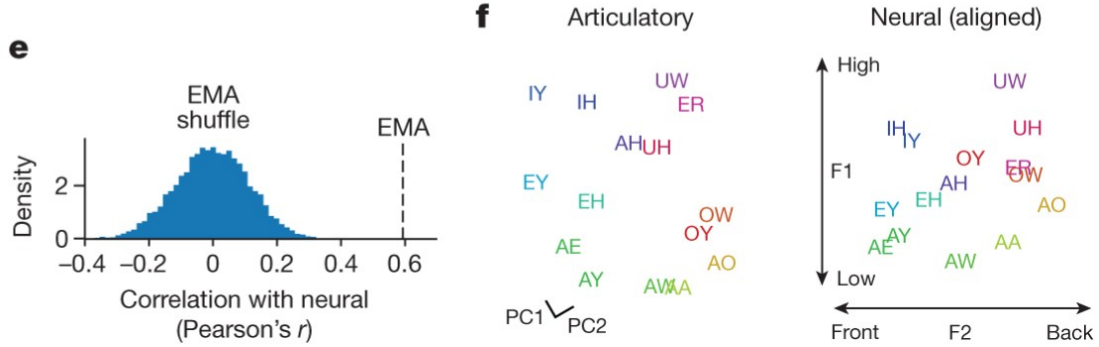

Willett et al. (2023), Supplementary Video 1.

# Related speech sounds evoke related cortical patterns

Willett and colleagues compared the neural responses to different speech sounds with measurements of articulatory movements. Sounds produced using similar mouth configurations tended to evoke related population patterns. This organization suggests that the recorded cortical activity retained aspects of the motor structure of speech rather than representing each attempted sound as a completely independent signal unrelated to the others in the participant's neural population.

A **phoneme representation** describes how the activity pattern differs across speech sounds. Comparing patterns can reveal whether lip-related sounds, tongue-related sounds, or other articulatory groups occupy nearby positions in a mathematical representation of the data. These positions summarize measured similarities; they do not correspond to literal locations of the sounds inside a small physical map of cortex.

The preservation of this articulatory structure helps explain why a decoder trained on sound components can generate many word sequences. The system does not need a separately recorded cortical pattern for every possible sentence. It combines information about the changing attempted sounds with a language model, allowing the same underlying sound distinctions to contribute to different words and phrases during communication through the interface.



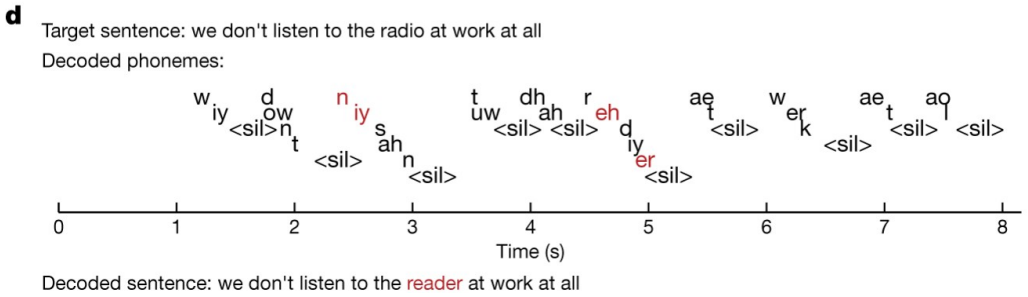

Willett et al. (2023), Fig. 3, selected panels.

# Silent speech attempts can drive the same interface

The speech study also tested a condition in which the participant silently mouthed the words rather than attempting to produce vocalizations. The implanted arrays continued recording cortical activity during these attempts. The source demonstration displays the resulting decoded text, showing that the computer's input came from the neural measurements rather than a microphone recognizing intelligible sounds produced by the participant.

The visual cue specified when to prepare and when to begin the attempted sentence. This timing separated the preparation interval from the active decoding interval. The network then interpreted changes in the recorded population over time, rather than assigning a word to a single isolated spike or treating every moment of the trial as an attempt to speak through the interface.

Errors could arise when the estimated sound sequence differed from the intended one or when the language model selected a different word sequence. The study evaluated word errors against the prompted sentence and examined performance with different vocabulary sizes. This directly measures the correspondence between the requested communication and the decoded output while preserving the distinction between cortical speech-related activity and the participant's actual acoustic output.




Willett et al. (2023), Fig. 2, selected panels.



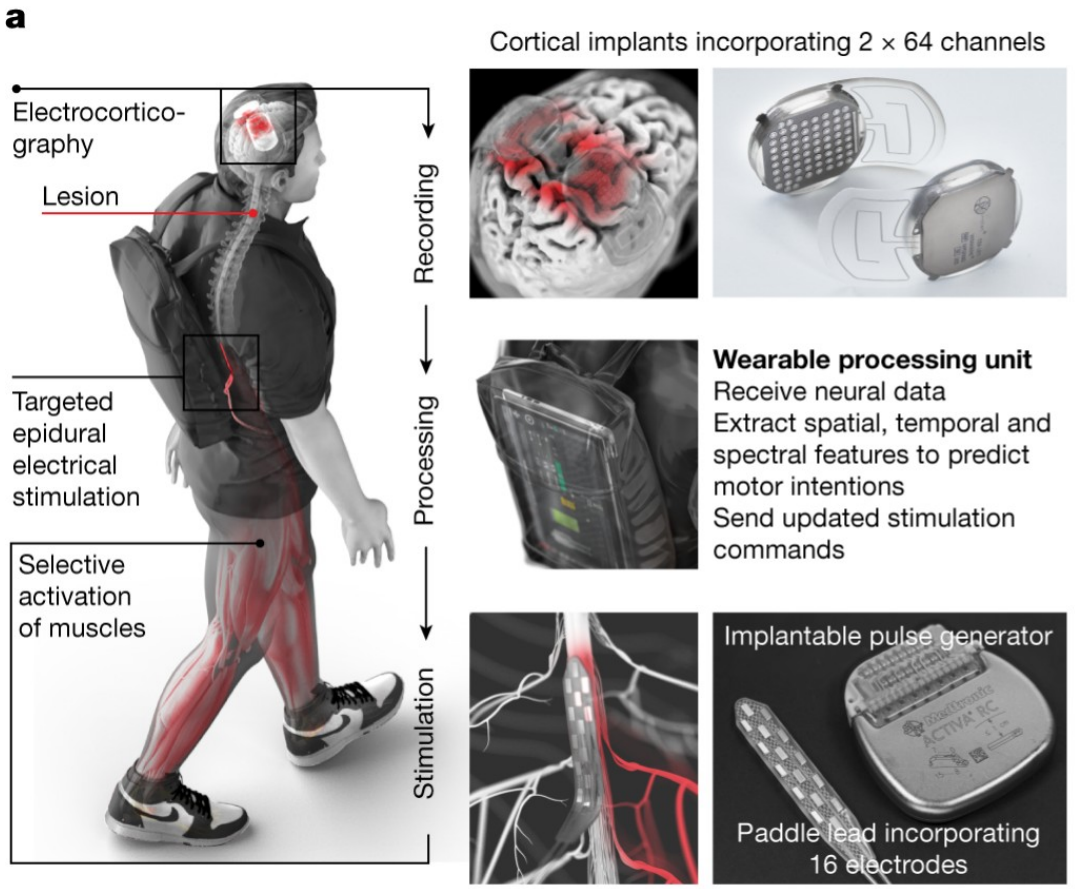

Willett et al. (2023), Supplementary Video 2.

# A brain–spine interface bridges an injured pathway

Lorach and colleagues developed a **brain–spine interface** for a participant with a chronic spinal cord injury. Cortical implants recorded signals over regions involved in controlling the legs. A processor decoded these signals and adjusted electrical stimulation delivered below the injury, providing a digital route between attempted movement in the brain and spinal circuits that could help activate the legs.

The cortical recording system used **electrocorticography**, or **ECoG**, which measures population electrical activity from electrodes positioned near the cortical surface. This differs from the penetrating microelectrode arrays used in the speech study. The interface decoded features of these surface signals associated with attempted leg actions, then translated the estimated action into a spinal stimulation program matched to that movement.

An implanted pulse generator delivered stimulation through electrodes over the lower spinal cord. This **epidural stimulation** reaches neural structures from outside the spinal cord itself. Adjusting the stimulated locations, pulse timing, and intensity helped recruit spinal pathways associated with particular leg movements, allowing the decoded cortical command to influence an output that the injury had made difficult to generate through the original descending route.



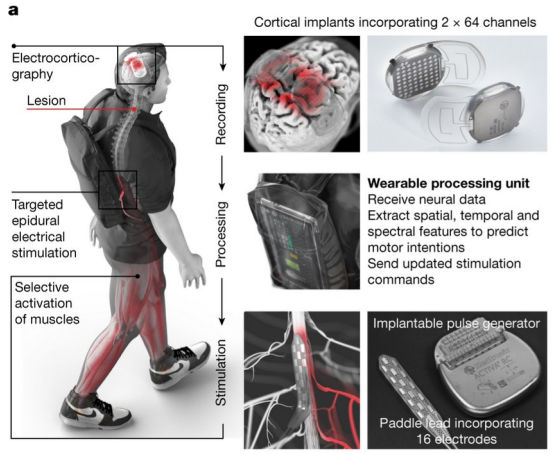

Lorach et al. (2023), Fig. 1, selected panels.

# Implanted devices exchange commands during movement

The brain–spine interface contained implanted cortical recording devices, a processing system, and an implanted spinal stimulator. Lorach and colleagues designed these components to exchange information during ongoing behavior. The cortical measurement was therefore not simply stored for later analysis; its decoded output changed the stimulation program while the participant attempted to stand or move the legs during use of the interface.

The processor estimated which movement was being attempted and how strongly to recruit the corresponding stimulation. This transformed a continuously changing neural measurement into a command with a physiological effect below the spinal injury. The source design demonstration shows how the recording, decoding, and stimulation components occupy different locations while operating as parts of the same functional system during movement.

Power, communication timing, and a stable signal affect whether the command arrives when the movement is needed. A delayed stimulation program could recruit a useful muscle group at the wrong part of the step. Coordinating the components allows the decoded leg intention to influence spinal activation in an order compatible with standing, initiating a step, and advancing the limb through the movement.




Lorach et al. (2023), Fig. 1, selected panels.



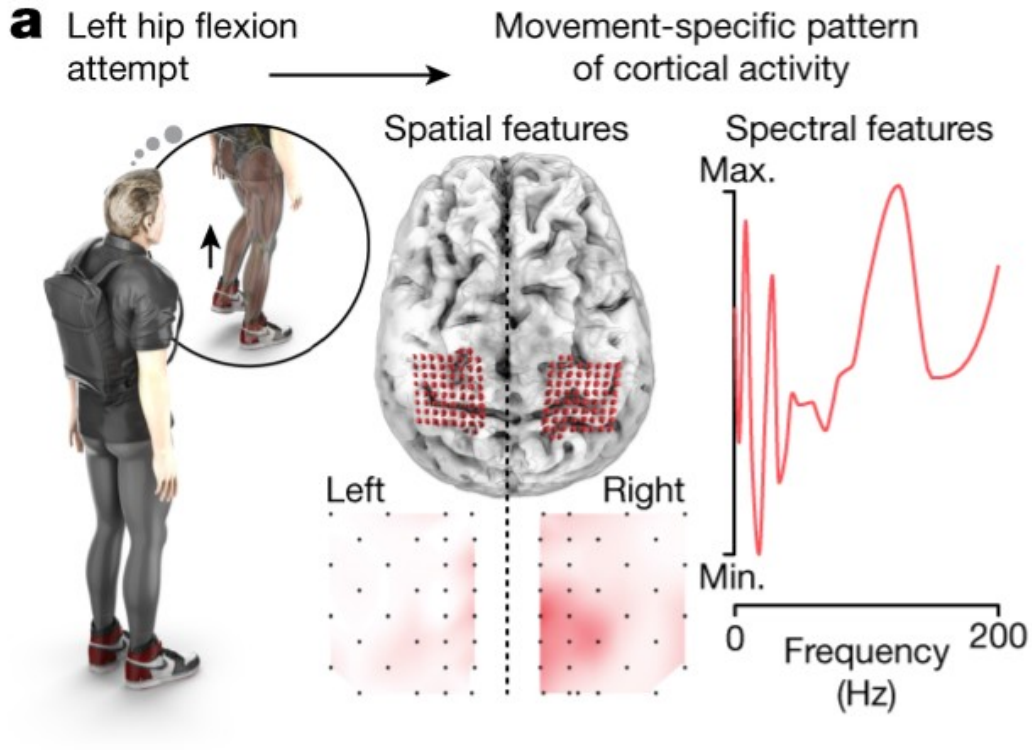

Lorach et al. (2023), Supplementary Video 1.

# Calibration pairs cortical patterns with spinal outputs

Lorach and colleagues calibrated the interface by recording cortical signals while the participant attempted selected leg movements. They identified signal features associated with those attempts and trained the decoder to distinguish them. This established a mapping from the recorded population activity to commands representing particular leg actions, rather than assuming that the same electrical pattern would mean the same movement for every individual.

The team separately tested spinal electrode configurations and stimulation parameters. Different settings recruited different combinations of muscles and movements. **Electromyography**, or **EMG**, measured electrical activity in the muscles, providing a physiological record of which outputs each program produced. This calibration paired a decoded cortical action with a stimulation pattern capable of helping generate a corresponding movement of the participant's leg.

The completed interface combined the two mappings during use. Cortical activity selected and modulated the spinal program, which changed muscle recruitment and limb motion. Measuring the resulting movement then revealed whether the command produced the intended action, allowing the system's neural input, stimulation output, and visible mechanical effect to be examined as distinct stages of the same attempted movement.



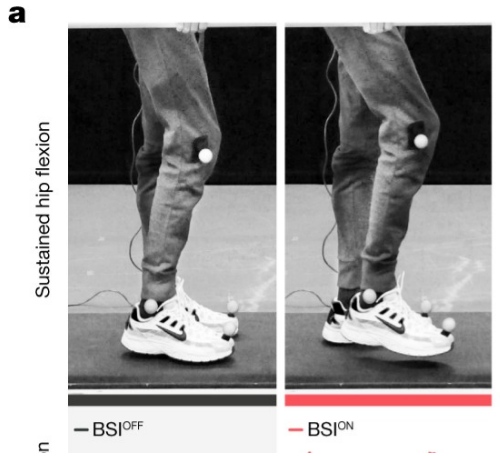

Lorach et al. (2023), Fig. 2, selected panels.

# Cortical commands help organize walking after injury

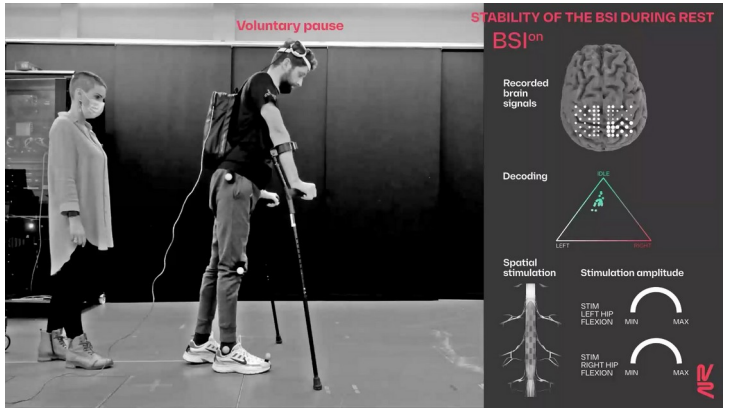
During interface use, the participant's cortical signals controlled spinal stimulation associated with leg movements. Lorach and colleagues measured changes in muscle activity and limb motion with the interface operating. The source demonstrations include standing and walking, showing how the decoded input and stimulation were used during recognizable actions rather than only during isolated contractions recorded while the participant remained stationary.

Walking requires the legs to alternate their supporting and advancing roles. The interface adjusted stimulation as the participant attempted successive movements, helping activate spinal pathways at the appropriate parts of that sequence. The resulting motion still depended on the participant's remaining nervous system, body mechanics, balance support, and sensory feedback from the moving limbs, rather than an external device moving the legs directly.

The participant also used the system for actions such as negotiating steps and changing the walking environment. These conditions require different combinations of joint movements from steady walking on a level surface. The demonstrations therefore examine whether cortical control can modulate stimulation as the intended action changes, rather than replaying one fixed sequence of stimulation pulses regardless of the participant's current movement attempt.




Lorach et al. (2023), Fig. 3, selected panels.



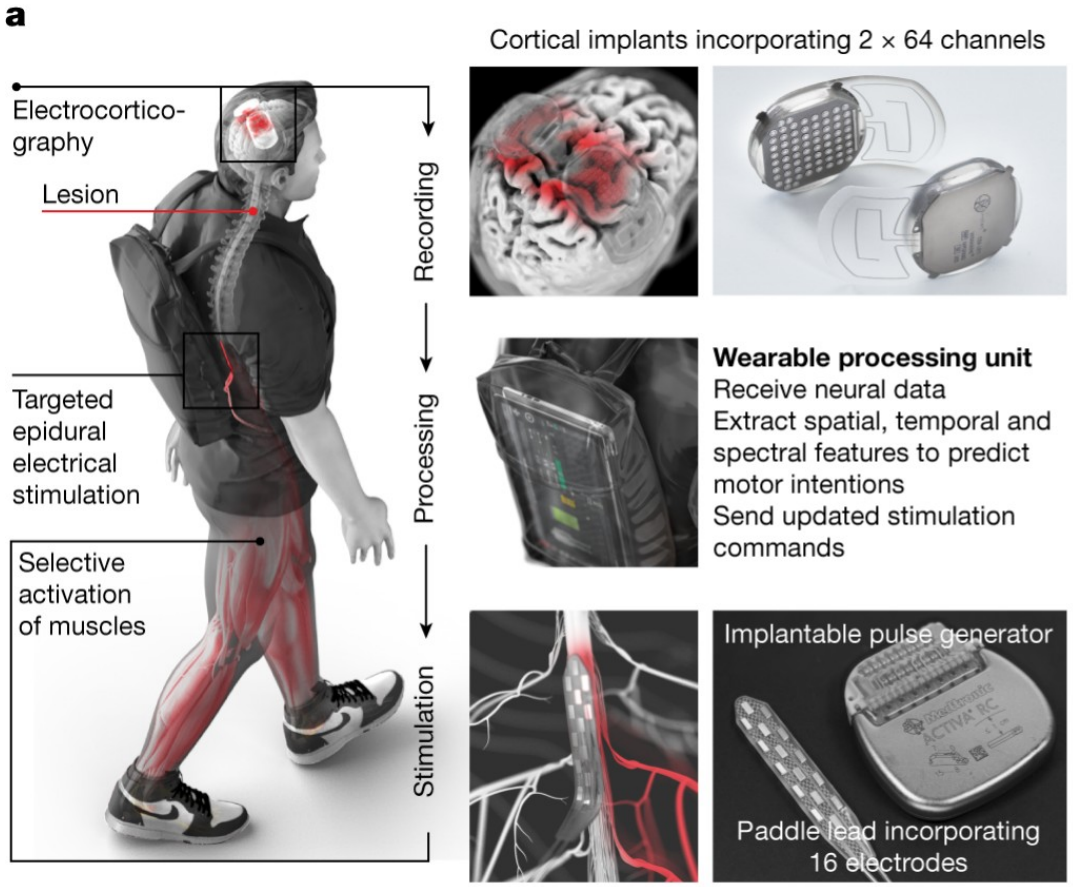

Lorach et al. (2023), Supplementary Video 2.

# Feedback closes the loop around neural control

An **open-loop controller** delivers a command without using the resulting outcome to adjust its next action. A fixed train of stimulation pulses can operate this way. In a **closed-loop system**, a measurement influences the command, and the consequences return through a feedback pathway that can change the following command during ongoing interaction with the body or the environment.

A brain–spine interface closes a loop through the participant. Neural activity changes stimulation, the legs move, and the participant receives visual and bodily sensory feedback about that movement. The participant can then adjust subsequent attempts. An automated controller can add another loop by using measured physiological or mechanical output to update the stimulation directly according to a specified control rule.

**Latency** is the delay between a measured event and the resulting command or response. Recording, data transmission, decoding, and stimulation each contribute part of this delay. A useful controller must deliver its output while the relevant state still persists, particularly when a short phase of movement requires one muscle group to be recruited before the next phase begins in the sequence.



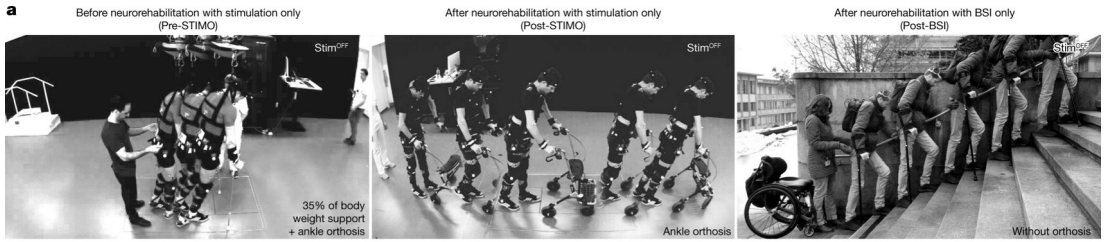

Lorach et al. (2023), Fig. 1, selected panels.

# Rehabilitation changes performance without stimulation

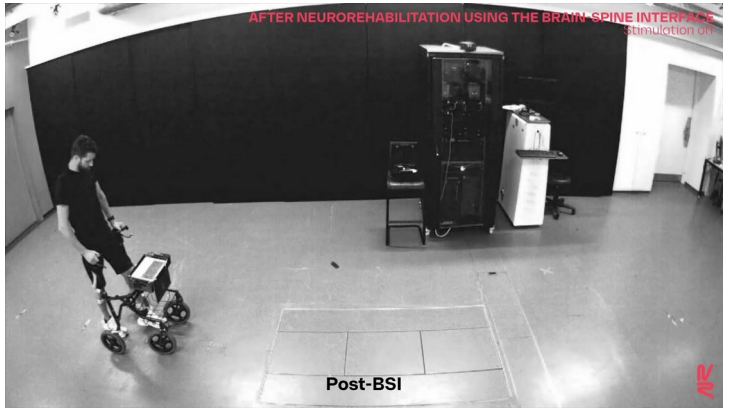
Lorach and colleagues followed the participant during rehabilitation supported by the brain–spine interface. They also assessed movement without stimulation, separating immediate assistance from changes that persisted outside active interface use. After training, the participant could perform some movements and walk with crutches overground without stimulation, demonstrating an improvement in the measured unstimulated performance relative to the earlier assessment in the study.

The source comparisons include muscle recordings and walking measures across assessment periods. These measurements examine what the participant's remaining biological system could produce when the digital bridge was not supplying stimulation. Comparing device-on and device-off conditions answers different questions: one measures performance with current assistance, while the other describes retained function after the period of practice and rehabilitation supported by the interface.

The study did not show that the original injury had disappeared or that normal spinal anatomy had been restored. The observed improvement concerns motor function available through surviving biological pathways after training. Distinguishing this retained ability from the immediate effects of stimulation specifies what changed in the participant's performance without inventing a particular cellular repair process that was not directly measured in the clinical study.




Lorach et al. (2023), Fig. 4, selected panels.




Lorach et al. (2023), Supplementary Video 3.

# Cursor motion and selection require different commands

Pandarinath and colleagues used an intracortical BCI to support communication by people with paralysis. The participants controlled a cursor to choose letters on an on-screen keyboard. Cursor motion specified where to point, while a separate selection rule determined when a key was activated, allowing pointing and committing to a choice to be handled as different parts of the interface task.

One selection method detected a neural pattern associated with an intended click. Another used **dwell time**, selecting a target after the cursor remained within it for a specified interval. These approaches place different demands on the user: one requires producing a distinguishable selection pattern, while the other requires keeping the cursor inside the desired target long enough for selection to occur.

Keyboard layout also changes the movements required to produce a message. Nearby frequent letters can reduce cursor travel, while familiar layouts can make the intended target easier to find. The investigators compared several layouts and selection conditions, measuring the resulting text output rather than assuming that an accurate movement decoder alone determines how quickly a person can communicate through the complete cursor-based interface.



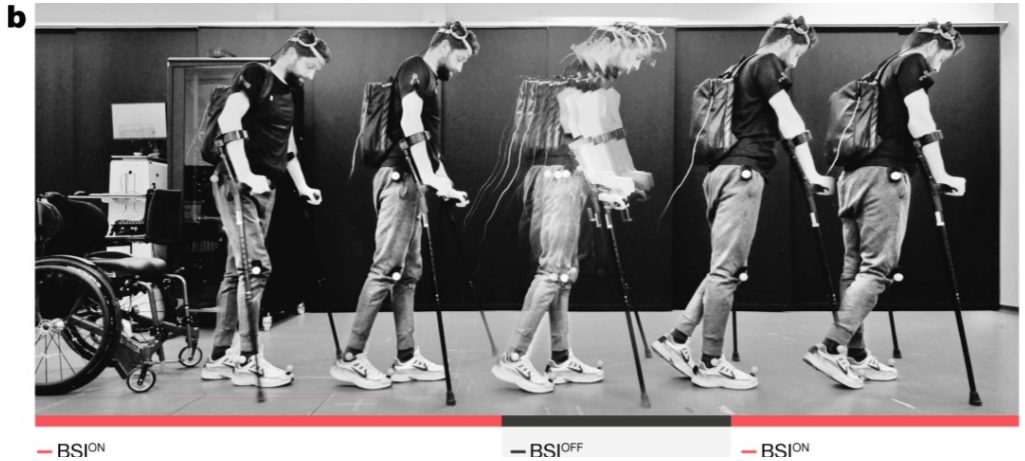

Pandarinath et al. (2017), Fig. 2, selected panels.

# A neural interface operates during independent daily use

The brain–spine study included independent use of the interface by the participant. This required recording and stimulation to remain usable while the participant prepared the equipment and moved outside a tightly repeated laboratory sequence. The source demonstration shows operation of the actual system in this setting, preserving the distinction between an implanted device's technical design and the actions a person can carry out using it.

The cortical decoder continued to translate attempted movements into stimulation commands as the participant changed activities. Stable calibration reduced the need to rebuild that mapping for every brief use. The investigators reported that the system remained reliable over a year, including use at home, allowing the contribution of the interface to be examined across repeated sessions rather than a single successful demonstration during initial testing.

Every intended action still passed through several physical stages: neural signals were measured, a movement was estimated, stimulation recruited spinal pathways, and the legs interacted with the supporting surface. A change in terrain altered the mechanical demands even when the same general walking intention persisted. The participant's continuing sensory feedback and adjustments were therefore part of operating the interface across these different movement conditions in daily use.




Lorach et al. (2023), Fig. 3, selected panels.



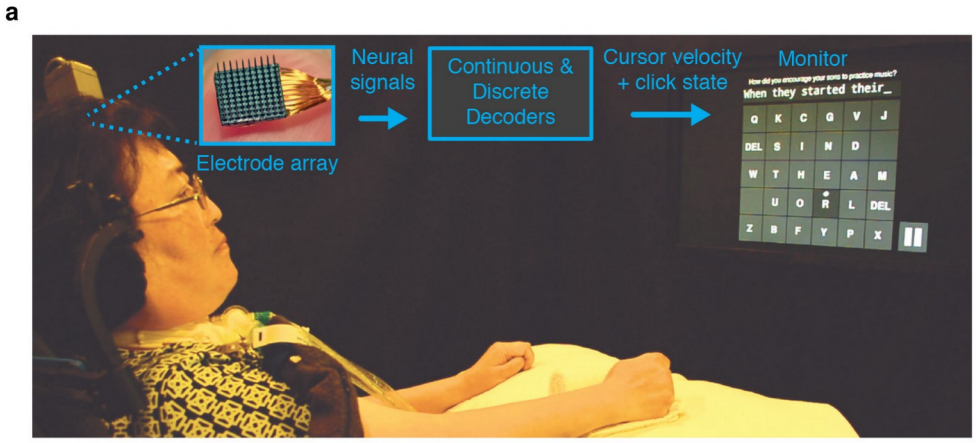

Lorach et al. (2023), Supplementary Video 4.

# Neural control supports self-generated written messages

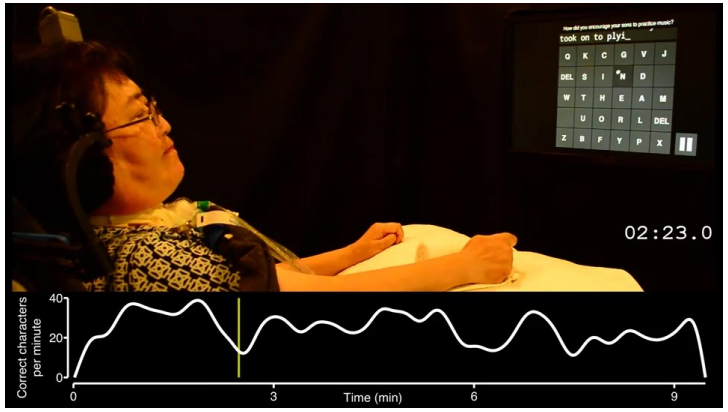
Pandarinath and colleagues tested **free-choice typing**, in which a participant composed an answer rather than copying a displayed sentence. The participant used recorded neural activity to move the cursor and select letters on the screen. The source recording follows this process through the production of an original response, showing the complete sequence from a question to a message created through the interface.

Visual feedback showed the cursor position, the selected letters, and the developing message. The participant could adjust the next movement or correct a mistake using this feedback. The decoder therefore operated within a continuing interaction between intention-related neural activity and the visible state of the keyboard, rather than generating an entire answer from one fixed measurement taken before typing began in the task.

In the free-choice example, participant T6 produced 255 characters in roughly nine minutes of task time. This measure counts entered characters rather than spoken words. A completed selection adds a character to the message, so the communication rate includes time spent locating a target, moving the cursor, and confirming the choice through the interface.




Pandarinath et al. (2017), Fig. 1, selected panels.




Pandarinath et al. (2017), Video 1.

# Key takeaways

**Perturbation tests a contribution.** Lesions, temporary inactivation, and targeted stimulation change neural activity; recordings measure the resulting signals and behavior.

**Every signal has a physical source.** Spikes, scalp currents, BOLD contrast, calcium fluorescence, and RNA abundance describe different processes with different timing.

**Anatomy needs directional evidence.** Anterograde and retrograde tracing identify projection direction; molecular labels and electron microscopy reveal different levels of circuit organization.

**Targeting has several steps.** Genetic identity, delivery location, expression, and illumination determine which cells or compartments a manipulation reaches.

**Gene tools act at different levels.** Reporters make processes visible, antisense changes RNA use, and CRISPR can alter DNA sequence or regulate transcription.

**Interfaces transform signals into actions.** Decoders map neural measurements to text, cursor movement, or stimulation; feedback allows users and controllers to adjust the next command.

Carlson & Birkett (2022); primary sources in slide notes